Kinetic Acceleration to Non-Equilibrium Steady States: An Operator-Theoretic Analysis

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We analyze the convergence rates of dissipative dynamical systems evolving towards Non-Equilibrium Steady States (NESS). We consider generators of the form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$, characterized by a timescale separation parameter γ and a non-equilibrium perturbation strength η. In the strong dissipation limit $\gamma \to \infty$ (adiabatic elimination / Quantum Zeno regime), the slow dynamics converge to an effective generator $\mathcal{L}_O$; in the lifting regime, the relevant macroscopic coercivity is captured by the self-adjoint reference operator Q_O associated with $\mathcal{V}_{\text{skew}}$. Using operator-modified Lyapunov functionals, we derive explicit lower bounds on the asymptotic decay rate (spectral bound) of $\mathcal{L}$ and show a quadratic improvement at an intermediate optimal friction scale, with rate $\Theta(\sqrt{\lambda(Q_O)})$. We also establish quantitative stability conditions: the acceleration persists under non-conservative perturbations whose skew and symmetric parts are controlled by the fast dissipation through the $(-\mathcal{R})^{1/2}$-energy norm, provided η remains below an explicit friction-dependent threshold. We illustrate the applicability of these bounds to nonequilibrium Langevin dynamics, a regularized Wigner–Fokker–Planck model, and a finite-dimensional boundary-driven toy model motivated by Zeno-limit/boundary-driving settings, where the effective slow generator is typically non-self-adjoint.

Keywords: Hypocoercivity, non-equilibrium steady states, spectral bound, adiabatic elimination, Quantum Zeno limit

I. INTRODUCTION

The relaxation dynamics of open quantum systems, describing entities coupled to uncontrollable environments, constitute a cornerstone of modern mathematical physics. In the regime where the environment is memoryless, the evolution of the system's state is governed by a quantum Markov semigroup (QMS) generated by a Lindbladian $\mathcal{L}$.[1,2] A central problem in the analysis of such systems is quantifying the rate of convergence to the stationary state. For systems coupled to a single thermal bath, the dynamics typically relax towards a Gibbs state satisfying the Detailed Balance Condition (DBC). This reversibility renders the generator $\mathcal{L}$ self-adjoint in an appropriate scalar product, permitting a robust convergence analysis via spectral gap estimates and functional inequalities such as the modified logarithmic Sobolev inequality.[3–5]

However, the most intriguing physical phenomena often emerge far from equilibrium. Driven-dissipative many-body systems, boundary-driven transport setups, and reservoir engineering protocols typically evolve towards a Non-Equilibrium Steady State (NESS).[6–8] In these scenarios, the generator $\mathcal{L}$ is inherently non-normal and violates detailed balance, sustaining non-trivial probability currents even in the steady state. Analyzing convergence to a NESS presents formidable analytical challenges: standard spectral tools for self-adjoint

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operators fail, and the interplay between coherent driving and dissipation can lead to complex dynamical phase transitions. Notably, in the limit of strong dissipation, the system may exhibit Zeno localization or critical slowing down, rendering equilibrium estimates overly pessimistic.[9–11]

A powerful paradigm for simplifying such multi-scale dynamics is adiabatic elimination (AE).[12,13] Consider a generator of the form $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$, where the dissipator $\mathcal{R}$ generates rapid relaxation towards a decoherence-free subspace $\mathcal{H}_S = \ker(\mathcal{R})$, and $\mathcal{V}$ represents a coupling interaction. AE rigorously derives an effective generator $\mathcal{L}_O$ acting solely on the slow subspace $\mathcal{H}_S$. The recent work of Carlen, Huse, and Lebowitz has placed the analysis of such systems on firm mathematical ground in the Zeno limit ($\gamma \to \infty$), including quantitative convergence to effective reduced dynamics and controlled asymptotic expansions under their structural assumptions.[14]

It is crucial, however, to distinguish the physical implications of this limit depending on the operational goal. In the context of quantum error correction (QEC), the suppression of transitions out of the code space serves as a mechanism for Zeno protection, preserving quantum memory against logical errors.[11,15] Conversely, for tasks requiring the rapid preparation of a specific NESS or the efficient extraction of information (such as dispersive qubit readout), this "freezing" of the dynamics constitutes a kinetic bottleneck. The convergence rate in the Zeno limit is enslaved by the timescale of the effective dynamics, which typically scales as $\Theta(\gamma^{-1})$.

To address the challenge of rapid convergence at finite friction, we turn to the technique of lifting, which is classical in kinetic theory and Markov chain Monte Carlo.[16,17] Lifting embeds a slow (typically diffusive) effective dynamics on $\mathcal{H}_S$ into a larger phase space by introducing auxiliary degrees of freedom so that the resulting dynamics explores state space ballistically at intermediate time scales. This juxtaposition reveals a significant theoretical gap. Zeno-limit analyses describe how the dynamics reduce to a general effective slow evolution as $\gamma \to \infty$, while standard lifting theory only provides sharp acceleration mechanisms in detailed-balance settings. A central question is whether the acceleration induced by the lifting structure remains stable under additional non-equilibrium perturbations, and how the admissible perturbation strength scales with the friction parameter.

In this paper, we establish a rigorous framework for non-equilibrium lifting, demonstrating that the acceleration provided by the lifting structure is robust against non-equilibrium perturbations, provided their strength is within an explicit range. Specifically, we analyze generators of the form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ as hypocoercive lifts bridging two asymptotic regimes: the Zeno limit ($\gamma \to \infty$), where mixing is suppressed, and the intermediate regime ($\gamma \approx \gamma_{\text{opt}}$), where convergence is optimally accelerated.

Our main theoretical contribution is a spectral analysis based on operator-modified Lyapunov functionals.[18–20] Concretely, we introduce an energy functional

$$\mathcal{E}_{\alpha,\mu}(f) := \frac{1}{2}\|f\|^2 + \alpha\,\Re\langle \mathcal{A}_\mu f, f\rangle, \tag{1}$$

where $\mathcal{A}_\mu$ is a macroscopic solver defined by a resolvent equation on the slow space. It is chosen so that differentiating the cross term along $\partial_t f_t = \mathcal{L} f_t$ produces a coercive contribution on the slow component.

We establish: (i) quantitative stability bounds for all frictions (Theorem V.1); (ii) an overdamped diffusive ceiling showing the intrinsic γ^{-1} bottleneck at high friction (Theorem V.4); (iii) applications to three representative examples.

The paper is organized as follows. Section II fixes notation and the $L^2(\sigma)$ framework. Section III recalls adiabatic elimination and the effective generator $\mathcal{L}_O$. Section IV states the non-equilibrium lifting assumptions and introduces the macroscopic solver. Section V proves explicit lower/upper bounds on the decay rate and identifies the optimal friction scale. Section VI provides representative examples. Section VII concludes.

II. PRELIMINARIES

We establish the operator-theoretic framework governing quantum Markov semigroups, with a specific focus on the geometry of the weighted L^2-spaces required to analyze convergence to nonequilibrium steady states.

A. Weighted Hilbert Spaces and Generator Decomposition

Let $\mathcal{H}$ be a finite-dimensional Hilbert space and $\mathcal{B}(\mathcal{H})$ be the algebra of linear operators on $\mathcal{H}$. We denote the identity element in $\mathcal{B}(\mathcal{H})$ by $\mathbf{1}$ and the identity map on $\mathcal{B}(\mathcal{H})$ by id. Given a full-rank quantum state $\sigma \in \mathcal{B}(\mathcal{H})$ (i.e., $\sigma \succ 0$, $\mathrm{tr}(\sigma) = 1$), we equip $\mathcal{B}(\mathcal{H})$ with the KMS (Kubo–Martin–Schwinger) inner product:

$$\langle X, Y\rangle_\sigma := \mathrm{tr}\big(\sigma^{1/2} X^\dagger \sigma^{1/2} Y\big), \qquad X, Y \in \mathcal{B}(\mathcal{H}), \tag{2}$$

and denote the associated Hilbert space by $L^2(\sigma)$ with norm $\|X\|_\sigma := \sqrt{\langle X, X\rangle_\sigma}$.

Let $(\mathcal{P}_t)_{t\geq 0}$ be a quantum Markov semigroup (QMS) acting on $\mathcal{B}(\mathcal{H})$ in the Heisenberg picture. The generator $\mathcal{L}$ is a Lindbladian of GKSL form

$$\mathcal{L}(X) = i[H, X] + \sum_\alpha \Big(L_\alpha X L_\alpha^\dagger - \tfrac{1}{2}\{L_\alpha^\dagger L_\alpha, X\}\Big), \tag{3}$$

where $H = H^\dagger$ is the Hamiltonian and $\{L_\alpha\}$ are the Lindblad jump operators. Let $\mathcal{L}^\dagger$ denote the adjoint of $\mathcal{L}$ with respect to the σ-inner product, i.e. $\langle X, \mathcal{L}Y\rangle_\sigma = \langle \mathcal{L}^\dagger X, Y\rangle_\sigma$.

We assume $\mathcal{P}_t$ is primitive, that is, its dual semigroup $\mathcal{P}_t^\dagger = \exp(t\mathcal{L}^\dagger)$ admits a unique full-rank invariant state σ, implying $\mathcal{L}^\dagger(\sigma) = 0$. We define the symmetric (dissipative) and antisymmetric (Hamiltonian-like) parts of the generator as

$$\mathcal{S} := \tfrac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger), \qquad \mathcal{A} := \tfrac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger). \tag{4}$$

By construction, $\mathcal{S}$ is self-adjoint and negative semidefinite ($\mathcal{S} \preceq 0$), while $\mathcal{A}$ is skew-adjoint ($\mathcal{A}^\dagger = -\mathcal{A}$).

Definition II.1 (Detailed balance vs. non-reversibility)**.** We say that $\mathcal{L}$ satisfies a detailed-balance condition (DBC) (with respect to $\langle\cdot,\cdot\rangle_\sigma$) if it is self-adjoint on $L^2(\sigma)$, i.e. $\mathcal{L} = \mathcal{L}^\dagger$, equivalently $\mathcal{A} = 0$. Otherwise $\mathcal{L}$ is non-reversible ($\mathcal{L} \neq \mathcal{L}^\dagger$, hence $\mathcal{A} \neq 0$) even though it may still have the same stationary state σ.

B. Spectral Gaps and Hypocoercivity

The convergence analysis relies on the spectral properties of $\mathcal{L}$ and its symmetric part $\mathcal{S}$ on the space of mean-zero observables. Let $\mathcal{H}_0 := \{X \in L^2(\sigma) : \mathrm{tr}(\sigma X) = 0\}$ be the subspace orthogonal to the fixed point $\mathbf{1}$. When needed, we denote the orthogonal projection onto $\mathbb{C}\mathbf{1}$ by $\mathbb{E}_\sigma(X) := \mathrm{tr}(\sigma X)\mathbf{1}$.

Definition II.2 (Two gap parameters)**.** On the mean-zero subspace $\mathcal{H}_0$, we will use two standard quantities.

The (symmetric) coercive gap of the dissipative part $-\mathcal{S}$ is

$$\lambda_S := \inf_{X\in\mathcal{H}_0, X\neq 0} \frac{-\langle X, \mathcal{S}X\rangle_\sigma}{\|X\|_\sigma^2}. \tag{5}$$

The asymptotic decay rate (spectral bound) of the full generator is

$$\lambda(\mathcal{L}) := \inf\left\{\Re(\mu) : \mu \in \mathrm{Spec}(-\mathcal{L}) \setminus \{0\}\right\}. \tag{6}$$

Definition II.2 is standard in semigroup theory: $\lambda(\mathcal{L})$ is the spectral bound of $-\mathcal{L}$ on the mean-zero subspace, and in finite dimensions it coincides with the exponential decay rate of the semigroup on $\mathcal{H}_0$.[21] In reversible systems (DBC), $\mathcal{L}$ is self-adjoint in $L^2(\sigma)$ and therefore $\lambda(\mathcal{L}) = \lambda_S$. In non-equilibrium systems, it can happen that $\lambda_S = 0$ (degenerate dissipation) while $\lambda(\mathcal{L}) > 0$: the skew-adjoint part can couple directions in $\ker(\mathcal{S}) \cap \mathcal{H}_0$ to dissipative directions, yielding exponential relaxation by hypocoercivity.[18,20]

More quantitatively, we will use the standard *hypocoercive estimate*.[18,20] Namely, there exist constants $C \geq 1$ and $\lambda > 0$ such that for all $X \in \mathcal{H}_0$ and $t \geq 0$,

$$\|\mathcal{P}_t X\|_\sigma \leq C\, e^{-\lambda t}\, \|X\|_\sigma. \tag{7}$$

If the estimate holds with $C = 1$, we call the dynamics *coercive* in $L^2(\sigma)$; otherwise, the need for $C > 1$ reflects the genuinely hypocoercive nature of the relaxation.

Lemma II.3 (Kernel characterization of coercivity vs. strict hypocoercivity)**.** *Assume* $\ker(\mathcal{L}) = \mathbb{C}\mathbf{1}$ *(primitivity). Then:*

1. *The semigroup is coercive in $L^2(\sigma)$ (i.e., the decay estimate holds with $C = 1$) if and only if* $\ker(\mathcal{S}) = \ker(\mathcal{L})$.

2. *If* $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$*, then no exponential decay estimate can hold with prefactor $C = 1$ and some rate $\lambda > 0$ on $\mathcal{H}_0$. In particular, whenever exponential decay does hold (e.g. for primitive finite-dimensional QMS, where the spectrum of $\mathcal{L}$ is discrete and 0 is a simple eigenvalue), any sharp estimate must be genuinely hypocoercive in the sense that it requires some $C > 1$.*

III. THE EFFECTIVE DYNAMICS AND THE ZENO LIMIT

In this section, we establish the rigorous connection between the full dynamics of the lifted system and the effective dynamics on the slow manifold. This procedure, physically known as *Adiabatic Elimination*, serves as the mathematical inverse of the lifting operation.[13,22]

We consider the generator $\mathcal{L}_\gamma$ acting on the full Hilbert space $L^2(\sigma)$, decomposed as:

$$\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}, \tag{8}$$

where $\gamma > 0$ is the timescale separation parameter. The self-adjoint **dissipator** $\mathcal{R}$ generates the fast relaxation, and the **interactor** $\mathcal{V}$ (not necessarily skew-adjoint) couples the fast and slow degrees of freedom.

We introduce the slow/fast decomposition

$$\mathcal{H}_S := \ker(\mathcal{R}), \qquad \mathcal{H}_F := \mathcal{H}_S^\perp. \tag{9}$$

Let P_S denote the orthogonal projection onto $\mathcal{H}_S$ and set $P_F := \mathrm{id} - P_S$. We assume $\mathcal{R}$ admits a spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$.

A. The Micro-Macro Decomposition

We decompose any observable $X \in L^2(\sigma)$ into macroscopic and microscopic parts:

$$X = u + v, \quad u := P_S X \in \mathcal{H}_S, \quad v := P_F X \in \mathcal{H}_F. \tag{10}$$

B. The Centered Coupling Condition

A critical structural requirement for diffusive lifting is that the interaction $\mathcal{V}$ does not generate net drift within the slow subspace to first order. Following standard centering/mean-zero (zero-average) conditions in overdamped/adiabatic-elimination limits, we enforce what

we called the **centered coupling condition**[14,23]

$$P_S \mathcal{V} P_S = 0. \tag{11}$$

This condition ensures that the effective dynamics on $\mathcal{H}_S$ arise from second-order fluctuations. More precisely, trajectories make excursions into the fast subspace $\mathcal{H}_F$ and then relax back to $\mathcal{H}_S$. Physically, this corresponds to the vanishing of the mean force in the overdamped limit or the Zeno limit.[14,23]

In the non-equilibrium lifting setting used later, we decompose

$$\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}, \tag{12}$$

where $\mathcal{V}_{\text{skew}} := \frac{1}{2}(\mathcal{V} - \mathcal{V}^\dagger)$ is anti-self-adjoint in $L^2(\sigma)$ (i.e., $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$) and $\mathcal{V}_{\text{pert}}$ is a (not necessarily normal) non-equilibrium perturbation with strength parameter η. To remain in the *diffusive* Zeno/AE scaling considered here (macroscopic time $\tau = t/\gamma$), we will enforce (11) by requiring separately that $P_S \mathcal{V}_{\text{skew}} P_S = 0$ and $P_S \mathcal{V}_{\text{pert}} P_S = 0$.

C. The Effective Generator

Under condition (11), the limiting dynamics as $\gamma \to \infty$ are governed by an effective operator acting on $\mathcal{H}_S$.

Definition III.1 (Effective Generator)**.** Let $\mathcal{R}^+$ denote the Moore-Penrose pseudoinverse of $\mathcal{R}$, which is a bounded operator on $\mathcal{H}_F$ with norm $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$. We define the **Effective Generator** $\mathcal{L}_O \in \mathcal{B}(\mathcal{H}_S)$ as:

$$\mathcal{L}_O := -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S. \tag{13}$$

In the context of lifting, $\mathcal{L}_O$ represents the *target* dynamics we wish to accelerate.

D. Convergence in the Zeno Limit

The physical justification for identifying $\mathcal{L}_O$ as the target dynamics lies in quantitative Zeno-limit convergence results. We consider the system on the *macroscopic* timescale τ, related to the microscopic timescale t by $\tau = \gamma^{-1} t$.

Carlen–Huse–Lebowitz[14] analyze boundary-driven quantum systems in the Schrödinger picture and measure errors in the trace norm $\|\cdot\|_1$. This choice is natural because completely positive trace-preserving (CPTP) maps are contractive in trace norm. Assuming that the fast boundary dissipator is ergodic and gapped, their results quantify the Zeno-limit approximation on time scales up to order γ. In particular, for initial data on the steady-state manifold, the projected evolution is approximated with error of order $T\gamma^{-2}$ uniformly for $t \in [0, T]$; for general initial data, one has an error bound of order $(\log(1+\gamma) + 1 + T)/\gamma$ uniformly for $t \in [\epsilon, \gamma T]$ (for fixed $0 < \epsilon < T$).

Here we work in the Heisenberg picture and in the weighted Hilbert space $L^2(\sigma)$, which is convenient for the operator-theoretic Lyapunov/hypocoercivity analysis developed later. We therefore record below a simplified bounded-operator $L^2(\sigma)$ variant (appropriate in finite dimensions), which makes the diffusive scaling in γ transparent in our notation.

Theorem III.2 (Convergence to Effective Dynamics)**.** *Assume* $\dim \mathcal{H} < \infty$*. Let* $X \in \mathcal{H}_S$ *be an initial slow observable. Let* $X^\gamma(\tau) := \exp(\tau\gamma\mathcal{L}_\gamma)X$ *denote the evolution under the full dynamics rescaled by* γ*. Let* $X^O(\tau) := \exp(\tau\mathcal{L}_O)X$ *denote the evolution under the effective generator.*

Assume the centered coupling condition (11) *and that the friction is sufficiently strong to ensure timescale separation, specifically* $\gamma\lambda_R > 2\|\mathcal{V}\|$*. Assume moreover that* $\mathcal{L}_O$ *generates*

a quantum Markov semigroup on $\mathcal{H}_S$ in the Heisenberg picture that leaves σ invariant in the Schrödinger picture. Define

$$C_{\mathrm{bd}} := \sqrt{\kappa(\sigma)} := \sqrt{\frac{\lambda_{\max}(\sigma)}{\lambda_{\min}(\sigma)}}\,. \tag{14}$$

Then, for any macroscopic time $\tau > 0$,

$$\|P_S X^\gamma(\tau) - X^O(\tau)\|_\sigma \le C_{\mathrm{bd}}^2 \left(\frac{1}{\gamma^2} C_{init} + \frac{\tau}{\gamma} C_{drift}\right) \|X\|_\sigma, \tag{15}$$

where

$$C_{init} = \frac{\|\mathcal{V}\|^2}{\lambda_R^2}, \quad C_{drift} = \frac{3\|\mathcal{V}\|^3}{\lambda_R^2}. \tag{16}$$

Proof. The rigorous proof is provided in Appendix B. The derivation explicitly tracks the error accumulation using Volterra–Grönwall estimates. The term scaling as γ^{-2} arises from the rapid decay of the initial boundary layer (transient fast excitations), while the term scaling as $\tau\gamma^{-1}$ represents the accumulation of higher-order drift corrections to the adiabatic manifold. □

We emphasize that Theorem III.2 is included only to give a concrete illustration of how the Zeno/AE approximation error depends on γ. Its assumptions (e.g. the finite-dimensional setting) are tailored to this simplified estimate and will not be used in our main convergence analysis below.

Theorem III.2 shows that in the strong-dissipation limit $\gamma \to \infty$ the decay rate of the full dynamics is enslaved by the effective generator, with scaling $\lambda(\mathcal{L}_\gamma) \sim \gamma^{-1}\lambda(\mathcal{L}_O)$. This is the overdamped (Zeno) regime. Our aim is to invert this logic: given a target slow dynamics $\mathcal{L}_O$, we construct a dilation $(\mathcal{H}, \mathcal{R}, \mathcal{V})$ satisfying (13) such that, at a finite friction scale γ, the convergence rate is much larger, ideally of kinetic order $\lambda(\mathcal{L}_\gamma) \sim \sqrt{\lambda(\mathcal{L}_O)}$.

IV. THE NON-EQUILIBRIUM LIFTING FRAMEWORK

Having established the physical context of the Zeno limit, we now construct the rigorous mathematical framework for acceleration at finite γ. Our approach generalizes the hypocoercivity methods developed for kinetic Langevin dynamics to the abstract setting of quantum Markov semigroups.[19]

A. Abstract Structural Assumptions

We consider a (possibly infinite-dimensional) Hilbert space $\mathcal{H}$ decomposed into a slow subspace $\mathcal{H}_S$ and a fast subspace $\mathcal{H}_F$, such that $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$; see Assumption IV.1 below. The dynamics are governed by a generator $\mathcal{L}$ depending on a timescale parameter $\gamma > 0$ and a non-equilibrium driving parameter $\eta \ge 0$:

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}. \tag{17}$$

The coupling term admits the decomposition $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$. We denote the orthogonal projections onto $\mathcal{H}_S$ and $\mathcal{H}_F$ by P_S and $P_F = \mathbf{1} - P_S$, respectively.

We make the following structural assumptions, which are essential for the validity of the Lyapunov analysis in Section V.

Assumption IV.1 (Structural Regularity of Non-Equilibrium Dynamics)**.** The system is defined on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. The generator is $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$. We assume the following structural conditions:

1. **Microscopic Spectral Gap.** The fast dissipator $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$. Moreover, it is strictly dissipative on $\mathcal{H}_F$: there exists $\lambda_R > 0$ such that

$$-\langle \mathcal{R}f, f\rangle \geq \lambda_R \|f\|^2, \qquad \forall f \in \mathcal{H}_F. \tag{18}$$

In particular, $(-\mathcal{R})^{1/2}$ is well-defined by functional calculus. This allows defining the Moore–Penrose pseudoinverse $\mathcal{R}^+$, which is a bounded self-adjoint operator on $\mathcal{H}_F$ with $-\mathcal{R}^+ \geq 0$.

2. **Centered Coupling.** We assume that the interaction does not generate an $\mathcal{O}(1)$ drift on the slow manifold in the macroscopic time scaling $\tau = t/\gamma$. Equivalently, the full interaction $\mathcal{V} := \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ satisfies

$$P_S \mathcal{V} P_S = 0. \tag{19}$$

Since η is a free parameter in our framework, we enforce this componentwise:

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \qquad P_S \mathcal{V}_{\text{pert}} P_S = 0. \tag{20}$$

This matches the centered coupling condition (11) used to define the effective generator $\mathcal{L}_O$.

3. **Fast-to-slow Regularity (Flux conditioning and flux regularity).** For $\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S$ (where Q_O is the reference effective generator defined in (29) below), define its *microscopic flux* induced by transport as

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{21}$$

In the bounded-operator (finite-dimensional) setting, Q_O is a bounded self-adjoint operator on $\mathcal{H}_S$ and thus $\text{dom}(Q_O) = \mathcal{H}_S$. In unbounded settings, the identities below are understood on a common invariant core $\mathcal{C}$ for the operators involved (specified in the examples/appendices) and extended to the relevant energy domains by density. We assume a **flux conditioning bound**: there exists a constant $M_R \in (0, \infty)$ such that

$$\|g_\psi\|^2 \leq M_R \, \langle g_\psi, -\mathcal{R}^+ g_\psi\rangle, \qquad \forall \psi \in \text{dom}(Q_O) \subset \mathcal{H}_S. \tag{22}$$

Since (22) is monotone in M_R, if it holds for some constant then it also holds for any larger one; throughout, M_R may be taken to be any convenient upper bound (at the cost of more conservative constants in subsequent estimates). In the bounded-operator (finite-dimensional) setting, since $-\mathcal{R} \geq 0$ is bounded on $\mathcal{H}_F$, one may take $M_R := \|\mathcal{R}\|$ (equivalently $\|-\mathcal{R}\|$ on $\mathcal{H}_F$). In particular, for all $h \in \mathcal{H}_F$,

$$\langle (-\mathcal{R})h, h\rangle \leq M_R \, \|h\|^2. \tag{23}$$

We also assume a **flux regularity bound**: applying the fast–fast component of the transport to the corrected flux is controlled by the induced slow operator. Concretely, there exists $\kappa_{\text{ell}} > 0$ such that for all $\psi \in \text{dom}(Q_O)$,

$$\|P_F \mathcal{V}_{\text{skew}} P_F \, (-\mathcal{R}^+) \, g_\psi\| \leq \kappa_{\text{ell}} \|P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi\| \; = \; \kappa_{\text{ell}} \|(-Q_O)\psi\|. \tag{24}$$

4. **Structure of Interactions (Decomposition of Perturbation).** The perturbation operator $\mathcal{V}_{\text{pert}}$ admits a decomposition into a skew-symmetric (possibly unbounded) part and a symmetric part controlled in the $(-\mathcal{R})^{1/2}$-energy norm $\|\cdot\|_{(1)}$:

$$\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}, \tag{25}$$

satisfying the following conditions:

(a) **Skew Perturbation Control:** The operator $\mathcal{K}_{\text{pert}}$ is skew-symmetric, i.e., for all $f \in \text{dom}(\mathcal{V}_{\text{pert}})$:

$$\Re\langle \mathcal{K}_{\text{pert}} f, f\rangle = 0. \tag{26}$$

Moreover, $\mathcal{K}_{\text{pert}}$ is relatively bounded with respect to $(-\mathcal{R})^{1/2}$: there exist constants $a_1, b_1 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{K}_{\text{pert}} f\| \leq a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|. \tag{27}$$

(b) **Symmetric Perturbation Control:** The operator $\mathcal{S}_{\text{pert}}$ is symmetric and defined at least on the energy domain $\text{dom}((-\mathcal{R})^{1/2})$. There exist constants $a_2, b_2 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{S}_{\text{pert}} f\| \leq a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|. \tag{28}$$

Equivalently, $\mathcal{S}_{\text{pert}}$ is bounded as a map from $(\text{dom}((-\mathcal{R})^{1/2}), \|\cdot\|_{(1)})$ to $\mathcal{H}$. (In PDE models, $\mathcal{S}_{\text{pert}}$ is often an unbounded multiplier on plain L^2; (28) is the standard replacement allowing it to be controlled by the fast dissipation.)

Under Assumption IV.1, we recall the definition of the effective generator from Section III, $L_O = -P_S \mathcal{V}\mathcal{R}^+\mathcal{V}P_S$. To construct the Lyapunov functional, we isolate the reversible contribution to the effective diffusion. We define the **Reference Effective Generator** Q_O as:

$$Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S. \tag{29}$$

From the skew-adjointness of $\mathcal{V}_{\text{skew}}$ and the negativity of $\mathcal{R}^+$, it follows that for any $f \in \mathcal{H}_S$,

$$\langle f, Q_O f\rangle = -\langle f, P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S f\rangle = -\langle \mathcal{V}_{\text{skew}}^\dagger f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle = \langle \mathcal{V}_{\text{skew}} f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle \leq 0. \tag{30}$$

Thus, Q_O is a self-adjoint, dissipative operator on $\mathcal{H}_S$.

Assumption IV.2 (Spectral Gap of Effective Dynamics)**.** Let $\mathcal{K} := \ker(Q_O) \subset \mathcal{H}_S$ be the kernel of the reference generator. We assume Q_O admits a **spectral gap** $\lambda(Q_O) > 0$ on the orthogonal complement $\mathcal{K}^\perp := \mathcal{H}_S \ominus \mathcal{K}$. That is, the Poincaré inequality holds:

$$-\langle f, Q_O f\rangle \geq \lambda(Q_O)\|f - P_{\mathcal{K}} f\|^2, \quad \forall f \in \mathcal{H}_S. \tag{31}$$

Consequently, for any shift parameter $\mu > 0$, the operator $(\mu - Q_O)$ is strictly positive definite on $\mathcal{H}_S$ and invertible, with operator norm $\|(\mu - Q_O)^{-1}\| \leq \mu^{-1}$.

In unbounded settings, we additionally assume (as part of the model-dependent analytic hypotheses) that $(\mu - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$.

B. The Macroscopic Solver

To control the mixing between the slow and fast subspaces, we introduce the operator $\mathcal{A}_\mu$. This operator acts as a "solver," inverting the transport generated by the skew-adjoint coupling $\mathcal{V}_{\text{skew}}$ relative to the diffusive timescale of Q_O.

Definition IV.3 (Macroscopic Solver)**.** Let $\mu > 0$ be a shift parameter (to be chosen, and this resolvent parameter μ should not be confused with the measures $\mu_0, \mu_\eta, \mu_{\text{ref}}$ used in the Examples.) We define the **macroscopic solver** $\mathcal{A}_\mu : \mathcal{H} \to \mathcal{H}_S$ as:

$$\mathcal{A}_\mu := (\mu - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} (-\mathcal{R}^+) P_F. \tag{32}$$

The key point is that the *plain* Lyapunov functional $\frac{1}{2}\|f\|^2$ cannot yield exponential decay whenever the dissipation is degenerate on the slow subspace.

Proposition IV.4 (Why the plain energy method fails)**.** *Assume $\eta = 0$ and $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}}$ with $\mathcal{V}_{\text{skew}}$ skew-adjoint and $\ker(\mathcal{R}) = \mathcal{H}_S \neq \{0\}$. Then there is no constant $\lambda > 0$ such that*

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2 \leq -\lambda\,\|f_t\|^2 \qquad \textit{for all solutions } \partial_t f_t = \mathcal{L} f_t. \tag{33}$$

Proof. For any $\phi \in \mathcal{H}_S$ with $\phi \neq 0$, one has $\mathcal{R}\phi = 0$ and $\Re\langle \mathcal{V}_{\text{skew}}\phi, \phi\rangle = 0$, hence

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2\Big|_{t=0,\ f_0=\phi} = \Re\langle \mathcal{L}\phi, \phi\rangle = \gamma\langle \mathcal{R}\phi, \phi\rangle + \Re\langle \mathcal{V}_{\text{skew}}\phi, \phi\rangle = 0. \tag{34}$$

If such a $\lambda > 0$ existed, we would get $0 \leq -\lambda\|\phi\|^2 < 0$, a contradiction. □

The role of $\mathcal{A}_\mu$ is precisely to introduce a cross term whose time derivative produces a negative contribution on $\mathcal{H}_S$ proportional to the effective operator Q_O.

The following lemma establishes the critical boundedness properties of $\mathcal{A}_\mu$, including the control over non-equilibrium perturbations.

Lemma IV.5 (Structural Estimates for the Macroscopic Solver)**.** *Let $\mu > 0$ be the resolvent shift parameter. We define the macroscopic solver $\mathcal{A}_\mu := (\mu - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F$, where Q_O is the reference effective dynamics.*

Under Assumption IV.1, the operator $\mathcal{A}_\mu$ satisfies the following structural estimates uniformly in μ. The constants defined below depend solely on the structural parameters $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$:

1. ***Solver Boundedness:***

$$\|\mathcal{A}_\mu\| \leq k_{\mathcal{A}}\mu^{-1/2}, \quad \textit{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}. \tag{35}$$

2. ***Regularity Estimates (Forward and Dual):*** *The solver acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$\|\mathcal{V}_{\text{skew}}\mathcal{A}_\mu\| \leq k_{\text{reg}}, \quad \textit{with } k_{\text{reg}} := \sqrt{M_R/\lambda_R}, \tag{36a}$$

$$\|\mathcal{A}_\mu \mathcal{V}_{\text{skew}} P_F\| \leq k_{\text{dual}}, \quad \textit{with } k_{\text{dual}} := \kappa_{\text{ell}}. \tag{36b}$$

The dual estimate (36b) *is a direct consequence of the Macroscopic Flux Regularity (Assumption IV.1.3).*

3. ***Perturbation Control:*** *The interaction with the non-equilibrium perturbation is controlled in both directions:*

$$\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu\| \leq k_P(\mu), \tag{37a}$$

$$\|\mathcal{A}_\mu \mathcal{V}_{\text{pert}}\| \leq \tilde{k}_P(\mu). \tag{37b}$$

with control constants

$$k_P(\mu) := (b_1 + b_2) k_{\mathcal{A}}\,\mu^{-1/2}, \tag{38}$$

$$\tilde{k}_P(\mu) := \Big((b_1 + b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\mu^{-1/2}. \tag{39}$$

Moreover, the centered perturbation coupling in Assumption IV.1.2 implies

$$P_S \mathcal{V}_{\text{pert}} P_S = 0 \qquad \Longrightarrow \qquad P_S \mathcal{V}_{\text{pert}}^\dagger P_S = 0, \tag{40}$$

and therefore

$$\mathcal{V}_{\text{pert}}\,\mathcal{H}_S \subset \mathcal{H}_F, \qquad \mathcal{V}_{\text{pert}}^\dagger\,\mathcal{H}_S \subset \mathcal{H}_F. \tag{41}$$

Proof. See Appendix C. □

C. The Modified Lyapunov Functional

We define a modified squared norm that explicitly incorporates the correlation between slow and fast degrees of freedom mediated by the macroscopic solver $\mathcal{A}_\mu$.

Definition IV.6 (Operator-Modified Lyapunov Functional)**.** For any $f \in \mathcal{H}$ and a parameter $\alpha \in (0,1)$, we define the functional:

$$\mathcal{E}_{\alpha,\mu}(f) := \frac{1}{2}\|f\|^2 + \alpha \Re\langle \mathcal{A}_\mu f, f\rangle. \tag{42}$$

This functional defines a norm equivalent to the standard $L^2(\sigma)$ norm, provided the weight α is sufficiently small relative to the bounded solver $\mathcal{A}_\mu$.

Proposition IV.7 (Norm Equivalence)**.** *If the parameter α satisfies $\alpha\|\mathcal{A}_\mu\| \leq 1/2$, then for all $f \in \mathcal{H}$:*

$$\frac{1}{2}(1 - 2\alpha\|\mathcal{A}_\mu\|)\|f\|^2 \leq \mathcal{E}_{\alpha,\mu}(f) \leq \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_\mu\|)\|f\|^2. \tag{43}$$

Proof. The proof follows directly from the Cauchy-Schwarz inequality $|\Re\langle \mathcal{A}_\mu f, f\rangle| \leq \|\mathcal{A}_\mu\|\|f\|^2$ and the triangle inequality. □

V. CONVERGENCE BOUNDS FOR THE LIFTED SEMIGROUP

Having established the non-equilibrium lifting structure and the modified Lyapunov functional, we now turn to the quantitative analysis of the convergence rate. In this section, we derive an explicit hypocoercive lower bound on the asymptotic decay rate $\lambda(\mathcal{L})$ (Definition II.2) and, in the overdamped regime, an upper bound that enforces a diffusive γ^{-1} barrier. Together, these results identify an intermediate friction scale $\gamma \asymp \sqrt{\lambda(Q_O)}$ at which the lower bound exhibits the kinetic $\Theta(\sqrt{\lambda(Q_O)})$ acceleration.

A. Lower Bound via Modified Hypocoercivity

In this subsection, we establish the explicit exponential decay rate of the evolution semigroup generated by $\mathcal{L}$. Our approach utilizes a quantitative hypocoercivity method based on a parameter-dependent Lyapunov functional. This construction allows us to optimize the convergence rate with respect to the friction parameter γ, rigorously bridging the overdamped and underdamped regimes, and to quantify the stability against non-equilibrium perturbations.

The detailed derivation, algebraic optimization, and stability analysis are provided in Appendix D. Here we present the main construction and the resulting explicit bound.

a. Main result: explicit convergence rate. The following theorem provides a rigorous lower bound on the decay rate. We emphasize that the structural properties derived in the appendix ensure a uniform norm equivalence, allowing us to fix the prefactor $C = \sqrt{3}$ independent of the friction regime.

Theorem V.1 (Explicit Quantitative Hypocoercivity)**.** *Suppose Assumptions IV.1 and IV.2 hold. Let λ_R be the microscopic spectral gap from* (18)*, and let $\lambda(Q_O)$ be the spectral gap of Q_O from Assumption IV.2. Define the structural constants*

$$\Theta := \frac{\sqrt{M_R}}{2}, \qquad K_{\text{geom}} := k_{\text{reg}} + \kappa_{\text{ell}} = \sqrt{M_R/\lambda_R} + \kappa_{\text{ell}}, \tag{44}$$

$$b_{2,\text{eff}} := b_2 + \frac{a_2}{2}\,(1 + \lambda_R), \tag{45}$$

where M_R is the flux-conditioning constant from (22)*, $k_{\rm reg}$ is defined in* (36a)*, and $\kappa_{\rm ell}$ is the flux-regularity constant from* (24)*. Let $\mathcal{Z}_*(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\rm geom}$. We define the following rate functions of the friction γ:*

1. ***Unperturbed algebraic gap** ($\eta = 0$)**.***

$$\Lambda_{\rm hypo}(\gamma) := \frac{\frac{1}{4}\,\gamma\,\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\rm reg} + \frac{1}{2}}. \tag{46}$$

2. ***Perturbation sensitivity bound.*** *Let $C_{\rm sing}$ be the singular constant associated with the perturbation interaction (defined in Appendix D). Define*

$$B_{\rm pert}(\gamma) := b_{2,\rm eff} + \underbrace{\frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\rm reg}}}_{\textit{Mixing weight } \alpha_*(\gamma)} C_{\rm sing}\,\lambda(Q_O)^{-1/2}. \tag{47}$$

(The factor $\lambda(Q_O)^{-1/2}$ reflects the singular behavior of the macroscopic solver as $\lambda(Q_O) \downarrow 0$.)

There exists a stability threshold $\eta^*(\gamma) := \Lambda_{\rm hypo}(\gamma)/(2B_{\rm pert}(\gamma))$. For any friction $\gamma > 0$, if the non-equilibrium perturbation strength satisfies $\eta \le \eta^*(\gamma)$, then the semigroup $e^{t\mathcal{L}}$ is exponentially contractive on the mean-zero subspace $\mathcal{H}_0$:

$$\|e^{t\mathcal{L}} f\| \le \sqrt{3}\, e^{-\lambda_\eta t}\|f\|, \quad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{48}$$

The global decay rate λ_η satisfies the strict lower bound:

$$\lambda_\eta \ge \frac{2}{3}\left(\Lambda_{\rm hypo}(\gamma) - \eta B_{\rm pert}(\gamma)\right). \tag{49}$$

Sketch of Proof. The proof is detailed in Appendix D. We optimize the parameters of the modified Lyapunov functional to maximize the algebraic dissipation. At the optimal choice, the distortion parameter admits a uniform bound

$$\delta(\gamma) = 2\alpha_*(\gamma)\,\|\mathcal{A}_{\mu_*}\| \le \delta_{\max} := \frac{\sqrt{\lambda_R}}{4\,\Theta\,K_{\rm geom}} \le \frac{1}{2}, \qquad \forall \gamma > 0, \tag{50}$$

which follows from the explicit formula $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\rm reg}}$, the solver bound $\|\mathcal{A}_{\mu_*}\| \le k_{\mathcal{A}}\,\mu_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$, and the elementary maximization

$$\sup_{x \ge 0} \frac{x}{(x + K_{\rm geom})^2} = \frac{1}{4K_{\rm geom}} \qquad \text{(attained at } x = K_{\rm geom}).$$

which guarantees the uniform norm equivalence $\frac{1}{3}\|f\|^2 \le \mathcal{E} \le \frac{3}{4}\|f\|^2$, yielding the prefactor $\sqrt{3}$ and the factor 2/3 in the rate estimate. □

b. Regime summary. In particular, Theorem V.1 implies diffusive scaling $\Lambda_{\rm hypo}(\gamma) = \Theta(\gamma^{-1})$ as $\gamma \to \infty$ and kinetic scaling at an intermediate friction.

Corollary V.2 (Kinetic Acceleration and Singularity Cancellation)**.** *Assume the setting of Theorem V.1. Assume furthermore that the structural constants Θ, $K_{\rm geom}$, $k_{\rm reg}$, λ_R, $b_{2,\rm eff}$, $C_{\rm sing}$ are independent of $\lambda(Q_O)$ (in particular, they remain $\mathcal{O}(1)$ as $\lambda(Q_O) \downarrow 0$), and that $b_{2,\rm eff} > 0$.*

Then the unperturbed rate function $\Lambda_{\rm hypo}(\gamma)$ defined in (46) *attains its global maximum at*

$$\gamma_{\rm opt} = \frac{\sqrt{\lambda(Q_O)}}{\Theta}\sqrt{K_{\rm geom}^2 + \left(2k_{\rm reg} + \tfrac{1}{2}\right)} \asymp \sqrt{\lambda(Q_O)}. \tag{51}$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$\max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}. \tag{52}$$

Moreover, the mixing weight in (47) *satisfies*

$$\alpha_*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}, \qquad \alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1), \tag{53}$$

so the solver singularity $\lambda(Q_O)^{-1/2}$ in $B_{\mathrm{pert}}(\gamma)$ is cancelled at the optimal friction. Consequently, $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$ and the stability threshold satisfies

$$\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{54}$$

Proof. Write

$$\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4}\,\frac{\gamma}{\left(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\mathrm{geom}}\right)^2 + \left(2k_{\mathrm{reg}} + \frac{1}{2}\right)}. \tag{55}$$

Set $x := \gamma\Theta\lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\mathrm{reg}} + \frac{1}{2}$. Then $\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4\Theta}\sqrt{\lambda(Q_O)}\,\frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$ is maximized at $x = \sqrt{K_{\mathrm{geom}}^2 + C_0}$, giving the stated γ_{opt} and the scaling $\max_\gamma \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (47), $B_{\mathrm{pert}}(\gamma) = b_{2,\mathrm{eff}} + \alpha_*(\gamma)\,C_{\mathrm{sing}}\,\lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\mathrm{opt}}$ we have $x = \mathcal{O}(1)$ and $\mathcal{Z}_*(\gamma_{\mathrm{opt}}) = x + K_{\mathrm{geom}} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\mathrm{opt}}) = \frac{\gamma_{\mathrm{opt}}\lambda_R}{\mathcal{Z}_*(\gamma_{\mathrm{opt}})^2+2k_{\mathrm{reg}}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, which implies $\alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$ and therefore $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$. Finally, $\eta^*(\gamma) = \Lambda_{\mathrm{hypo}}(\gamma)/(2B_{\mathrm{pert}}(\gamma))$ yields $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$. □

Define the (rigorous) L^2 relaxation time as in[24] by

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{t \geq 0\,;\ \|e^{t\mathcal{L}}f\| \leq e^{-1}\,\|f\| \text{ for all } f \in \mathcal{H}_0\Big\}. \tag{56}$$

The following holds:

Corollary V.3 (Acceleration in the L^2 relaxation time)**.** *Assume the setting of Theorem V.1. Then for $\eta \leq \eta^*(\gamma)$, the exponential decay estimate from Theorem V.1 implies*

$$t_{\mathrm{rel}}(\mathcal{L}) \leq \frac{1+\log\sqrt{3}}{\lambda_\eta}. \tag{57}$$

In particular, at the optimal friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$ (Corollary V.2) and for perturbations within the stability threshold, one has

$$t_{\mathrm{rel}}(\mathcal{L}) = \mathcal{O}\big(\lambda(Q_O)^{-1/2}\big). \tag{58}$$

Proof. By Theorem V.1, for all $f \in \mathcal{H}_0$ and $t \geq 0$ one has $\|e^{t\mathcal{L}}f\| \leq \sqrt{3}\,e^{-\lambda_\eta t}\,\|f\|$. Choose $t = (1+\log\sqrt{3})/\lambda_\eta$ so that $\sqrt{3}\,e^{-\lambda_\eta t} = e^{-1}$. □

B. Overdamped diffusive ceiling

In the previous subsection, we established a lower bound on the decay rate that scales as $\mathcal{O}(\gamma^{-1})$ in the high-friction limit and exhibits a quadratic speedup at intermediate friction. To justify that the overdamped γ^{-1} scaling is *intrinsic* (and not a technical artifact), we now prove a matching *upper bound on any L^2 exponential decay rate.*

Define *singular value gap*:

$$s(\mathcal{L}) := \inf_{f \in \mathcal{H}_0 \cap \mathrm{dom}(\mathcal{L}) \setminus \{0\}} \frac{\|\mathcal{L} f\|}{\|f\|}, \tag{59}$$

which is the smallest singular value of $\mathcal{L}|_{\mathcal{H}_0}$ in finite dimensions, and remains meaningful as an infimum over the operator domain in unbounded settings. Quantities of this type are continuous-time analogues of *multiplicative reversiblizations* and the *pseudo spectral gap* used in non-reversible Markov chain theory; see, e.g.,[25]. In the discrete-time finite-state setting, a closely related point of view is developed by Chatterjee, who defines a nonreversible "spectral gap" via singular values of the generator $I - P$.[26]

Theorem V.4 (Overdamped diffusive ceiling via singular value gap)**.** *Assume Assumptions IV.1.1–4 and Assumption IV.2. Assume also* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *Let* $C \geq 1$ *and* $\nu > 0$ *be such that*

$$\|e^{t\mathcal{L}} f\| \leq C\, e^{-\nu t}\, \|f\|, \qquad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{60}$$

Then for every $\gamma > 0$,

$$\nu \leq (1 + \log C)\, s(\mathcal{L}) \leq (1 + \log C) \left[\frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma} \right], \tag{61}$$

where κ_{ell} *is the flux-regularity constant from Assumption IV.1.3 and* a_1, a_2, b_1, b_2 *are from Assumption IV.1.4. In particular, for* $\eta \leq c/\gamma$ *one obtains the diffusive overdamped scaling* $\nu = \mathcal{O}(\gamma^{-1})$ *as* $\gamma \to \infty$.

Sketch of Proof. The proof is given in Appendix E. The key point is that one can test the singular value gap on a *first-order corrector* $f_\gamma := \phi - \gamma^{-1}\psi$ built from a slow eigenfunction $Q_O\phi = -\lambda(Q_O)\phi$ and $\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$. The flux regularity bound (24) controls the only nontrivial fast remainder term $P_F\mathcal{V}_{\mathrm{skew}}\psi$ without ever invoking $\|\mathcal{V}_{\mathrm{skew}}\|$. The perturbation contribution is controlled by the $(-\mathcal{R})^{1/2}$-energy bounds in Assumption IV.1.4 and the explicit identity $\|(-\mathcal{R})^{1/2}\psi\|^2 = \lambda(Q_O)$. □

Theorem V.4 shows that in the overdamped regime the best achievable L^2 exponential decay rate (with any finite prefactor C) cannot beat the macroscopic diffusion scale $\lambda(Q_O)/\gamma$, up to explicit perturbative contributions. Combined with the lower bound in Theorem V.1, this confirms that the accelerated $\sqrt{\lambda(Q_O)}$ regime must occur at *intermediate* friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, rather than by taking $\gamma \to \infty$.

Corollary V.5 (Diffusive ceiling for the L^2 relaxation time)**.** *Assume the setting of Theorem V.4. Define* $t_{\mathrm{rel}}(\mathcal{L})$ *as in Corollary V.3. Then the singular-value gap satisfies the explicit ceiling*

$$s(\mathcal{L}) \leq \frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{62}$$

Moreover, the relaxation time is bounded below in terms of the singular-value gap (see Lemma E.2):

$$t_{\mathrm{rel}}(\mathcal{L}) \geq \frac{1}{2\, s(\mathcal{L})}. \tag{63}$$

Consequently, if $\eta \leq c/\gamma$, *then* $s(\mathcal{L}) = \mathcal{O}(\gamma^{-1})$ *as* $\gamma \to \infty$ *and hence*

$$t_{\mathrm{rel}}(\mathcal{L}) = \Omega(\gamma) \qquad (\gamma \to \infty). \tag{64}$$

In particular, the kinetic scaling $t_{\mathrm{rel}}(\mathcal{L}) \asymp \lambda(Q_O)^{-1/2}$ *is achieved at optimal friction, combined with Corollary V.3.*

VI. EXAMPLES

In this section, we demonstrate the universality of the non-equilibrium lifting framework by analyzing three representative dynamical systems spanning classical statistical mechanics, semiclassical phase-space dynamics, and a finite-dimensional boundary-driven toy model. These examples elucidate how the abstract operator-theoretic structure of Theorem V.1 manifests in physical regimes ranging from kinetic theory to quantum transport.

We begin with the classical **Nonequilibrium Langevin Dynamics** (Section VI A), establishing the baseline correspondence between kinetic lifting and the overdamped Smoluchowski limit.[19,23] We then extend this geometric insight to the quantum phase space via the **Wigner–Fokker–Planck Equation** (Section VI B).[27,28] Here, we show that the strong friction limit yields the **Quantum Smoluchowski Equation**, capturing non-trivial quantum corrections to diffusion coefficients—such as vacuum fluctuations—within our rigorous spectral framework. Finally, we present a **finite-dimensional boundary-driven toy model** (Section VI C) motivated by Zeno-limit/boundary-driving settings, for which the effective slow generator is generally non-self-adjoint on the slow space and thus falls outside the reversible setting[24].

A. Example 1: Classical Nonequilibrium Langevin Dynamics

The underdamped Langevin equation with a non-gradient drift serves as a standard testbed for non-equilibrium kinetic theory. While the overdamped limit ($\gamma \to \infty$) is classical, we apply our lifting framework to demonstrate the quadratic speedup of the convergence rate at finite, optimal friction.

Setup.
Phase space and reference measure. Consider a particle in the phase space $\mathcal{X} = \mathbb{T}^d \times \mathbb{R}^d$ ($d \geq 1$), where the position space is the torus $\mathbb{T}^d$. The system is subject to a smooth conservative potential $U \in C^\infty(\mathbb{T}^d)$ and a non-conservative external force field $F \in C^\infty(\mathbb{T}^d; \mathbb{R}^d)$. Let μ_0 be the Gibbs measure of the unperturbed Langevin dynamics:

$$d\mu_0(q,p) = Z^{-1} e^{-\beta(\frac{1}{2}|p|^2 + U(q))} dq\, dp, \tag{65}$$

where $\beta > 0$ is the inverse temperature and Z is the partition function. Write $d\mu_0(q,p) = d\nu(q)\, d\nu_p(p)$, where $d\nu(q) = Z_q^{-1} e^{-\beta U(q)} dq$ and $d\nu_p(p) = Z_p^{-1} e^{-\frac{\beta}{2}|p|^2} dp$.

For the forced (nonequilibrium) dynamics ($\eta \neq 0$), the invariant measure is generally *not* μ_0. Under standard smoothness/non-degeneracy assumptions, there exists a unique invariant measure μ_η which is absolutely continuous with respect to μ_0 and can be written as $d\mu_\eta = h_\eta\, d\mu_0$, as in standard analyses of forced Langevin dynamics.[19,23] In this example we formulate the operator bounds in the reference space $L^2(\mu_0)$ and measure convergence toward the stationary density h_η (as is common in the hypocoercivity literature for nonequilibrium Langevin).

Functional-analytic setting. For $\eta \neq 0$ we work in the reference space $L^2(\mu_0)$ and interpret "convergence to the NESS" as $L^2(\mu_0)$-convergence of the forward evolution toward the stationary density h_η, i.e. $\|e^{t\mathcal{L}^\dagger} f - h_\eta\|_{L^2(\mu_0)} \leq C e^{-\lambda t} \|f - h_\eta\|_{L^2(\mu_0)}$, where $\dagger$ denotes the $L^2(\mu_0)$-adjoint.

Remark VI.1 (Applicability of main theorems). Since h_η is the unique stationary density, it satisfies $\mathcal{L}^\dagger h_\eta = 0$. Therefore the translated variable $g := f - h_\eta$ evolves by the same semigroup, $e^{t\mathcal{L}^\dagger} g = e^{t\mathcal{L}^\dagger} f - h_\eta$. Consequently, exponential convergence to h_η is equivalent to exponential decay of g to 0 on the mean-zero subspace (with respect to μ_η, or equivalently after centering in $L^2(\mu_0)$). Thus, when applying Theorem V.1 in this example, one simply applies it to the centered function $g = f - h_\eta$; the theorem's "$e^{t\mathcal{L}}$ decays to 0" formulation is exactly the same statement after shifting by the stationary state.

Generator decomposition. The evolution is governed by the generator $\mathcal{L}$, which we decompose according to the hypocoercive structure $\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\text{pert}}$:

1. **Hamiltonian Transport ($\mathcal{V}_{\text{skew}}$):** The generator of the Hamiltonian flow for $H(q,p) = \frac{1}{2}|p|^2 + U(q)$:

$$\mathcal{V}_{\text{skew}} := p \cdot \nabla_q - \nabla U(q) \cdot \nabla_p. \tag{66}$$

2. **Dissipation ($\mathcal{R}$):** The Ornstein-Uhlenbeck operator acting on momenta:

$$\mathcal{R} := -p \cdot \nabla_p + \beta^{-1}\Delta_p. \tag{67}$$

3. **Non-Equilibrium Perturbation ($\mathcal{V}_{\text{pert}}$):** We take the standard (unregularized) forcing

$$\mathcal{V}_{\text{pert}} f := F(q) \cdot \nabla_p f. \tag{68}$$

Effective slow generator (Smoluchowski limit). We compute the effective generator on slow observables by the adiabatic-elimination formula $\mathcal{L}_O = -P_S(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})\,\mathcal{R}^+\,(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})P_S$. Let $\phi \in \mathcal{H}_S$ be a smooth slow observable, i.e. $\phi = \phi(q)$. First, $\mathcal{V}_{\text{pert}}\phi = 0$ and $\mathcal{V}_{\text{skew}}\phi = p \cdot \nabla_q\phi \in \mathcal{H}_F$. Since $\mathcal{R}(p_i) = -p_i$, we have $\mathcal{R}^+(p_i) = -p_i$ on $\mathcal{H}_F$, hence

$$\mathcal{R}^+\mathcal{V}_{\text{skew}}\phi = -p \cdot \nabla_q\phi. \tag{69}$$

Writing the combined transport as $\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}} = p \cdot \nabla_q - (\nabla U - \eta F) \cdot \nabla_p$, we compute

$$(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})\big(-p \cdot \nabla_q\phi\big) = -\sum_{i,j} p_i p_j\, \partial_{q_i q_j}\phi + (\nabla U - \eta F) \cdot \nabla_q\phi. \tag{70}$$

Projecting onto $\mathcal{H}_S$ amounts to averaging in p against the centered Gaussian, for which $\int p_i p_j\, d\nu_p = \beta^{-1}\delta_{ij}$. Therefore the effective slow generator on observables is

$$\mathcal{L}_O\phi = \beta^{-1}\Delta_q\phi - (\nabla U - \eta F) \cdot \nabla_q\phi. \tag{71}$$

In particular, in the reversible case $\eta = 0$ the corresponding reference operator can be taken as

$$Q_O := \mathcal{L}_O\big|_{\eta=0} = \beta^{-1}\Delta_q - \nabla U \cdot \nabla_q, \tag{72}$$

viewed as a self-adjoint operator on $L^2(\nu)$ with $d\nu \propto e^{-\beta U(q)}dq$. Equivalently, setting $D := \nabla_q + (\beta/2)\nabla U$ and letting $D^\dagger$ denote the adjoint in $L^2(\nu)$, one has the Dirichlet-form representation

$$Q_O = -(1/\beta)\, D^\dagger D, \tag{73}$$

which is exactly the form implemented in our numerical computation of $\lambda(Q_O)$. Equivalently, the adjoint effective dynamics for the slow density $\rho(q,t)$ is the (forced) Smoluchowski equation

$$\partial_t\rho = \beta^{-1}\Delta_q\rho + \nabla_q \cdot \big(\rho(\nabla U - \eta F)\big). \tag{74}$$

Verification of the structural assumptions.

We verify that the tuple $(\mathcal{R}, \mathcal{V}_{\text{skew}}, \mathcal{V}_{\text{pert}})$ satisfies Assumption IV.1.1–4 on a natural smooth core in $L^2(\mu_0)$.

Proposition VI.2 (Verification for Nonequilibrium Langevin Dynamics)**.** *The operators defined above satisfy Assumption IV.1.1–4 in $L^2(\mu_0)$.*

Proof. We address each condition systematically.

1. Microscopic Spectral Gap ($\mathcal{R}$). The operator $\mathcal{R}$ acts only on the p-variables. It is classical that the Ornstein–Uhlenbeck operator admits a spectral gap $\lambda_R = 1$ in the Gaussian-weighted space $L^2(e^{-\beta p^2/2}dp)$. Specifically, for any $f \perp \ker(\mathcal{R})$ (where $\ker(\mathcal{R})$ consists of functions of q only), we have $-\langle \mathcal{R}f, f\rangle_{\mu_0} \geq \|f\|^2$.

2. Skew-Adjointness and Centered Coupling ($\mathcal{V}_{\text{skew}}$). The Liouville operator $\mathcal{V}_{\text{skew}}$ satisfies $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$ in $L^2(\mu_0)$ because the Hamiltonian flow preserves the Gibbs measure. The slow subspace $\mathcal{H}_S$ consists of functions $\phi(q)$ independent of p. The projection is $P_S f = \int f d\nu_p$, where $d\nu_p$ is the marginal Gaussian measure on momenta. For any $\phi \in \mathcal{H}_S$, $\mathcal{V}_{\text{skew}}\phi = p \cdot \nabla_q \phi$. Then, $P_S \mathcal{V}_{\text{skew}} P_S \phi = \int (p \cdot \nabla_q \phi) d\nu_p = (\int p\, d\nu_p) \cdot \nabla_q \phi = 0$, since the Gaussian is centered at zero.

Moreover, for any slow observable $\phi(q)$ we have $\mathcal{V}_{\text{pert}}\phi = F(q) \cdot \nabla_p \phi = 0$. Therefore $P_S \mathcal{V}_{\text{pert}} P_S = 0$, and hence $P_S \mathcal{V} P_S = 0$ for $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$.

3. Fast-to-slow Regularity (Assumption IV.1.3). Let $\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S$ be a smooth function of q only. Then

$$g_\psi = P_F \mathcal{V}_{\text{skew}} P_S \psi = p \cdot \nabla_q \psi, \tag{75}$$

which is centered in p and therefore lies in $\mathcal{H}_F$. Since the Ornstein–Uhlenbeck operator satisfies $\mathcal{R}(p_i) = -p_i$, we have $\mathcal{R} g_\psi = -g_\psi$ and hence $(-\mathcal{R}^+) g_\psi = g_\psi$ (on this first Hermite sector). Therefore

$$\langle g_\psi, -\mathcal{R}^+ g_\psi \rangle_{\mu_0} = \|g_\psi\|_{\mu_0}^2, \tag{76}$$

so the flux conditioning bound (22) holds with $M_R = 1$.

It remains to check the flux regularity bound (24) in a self-contained way. Since $(-\mathcal{R}^+) g_\psi = g_\psi$ and $P_F(p \cdot \nabla_q \psi) = p \cdot \nabla_q \psi$, a direct computation gives

$$P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi = P_F \mathcal{V}_{\text{skew}} \left(p \cdot \nabla_q \psi\right) = \sum_{i,j} \left(p_i p_j - \beta^{-1}\delta_{ij}\right) \partial_{q_i q_j} \psi. \tag{77}$$

Step 1: explicit reduction to a second-derivative norm. Define the centered quadratic polynomials

$$Z_{ij}(p) := p_i p_j - \beta^{-1}\delta_{ij}. \tag{78}$$

For each fixed q, Cauchy–Schwarz in the finite sum over i, j yields

$$\left| \sum_{i,j} Z_{ij}(p)\, \partial_{q_i q_j} \psi(q) \right|^2 \leq \left(\sum_{i,j} |Z_{ij}(p)|^2 \right) \left(\sum_{i,j} |\partial_{q_i q_j} \psi(q)|^2 \right). \tag{79}$$

Integrating first in p and then in q gives

$$\left\| P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi \right\|_{L^2(\mu_0)} \leq c_{d,\beta} \, \|\nabla_q^2 \psi\|_{L^2(\nu)}, \tag{80}$$

where the constant $c_{d,\beta}$ is defined explicitly by

$$c_{d,\beta}^2 := \int_{\mathbb{R}^d} \sum_{i,j} |Z_{ij}(p)|^2 \, d\nu_p(p). \tag{81}$$

Since the coordinates of p are i.i.d. $N(0, \beta^{-1})$ under ν_p, we have $\mathbb{E}[p_i^4] = 3\beta^{-2}$ and $\mathbb{E}[p_i^2 p_j^2] = \beta^{-2}$ for $i \neq j$. Therefore

$$\|Z_{ii}\|_{L^2(\nu_p)}^2 = \mathbb{E}\big[(p_i^2 - \beta^{-1})^2\big] = 2\beta^{-2}, \qquad \|Z_{ij}\|_{L^2(\nu_p)}^2 = \mathbb{E}[p_i^2 p_j^2] = \beta^{-2} \quad (i \neq j), \tag{82}$$

and hence

$$c_{d,\beta}^2 = d\,(2\beta^{-2}) + d(d-1)\,\beta^{-2} = \frac{d(d+1)}{\beta^2}, \qquad c_{d,\beta} = \frac{\sqrt{d(d+1)}}{\beta}. \tag{83}$$

Step 2: control of $\|\nabla_q^2\psi\|_{L^2(\nu)}$ by $\|(-Q_O)\psi\|_{L^2(\nu)}$ with explicit structural constants. Recall that the reversible overdamped generator acting on q-observables is

$$Q_O\psi = \beta^{-1}\Delta_q\psi - \nabla U\cdot\nabla_q\psi, \tag{84}$$

so that

$$\Delta_q\psi = \beta\big(Q_O\psi + \nabla U\cdot\nabla_q\psi\big). \tag{85}$$

Let $M_U := \|\nabla U\|_{L^\infty(\mathbb{T}^d)}$.

(i) Hessian vs. Laplacian on the torus in $L^2(\nu)$. Write $d\nu(q) = \rho(q)\,dq$ with $\rho(q) = Z_q^{-1}e^{-\beta U(q)}$. Since ρ is smooth and strictly positive on the compact torus, we may set $m := \inf_{\mathbb{T}^d}\rho > 0$ and $M := \sup_{\mathbb{T}^d}\rho < \infty$.

Fourier/Parseval step (in $L^2(dq)$). Assume $\int_{\mathbb{T}^d}\psi\,dq = 0$. Write the Fourier series

$$\psi(q) = \sum_{k\in\mathbb{Z}^d\setminus\{0\}} \hat{\psi}_k\, e^{ik\cdot q}. \tag{86}$$

Then

$$\partial_{q_iq_j}\psi(q) = \sum_{k\neq 0}(-k_ik_j)\,\hat{\psi}_k\, e^{ik\cdot q}, \tag{87}$$

$$\Delta_q\psi(q) = \sum_{k\neq 0}(-|k|^2)\,\hat{\psi}_k\, e^{ik\cdot q}. \tag{88}$$

By Parseval's identity,

$$\|\nabla_q^2\psi\|_{L^2(dq)}^2 = \sum_{i,j=1}^d \|\partial_{q_iq_j}\psi\|_{L^2(dq)}^2 = \sum_{k\neq 0}\left(\sum_{i,j=1}^d k_i^2k_j^2\right)|\hat{\psi}_k|^2 = \sum_{k\neq 0}|k|^4|\hat{\psi}_k|^2 = \|\Delta_q\psi\|_{L^2(dq)}^2. \tag{89}$$

Hence $\|\nabla_q^2\psi\|_{L^2(dq)} = \|\Delta_q\psi\|_{L^2(dq)}$ for mean-zero ψ.

Norm-equivalence step (switching to $L^2(\nu)$). For any measurable f, the bounds $m \le \rho \le M$ imply

$$m\,\|f\|_{L^2(dq)}^2 \le \|f\|_{L^2(\nu)}^2 := \int_{\mathbb{T}^d}|f|^2\,d\nu = \int_{\mathbb{T}^d}|f|^2\,\rho\,dq \le M\,\|f\|_{L^2(dq)}^2. \tag{90}$$

Applying this to $f = \nabla_q^2\psi$ and $f = \Delta_q\psi$ gives

$$\|\nabla_q^2\psi\|_{L^2(\nu)}^2 \le M\,\|\nabla_q^2\psi\|_{L^2(dq)}^2 = M\,\|\Delta_q\psi\|_{L^2(dq)}^2 \le \frac{M}{m}\,\|\Delta_q\psi\|_{L^2(\nu)}^2, \tag{91}$$

hence

$$\|\nabla_q^2\psi\|_{L^2(\nu)} \le \sqrt{\tfrac{M}{m}}\,\|\Delta_q\psi\|_{L^2(\nu)}. \tag{92}$$

(The factor $(M/m)^{1/2}$ is structural, depending only on U and β.)

(ii) Bounding $\|\Delta_q\psi\|$ by $\|(-Q_O)\psi\|$. From the identity for $\Delta_q\psi$ and $\|\nabla U\cdot\nabla_q\psi\| \le M_U\,\|\nabla_q\psi\|$, we have

$$\|\Delta_q\psi\|_{L^2(\nu)} \le \beta\,\|(-Q_O)\psi\|_{L^2(\nu)} + \beta M_U\,\|\nabla_q\psi\|_{L^2(\nu)}. \tag{93}$$

To estimate the remaining gradient term with a structural constant, we use the Dirichlet form of Q_O. For mean-zero ψ (equivalently $\psi \in \ker(Q_O)^\perp$), the Poincaré inequality for ν gives a constant $C_{\mathrm{P}}(\nu) \in (0,\infty)$ such that

$$\|\psi\|_{L^2(\nu)}^2 \le C_{\mathrm{P}}(\nu)\, \|\nabla_q \psi\|_{L^2(\nu)}^2. \tag{94}$$

Moreover, since Q_O is reversible with invariant measure ν, we can compute the corresponding Dirichlet form explicitly. Writing $d\nu = \rho\, dq$ with $\rho \propto e^{-\beta U}$, we first rewrite Q_O in divergence form:

$$Q_O\psi = \beta^{-1}\Delta_q\psi - \nabla U \cdot \nabla_q \psi = \beta^{-1}\, \rho^{-1}\, \nabla_q \cdot \big(\rho\, \nabla_q \psi\big), \tag{95}$$

since $\nabla_q \rho = -\beta\rho\, \nabla U$. Therefore, using periodicity on $\mathbb{T}^d$ (so boundary terms vanish), integration by parts gives

$$\langle (-Q_O)\psi, \psi\rangle_{L^2(\nu)} = -\int_{\mathbb{T}^d} (Q_O\psi)\, \psi\, d\nu = -\beta^{-1} \int_{\mathbb{T}^d} \rho^{-1}\, \nabla_q \cdot (\rho \nabla_q \psi)\, \psi\, \rho\, dq \tag{96}$$

$$= -\beta^{-1} \int_{\mathbb{T}^d} \nabla_q \cdot (\rho\nabla_q\psi)\, \psi\, dq = \beta^{-1} \int_{\mathbb{T}^d} \rho\, \nabla_q\psi \cdot \nabla_q \psi\, dq = \beta^{-1}\, \|\nabla_q \psi\|_{L^2(\nu)}^2. \tag{97}$$

Combining this with Cauchy–Schwarz and (94),

$$\begin{aligned}\beta^{-1}\, \|\nabla_q\psi\|_{L^2(\nu)}^2 = \langle (-Q_O)\psi, \psi\rangle_{L^2(\nu)} &\le \|(-Q_O)\psi\|_{L^2(\nu)}\, \|\psi\|_{L^2(\nu)} \\ &\le \sqrt{C_{\mathrm{P}}(\nu)}\, \|(-Q_O)\psi\|_{L^2(\nu)}\, \|\nabla_q\psi\|_{L^2(\nu)},\end{aligned}$$

so that

$$\|\nabla_q \psi\|_{L^2(\nu)} \le \beta\sqrt{C_{\mathrm{P}}(\nu)}\, \|(-Q_O)\psi\|_{L^2(\nu)}. \tag{98}$$

Substituting (98) into (93) yields

$$\|\Delta_q\psi\|_{L^2(\nu)} \le \beta\big(1 + \beta M_U \sqrt{C_{\mathrm{P}}(\nu)}\big)\, \|(-Q_O)\psi\|_{L^2(\nu)}. \tag{99}$$

Finally, combining with (92) gives a purely structural constant

$$C_{\mathrm{H}^2}(U,\beta,d) := \sqrt{\tfrac{M}{m}}\, \beta\big(1 + \beta M_U \sqrt{C_{\mathrm{P}}(\nu)}\big) \tag{100}$$

such that

$$\|\nabla_q^2 \psi\|_{L^2(\nu)} \le C_{\mathrm{H}^2}(U,\beta,d)\, \|(-Q_O)\psi\|_{L^2(\nu)}. \tag{101}$$

Conclusion. Combining (80) with the previous display yields

$$\big\|P_F \mathcal{V}_{\mathrm{skew}} P_F(-\mathcal{R}^+) g_\psi\big\|_{L^2(\mu_0)} \le \kappa_{\mathrm{ell}}\, \|(-Q_O)\psi\|_{L^2(\nu)}, \tag{102}$$

where we may take the explicit structural constant

$$\begin{aligned}\kappa_{\mathrm{ell}} := c_{d,\beta}\, C_{\mathrm{H}^2}(U,\beta,d) &= \frac{\sqrt{d(d+1)}}{\beta}\, \sqrt{\tfrac{M}{m}}\, \beta\big(1 + \beta M_U \sqrt{C_{\mathrm{P}}(\nu)}\big) \\ &= \sqrt{d(d+1)}\, \sqrt{\tfrac{M}{m}}\, \big(1 + \beta \|\nabla U\|_{L^\infty(\mathbb{T}^d)} \sqrt{C_{\mathrm{P}}(\nu)}\big).\end{aligned}$$

This verifies the flux regularity bound (24) with constants that are independent of treating $\lambda(Q_O)$ as a variable in the main theorems.

4(a)–4(b). Perturbation decomposition and energy bounds. We compute the $L^2(\mu_0)$-adjoint $\mathcal{V}_{\mathrm{pert}}^\dagger$ and split $\mathcal{V}_{\mathrm{pert}}$ into its skew and symmetric parts. Let $f, g \in C_c^\infty(\mathbb{T}^d \times$

$\mathbb{R}^d$). Write $d\mu_0(q,p) = Z^{-1}e^{-\beta U(q)}e^{-\frac{\beta}{2}|p|^2}\,dq\,dp$. Then, using that $F = F(q)$ is independent of p and integrating by parts in p (no boundary term since f, g are compactly supported in p),

$$\langle \mathcal{V}_{\rm pert} f, g\rangle_{\mu_0} = \int_{\mathbb{T}^d} e^{-\beta U(q)} dq \int_{\mathbb{R}^d} \big(F(q)\cdot\nabla_p f(q,p)\big)\,\overline{g(q,p)}\,e^{-\frac{\beta}{2}|p|^2}\,dp \tag{103}$$

$$= -\int_{\mathbb{T}^d} e^{-\beta U(q)} dq \int_{\mathbb{R}^d} f(q,p)\,\nabla_p\cdot\Big(F(q)\,\overline{g(q,p)}\,e^{-\frac{\beta}{2}|p|^2}\Big)\,dp \tag{104}$$

$$= -\int_{\mathbb{T}^d} e^{-\beta U(q)} dq \int_{\mathbb{R}^d} f(q,p)\,\Big(F(q)\cdot\nabla_p\overline{g(q,p)}\Big)e^{-\frac{\beta}{2}|p|^2}\,dp \tag{105}$$

$$-\int_{\mathbb{T}^d} e^{-\beta U(q)} dq \int_{\mathbb{R}^d} f(q,p)\,\overline{g(q,p)}\,F(q)\cdot\nabla_p\Big(e^{-\frac{\beta}{2}|p|^2}\Big)\,dp. \tag{106}$$

Since $\nabla_p\big(e^{-\frac{\beta}{2}|p|^2}\big) = -\beta p\,e^{-\frac{\beta}{2}|p|^2}$, this becomes

$$\langle \mathcal{V}_{\rm pert} f, g\rangle_{\mu_0} = \int f\,\overline{\Big(-F\cdot\nabla_p g + \beta(F\cdot p)g\Big)}\,d\mu_0. \tag{107}$$

Hence the $L^2(\mu_0)$-adjoint is

$$\mathcal{V}_{\rm pert}^\dagger = -F\cdot\nabla_p + \beta(F\cdot p). \tag{108}$$

We then define the skew/symmetric parts in $L^2(\mu_0)$ by

$$\mathcal{K}_{\rm pert} := \tfrac{1}{2}\big(\mathcal{V}_{\rm pert} - \mathcal{V}_{\rm pert}^\dagger\big) = F\cdot\nabla_p - \tfrac{\beta}{2}(F\cdot p), \qquad \mathcal{S}_{\rm pert} := \tfrac{1}{2}\big(\mathcal{V}_{\rm pert} + \mathcal{V}_{\rm pert}^\dagger\big) = \tfrac{\beta}{2}(F\cdot p). \tag{109}$$

By definition one has $(\mathcal{K}_{\rm pert})^\dagger = -\mathcal{K}_{\rm pert}$ (hence $\Re\langle\mathcal{K}_{\rm pert} f, f\rangle_{\mu_0} = 0$) and $(\mathcal{S}_{\rm pert})^\dagger = \mathcal{S}_{\rm pert}$.

Energy control of $\mathcal{K}_{\rm pert}$ (Assumption IV.1.4(a)). Since $F \in L^\infty(\mathbb{T}^d)$,

$$\|\mathcal{K}_{\rm pert} f\| \le \|F\|_{L^\infty}\,\|\nabla_p f\| + \tfrac{\beta}{2}\,\|F\|_{L^\infty}\,\||p|f\|. \tag{110}$$

For the Ornstein–Uhlenbeck generator on $L^2(\mu_0)$ one has $\langle f, -\mathcal{R}f\rangle_{\mu_0} = \beta^{-1}\|\nabla_p f\|^2$, hence $\|\nabla_p f\| \le \sqrt{\beta}\,\|(-\mathcal{R})^{1/2} f\|$. Moreover, one can bound the (unbounded) multiplier p by a Gaussian integration-by-parts estimate. Let $w(p) := e^{-\frac{\beta}{2}|p|^2}$. For each coordinate i and $f \in C_c^\infty(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} p_i^2|f|^2\,w\,dp = -\frac{1}{\beta}\int_{\mathbb{R}^d} p_i|f|^2\,\partial_{p_i} w\,dp = \frac{1}{\beta}\int_{\mathbb{R}^d}\partial_{p_i}(p_i|f|^2)\,w\,dp \tag{111}$$

$$= \frac{1}{\beta}\int_{\mathbb{R}^d}|f|^2\,w\,dp + \frac{2}{\beta}\Re\int_{\mathbb{R}^d} p_i f\,\overline{\partial_{p_i} f}\,w\,dp. \tag{112}$$

We estimate the cross term by Young's inequality. For any $\varepsilon \in (0,1)$,

$$\frac{2}{\beta}\Big|\Re\int_{\mathbb{R}^d} p_i f\,\overline{\partial_{p_i} f}\,w\,dp\Big| \le \varepsilon\,\|p_i f\|_{L^2(wdp)}^2 + \frac{1}{\varepsilon\beta^2}\,\|\partial_{p_i} f\|_{L^2(wdp)}^2. \tag{113}$$

Choosing $\varepsilon = \frac{1}{2}$ and inserting this into the previous inequality yields, for each $i \in \{1,\dots,d\}$,

$$\|p_i f\|_{L^2(wdp)}^2 \le \frac{2}{\beta}\,\|f\|_{L^2(wdp)}^2 + \frac{4}{\beta^2}\,\|\partial_{p_i} f\|_{L^2(wdp)}^2. \tag{114}$$

Summing over i gives

$$\||p|f\|_{L^2(wdp)}^2 = \sum_{i=1}^d \|p_i f\|_{L^2(wdp)}^2 \le \frac{2d}{\beta}\,\|f\|_{L^2(wdp)}^2 + \frac{4}{\beta^2}\,\|\nabla_p f\|_{L^2(wdp)}^2. \tag{115}$$

Taking square roots and using $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ yields the explicit multiplier bound

$$\||p|f\|_{L^2(\mu_0)} \le \sqrt{\tfrac{2d}{\beta}}\,\|f\|_{L^2(\mu_0)} + \frac{2}{\beta}\,\|\nabla_p f\|_{L^2(\mu_0)} =: C_0\,\|f\|_{L^2(\mu_0)} + C_1\,\|\nabla_p f\|_{L^2(\mu_0)}. \quad (116)$$

Combining this with $\|\nabla_p f\| \le \sqrt{\beta}\,\|(-\mathcal{R})^{1/2} f\|$ and $\|\mathcal{K}_{\text{pert}} f\| \le \|F\|_{L^\infty}\,\|\nabla_p f\| + \frac{\beta}{2}\,\|F\|_{L^\infty}\,\||p|f\|$ (from the definition of $\mathcal{K}_{\text{pert}}$), we obtain

$$\|\mathcal{K}_{\text{pert}} f\| \le a_1\,\|(-\mathcal{R})^{1/2} f\| + b_1\,\|f\|, \quad (117)$$

for some explicit constants $a_1, b_1 \ge 0$ depending only on β, d and $\|F\|_{L^\infty}$.

Energy control of $\mathcal{S}_{\text{pert}}$ (Assumption IV.1.4(b)). Recall

$$\mathcal{S}_{\text{pert}} = \frac{\beta}{2}\,(F(q)\cdot p), \quad (118)$$

so by $|F\cdot p| \le \|F\|_{L^\infty}|p|$ and (116),

$$\|\mathcal{S}_{\text{pert}} f\|_{L^2(\mu_0)} \le \frac{\beta}{2}\,\|F\|_{L^\infty}\,\||p|f\|_{L^2(\mu_0)} \le \frac{\beta}{2}\,\|F\|_{L^\infty}\left(C_0\,\|f\|_{L^2(\mu_0)} + C_1\,\|\nabla_p f\|_{L^2(\mu_0)}\right) \quad (119)$$

$$\le a_2\,\|(-\mathcal{R})^{1/2} f\|_{L^2(\mu_0)} + b_2\,\|f\|_{L^2(\mu_0)}, \quad (120)$$

for suitable constants $a_2, b_2 \ge 0$.[19] This verifies Assumption IV.1.4(b) in the natural hypocoercive functional setting. □

Consistency with sharp kinetic estimates.

Despite the regularization required for the strict L^2 proof, our derived rate function $\Lambda_{\text{hypo}}(\gamma)$ captures the fundamental scaling laws of kinetic theory. We compare our result with the benchmark established in the literature.

A precise hypocoercive estimate for the nonequilibrium Langevin dynamics is proved in[19] (Theorem 3). Mapping their notation to ours (friction $\xi \leftrightarrow \gamma$, forcing $\tau \leftrightarrow \eta$), their result states that for sufficiently small perturbations, the convergence to the steady state h_η satisfies:

$$\left\|e^{t\mathcal{L}^\dagger_{\gamma,\eta}} f - h_\eta\right\|_{L^2(\mu_0)} \le C e^{-\overline{\lambda}_\delta \min(\gamma,\gamma^{-1})t}\|f - h_\eta\|_{L^2(\mu_0)}. \quad (121)$$

The crucial feature of this result is the specific scaling of the decay rate $\asymp \min(\gamma, \gamma^{-1})$, which encapsulates the crossover between the underdamped and overdamped regimes, peaking at $\gamma \sim 1$.

We observe that the specific scaling law (121) is an exact realization of the general rate function $\Lambda_{\text{hypo}}(\gamma)$ derived in our Theorem V.1:

1. **The Overdamped Regime ($\gamma \to \infty$):** The benchmark result (121) dictates a rate scaling of $\mathcal{O}(\gamma^{-1})$. This matches our asymptotic analysis in Section V A, where we established that the hypocoercive rate degenerates to the macroscopic spectral gap scaling:

$$\Lambda_{\text{hypo}}(\gamma) \asymp \gamma^{-1}\lambda(Q_O). \quad (122)$$

   Our framework thus correctly identifies that in the high-friction limit, the convergence is enslaved by the slow effective diffusion process.

2. **The Underdamped Regime ($\gamma \to 0$):** The benchmark result scales as $\mathcal{O}(\gamma)$. Similarly, the denominator of our general rate function $\Lambda_{\text{hypo}}(\gamma)$ is dominated by constant structural terms as $\gamma \to 0$, yielding a linear dependence $\Lambda_{\text{hypo}}(\gamma) \asymp \gamma$. This reflects the regime where insufficient dissipation limits the entropy production.

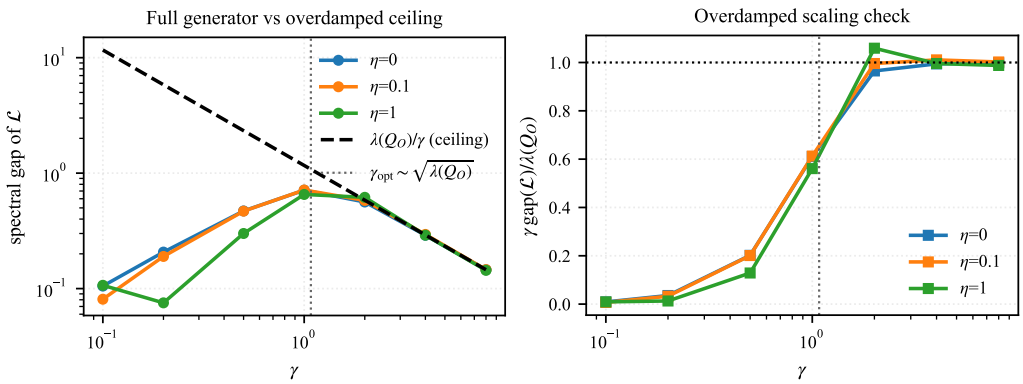

Figure 1. Numerical verification of Example VI A (Galerkin spectral gap). *Left:* $\mathrm{gap}(\mathcal{L})$ vs. γ with the overdamped ceiling $\lambda(Q_O)/\gamma$ and $\gamma_{\rm opt} \sim \sqrt{\lambda(Q_O)}$. *Right:* scaled gap $\gamma\,\mathrm{gap}(\mathcal{L})/\lambda(Q_O)$ approaching a constant for large γ. Parameters: $U_0 = 1$, $m = \beta = 1$, $K = 8$, $N = 16$, $K_{\rm ovd} = 24$, and $\eta \in \{0, 0.1, 1\}$.

3. **Optimal Lift ($\gamma \sim 1$):** The function $\min(\gamma, \gamma^{-1})$ attains its global maximum at $\gamma = 1$. Our Corollary V.2 predicts an optimal friction of the form $\gamma_{\rm opt} \asymp \sqrt{\lambda(Q_O)}$. In the standard Langevin setting on the torus with unit mass and temperature considered in[19], the macroscopic operator on the slow variables has a spectral gap bounded away from 0 (and from ∞) by constants depending only on the model parameters; in particular, $\lambda(Q_O)$ is of order one. Therefore $\gamma_{\rm opt}$ is also of order one, consistent with the benchmark rate peaking at $\gamma \sim 1$.

   We emphasize that the quadratic improvement interpretation of the scaling $\gamma_{\rm opt} \asymp \sqrt{\lambda(Q_O)}$ is most meaningful in families of problems where $\lambda(Q_O) \downarrow 0$ (slow macroscopic mixing); when $\lambda(Q_O) = \mathcal{O}(1)$, the optimal lift still occurs at $\gamma = \mathcal{O}(1)$ but the improvement is at the level of constants.

The scaling $\min(\gamma, \gamma^{-1})$ can also be derived via specific commutator relations.[19] It is fully captured by our abstract rate function $\Lambda_{\rm hypo}(\gamma)$. This comparison confirms that the operator-modified Lyapunov framework is sharp: it recovers the exact optimal scaling behaviors for classical kinetic models while extending the validity of the quadratic speedup to general non-equilibrium quantum systems.

Numerical verification (Galerkin spectral gap). We now corroborate the lower/upper-bound corollaries by a direct spectral computation, following the Galerkin discretization in[19]. We work in $d = 1$ with the rotor potential $U(q) = U_0(1 - \cos q)$ (here $U_0 = 1$) and parameters $\beta = m = 1$. The basis is the Fourier–Hermite family $\psi_{nk}(q, p) = G_k(q)H_n(p)$ with $G_k(q) \propto e^{ikq+\beta U(q)/2}$ and Hermite polynomials H_n orthonormal in the Gaussian momentum marginal. The generator $\mathcal{L}$ is represented by the finite matrix of the exact coefficients in[19]; the spectral gap is computed as the smallest positive real part of $-\lambda$ over eigenvalues λ of this matrix. For the effective operator, we use the overdamped Fourier representation $Q_O = -(1/\beta)D^\dagger D$ on $L^2(\nu)$, with $D = \partial_q + (\beta/2)U'(q)$ in the weighted Fourier basis.

We take truncations $K = 8$ (Fourier) and $N = 16$ (Hermite) for $\mathcal{L}$, and a larger Fourier cutoff $K_{\rm ovd} = 24$ for the overdamped operator Q_O on $L^2(\nu)$. Friction is scanned over $\gamma \in \{0.1, 0.2, 0.5, 1, 2, 4, 8\}$ and forcing over $\eta \in \{0, 0.1, 1\}$. The computed macroscopic gap is $\lambda(Q_O) \approx 1.1655$, hence the predicted optimal friction is $\gamma_{\rm opt} \approx \sqrt{\lambda(Q_O)} \approx 1.08$.

The two panels directly verify the scaling statements from the lower/upper-bound corollaries. For $\eta = 0$, the underdamped ratio $\mathrm{gap}(\mathcal{L})/\gamma$ is nearly constant at small friction (e.g., 1.05 at $\gamma = 0.1$ and 1.03 at $\gamma = 0.2$), while the overdamped ratio $\gamma\,\mathrm{gap}(\mathcal{L})/\lambda(Q_O)$ approaches 1 at large friction (0.994 at $\gamma = 4$ and 0.999 at $\gamma = 8$), confirming the diffusive ceiling. The spectral gap peaks near $\gamma \approx 1$, consistent with $\gamma_{\rm opt} \approx 1.08$. For $\eta = 0.1$ the

same trends persist with only mild shifts in the prefactors. For $\eta = 1$ the two smallest frictions show positive real eigenvalues (loss of stability outside the perturbative regime), while the remaining points still exhibit the overdamped γ^{-1} scaling.

As a consistency check of the discretization, we consider the free case $U_0 = 0$ (and $\eta = 0$). In this situation the generator is explicitly diagonalizable in the Fourier–Hermite basis, and the eigenvalues are known in closed form:[29]

$$\lambda_{n,k} = -\frac{n\gamma}{m} - \frac{k^2}{\beta\gamma}, \qquad n \in \mathbb{N},\ k \in \mathbb{Z}, \tag{123}$$

so the spectral gap equals

$$\min\Big(\frac{\gamma}{m}, \frac{1}{\beta\gamma}\Big), \tag{124}$$

with the crossover at $\gamma = \sqrt{m/\beta}$. For our choice $m = \beta = 1$, this reduces to $\min(\gamma, \gamma^{-1})$, which our Galerkin discretization recovers to relative error $\sim 10^{-13}$.

B. Example 2: A regularized Wigner–Fokker–Planck model

We briefly outline how the abstract lifting structure applies to a (regularized) Wigner–Fokker–Planck (WFP) equation in phase space. In particular, the strong-friction ($\gamma \to \infty$) reduction yields an effective (quantum) Smoluchowski dynamics on the slow variable, and the accelerated regime again occurs at intermediate friction $\gamma \asymp \sqrt{\lambda(Q_O)}$.

Phase space, reference measure, and a genuinely quantum correction. For simplicity we take $\mathcal{X} = \mathbb{T}^d \times \mathbb{R}^d$ with coordinates (q, p). We work in a weighted space $L^2(\mu_{\mathrm{ref}})$ with a smooth, strictly positive *reference* measure of Gibbs type

$$d\mu_{\mathrm{ref}}(q,p) \propto \exp\Big(-\frac{m}{D_{pp}}\big(\tfrac{|p|^2}{2m} + U(q)\big)\Big)\, dq\, dp = \exp\Big(-\frac{|p|^2}{2D_{pp}} - \frac{m}{D_{pp}}U(q)\Big)\, dq\, dp, \tag{125}$$

so that the Hamiltonian transport part is skew-adjoint on a natural smooth core. Equivalently,

$$d\mu_{\mathrm{ref}}(q,p) = Z^{-1}\, e^{-\frac{m}{D_{pp}}U(q)}\, dq\, M(p)\, dp, \qquad M(p) := \frac{1}{Z_p}\exp\Big(-\frac{|p|^2}{2D_{pp}}\Big). \tag{126}$$

As in Example VI A, this reference weight is *not* assumed to be the stationary distribution of the full nonequilibrium dynamics when $\eta \neq 0$. Rather, the nonequilibrium steady state (when it exists and is unique) is represented as a density $d\mu_{\eta,\ell,\hbar} = h_{\eta,\ell,\hbar}\, d\mu_{\mathrm{ref}}$, and our L^2 bounds quantify convergence toward $h_{\eta,\ell,\hbar}$ in the reference space. The role of μ_{ref} is structural: it provides a convenient Hilbert setting in which the Hamiltonian transport part is skew-adjoint on a natural smooth core and the fast operator is dissipative.

In order for this example to be *genuinely different* from the classical Langevin/Fokker–Planck setting at the PDE level, we incorporate a standard quantum/disorder mechanism: *local system–bath coupling with finite correlation length.* In the Wigner representation this leads to friction/noise operators that are *nonlocal in momentum* (equivalently, involve arbitrarily high p-derivatives when expanded), even in the high-temperature Markovian regime; see[28]. Concretely, for a Gaussian bath correlation profile with correlation length $\ell > 0$, the corresponding momentum-space kernels are smoothing operators. In the weighted space $L^2(M(p)dp)$ it is natural to express these kernels through the self-adjoint momentum Dirichlet operator

$$\mathcal{A}_p := \nabla_p^\dagger \nabla_p, \qquad \nabla_p^\dagger := -\nabla_p + \frac{p}{D_{pp}}, \tag{127}$$

where † denotes the adjoint in $L^2(M(p)dp)$. We model the finite-correlation-length quantum correction by a *dissipative* higher-order modification of the momentum Dirichlet form. Concretely, we take the (self-adjoint, non-positive) operator

$$\mathcal{N}_{\ell,\hbar} := -\mathcal{A}_p - \frac{\hbar^2}{4\ell^2}\,\mathcal{A}_p^2. \tag{128}$$

This keeps the leading Ornstein–Uhlenbeck/Dirichlet part $-\mathcal{A}_p$ (hence retains the correct energy norm controlling one momentum derivative) while adding an $\mathcal{O}((\hbar/\ell)^2)$ higher-order dissipative correction. In the Gaussian limit $\ell \to \infty$ (or formally $\hbar \to 0$), one has $\mathcal{N}_{\ell,\hbar} \to -\mathcal{A}_p$ and therefore the fast operator reduces exactly to the standard Ornstein–Uhlenbeck dissipator on $L^2(M(p)dp)$.

Generator decomposition. We consider the hypocoercive splitting

$$\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\,\mathcal{R}_{\ell,\hbar} + \eta\,\mathcal{V}_{\text{pert}}, \tag{129}$$

with

$$\mathcal{V}_{\text{skew}} := \frac{p}{m}\cdot\nabla_q - \nabla_q U(q)\cdot\nabla_p, \qquad \mathcal{R}_{\ell,\hbar} := D_{pp}\,\mathcal{N}_{\ell,\hbar}, \qquad \mathcal{V}_{\text{pert}} := F(q)\cdot\nabla_p. \tag{130}$$

The slow subspace $\mathcal{H}_S = \ker(\mathcal{R}_{\ell,\hbar})$ consists of p-independent functions.

Here $D_{pp} > 0$ is the (effective) momentum diffusion/variance parameter; in semiclassical quantum Brownian-motion models it generally receives non-classical corrections and may remain bounded away from 0 as $T \to 0$ due to vacuum fluctuations; see, e.g.,[27]. The orthogonal projection P_S is the conditional expectation in p with respect to $M(p)\,dp$ (i.e. orthogonal projection onto p-independent functions in $L^2(\mu_{\text{ref}})$).

Remark VI.3 (Why this is not just "Example 1 again"). If one sets $\ell = \infty$ (global bath coupling), then $\mathcal{N}_{\ell,\hbar} \to -\mathcal{A}_p$ and hence $\mathcal{R}_{\ell,\hbar} \to -D_{pp}\,\mathcal{A}_p = -p\cdot\nabla_p + D_{pp}\Delta_p$, so the model collapses to the classical Kramers/Fokker–Planck structure. For finite ℓ, however, the fast dissipator contains the full hierarchy of higher momentum derivatives (or, equivalently, a nonlocal convolution kernel in p), which leads in the overdamped limit to a *Kramers–Moyal/quantum Smoluchowski* effective equation with non-Gaussian Markovian noise contributions[28]. This provides a concrete phase-space quantum example in which the lifted generator is not merely a re-parameterized classical Langevin operator.

Recovery of the (quantum) Smoluchowski effective dynamics. We now derive the effective slow generator explicitly from the abstract formula. Let $\phi \in \mathcal{H}_S$ be a slow observable, i.e. $\phi = \phi(q)$ independent of p.

Under our general AE reduction (including forcing),

$$\mathcal{L}_O = -P_S(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})\,\mathcal{R}_{\ell,\hbar}^{+}\,(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})P_S. \tag{131}$$

Since $\mathcal{V}_{\text{pert}}\phi = 0$ for $\phi = \phi(q)$, all terms with $\mathcal{V}_{\text{pert}}$ acting on the *right* vanish on ϕ; the only $\mathcal{O}(\eta)$ contribution is the left mixed term $-P_S\mathcal{V}_{\text{pert}}\,\mathcal{R}_{\ell,\hbar}^{+}\,\mathcal{V}_{\text{skew}}P_S$.

Since $\nabla_p\phi = 0$, we have

$$\mathcal{V}_{\text{skew}}\phi = \frac{p}{m}\cdot\nabla_q\phi \in \mathcal{H}_F. \tag{132}$$

Here we keep the *full* $\mathcal{R}_{\ell,\hbar}$. Since $\mathcal{A}_p p_i = D_{pp}^{-1} p_i$ on the first Hermite sector, we have

$$\mathcal{R}_{\ell,\hbar}p_i = D_{pp}\Big(-\mathcal{A}_p - \frac{\hbar^2}{4\ell^2}\,\mathcal{A}_p^2\Big)p_i = -\kappa_1(\ell,\hbar)\,p_i, \tag{133}$$

with the explicit coefficient

$$\kappa_1(\ell,\hbar) := 1 + \frac{\hbar^2}{4\ell^2 D_{pp}}. \tag{134}$$

Therefore $\mathcal{R}^+_{\ell,\hbar} p_i = -(1/\kappa_1(\ell,\hbar))\, p_i$ on the fast subspace, and hence

$$\mathcal{R}^+_{\ell,\hbar}\mathcal{V}_{\rm skew}\phi = -\frac{1}{\kappa_1(\ell,\hbar)}\,\frac{p}{m}\cdot\nabla_q\phi. \tag{135}$$

We compute

$$\mathcal{V}_{\rm skew}\Big(-\frac{1}{\kappa_1(\ell,\hbar)}\,\frac{p}{m}\cdot\nabla_q\phi\Big) = -\frac{1}{\kappa_1(\ell,\hbar)}\,\frac{p}{m}\cdot\nabla_q\Big(\frac{p}{m}\cdot\nabla_q\phi\Big) + \frac{1}{\kappa_1(\ell,\hbar)}\,\nabla_q U\cdot\nabla_p\Big(\frac{p}{m}\cdot\nabla_q\phi\Big) \tag{136}$$

$$= -\frac{1}{\kappa_1(\ell,\hbar)}\,\frac{1}{m^2}\sum_{i,j} p_i p_j\,\partial_{q_i q_j}\phi + \frac{1}{\kappa_1(\ell,\hbar)}\,\frac{1}{m}\,\nabla_q U\cdot\nabla_q\phi. \tag{137}$$

Projecting onto $\mathcal{H}_S$ amounts to averaging in p against $M(p)\,dp$, for which $\int p_i p_j\, M(p)dp = D_{pp}\,\delta_{ij}$. Therefore

$$P_S\mathcal{V}_{\rm skew}\mathcal{R}^+_{\ell,\hbar}\mathcal{V}_{\rm skew}\phi = -\frac{1}{\kappa_1(\ell,\hbar)}\,\frac{D_{pp}}{m^2}\,\Delta_q\phi + \frac{1}{\kappa_1(\ell,\hbar)}\,\frac{1}{m}\,\nabla_q U\cdot\nabla_q\phi. \tag{138}$$

For the forcing contribution, we compute

$$\mathcal{V}_{\rm pert}\Big(-\frac{1}{\kappa_1(\ell,\hbar)}\,\frac{p}{m}\cdot\nabla_q\phi\Big) = F(q)\cdot\nabla_p\Big(-\frac{1}{\kappa_1(\ell,\hbar)}\,\frac{p}{m}\cdot\nabla_q\phi\Big) = -\frac{1}{\kappa_1(\ell,\hbar)}\,\frac{1}{m}\,F(q)\cdot\nabla_q\phi, \tag{139}$$

which is independent of p and hence fixed by P_S. Therefore the explicit slow generator on observables is

$$\mathcal{L}_O\phi = \frac{1}{\kappa_1(\ell,\hbar)}\left(\frac{D_{pp}}{m^2}\,\Delta_q\phi - \frac{1}{m}\,\nabla_q U\cdot\nabla_q\phi + \frac{\eta}{m}\,F(q)\cdot\nabla_q\phi\right). \tag{140}$$

Equivalently, the adjoint dynamics for the slow density $\rho(q,t)$ is the Smoluchowski equation

$$\partial_t\rho = \frac{1}{\kappa_1(\ell,\hbar)}\left(\frac{D_{pp}}{m^2}\,\Delta_q\rho + \frac{1}{m}\,\nabla_q\cdot\big(\rho\,\nabla_q U\big) - \frac{\eta}{m}\,\nabla_q\cdot\big(\rho\,F\big)\right), \tag{141}$$

with D_{pp} encoding the quantum diffusion coefficient.[27,30]

The computation above already uses the full $\mathcal{R}_{\ell,\hbar}$ and exhibits a *leading* quantum renormalization through the explicit factor $\kappa_1(\ell,\hbar)^{-1}$. Expanding κ_1 for $\hbar/\ell \ll 1$ gives

$$\frac{1}{\kappa_1(\ell,\hbar)} = 1 - \frac{\hbar^2}{4\ell^2 D_{pp}} + \mathcal{O}\big((\hbar/\ell)^4\big), \tag{142}$$

so the effective diffusion and drift coefficients acquire an explicit, controlled $\mathcal{O}((\hbar/\ell)^2)$ correction.

Beyond this leading correction, the expansion of $\mathcal{R}_{\ell,\hbar}$ in powers of $\mathcal{A}_p$ generates a full hierarchy of higher-order terms in the strong-friction/adiabatic-elimination expansion. These come from repeated excursions into higher Hermite sectors, for which $\mathcal{A}_p$ has eigenvalues n/D_{pp} (so the corresponding fast rates are $\kappa_n(\ell,\hbar) = n + \frac{\hbar^2}{4\ell^2 D_{pp}}\,n^2$). In the Wigner picture, this produces an even-derivative Kramers–Moyal-type expansion for the slow marginal ρ with coefficients scaling as $\mathcal{O}((\hbar/\ell)^{2n-2})$; see[28].

Our framework then predicts an optimal friction scale $\gamma_{\rm opt} \asymp \sqrt{\lambda(Q_O)}$ for the accelerated convergence. For the Gaussian-limit (global-coupling) Wigner–Fokker–Planck equation, the existence of spectral gaps and large-time behavior in weighted spaces is treated in detail in[31]; for the strong-damping/Kramers–Moyal reduction induced by finite correlation length, see[28].

Verification of the structural assumptions.

Proposition VI.4 (Structural Assumptions for WFP)**.** *The WFP operator satisfies Assumption IV.1.1–4 in the weighted space $L^2(\mu_{\rm ref})$.*

Proof. **Conditions 1–3:** On $L^2(\mu_{\rm ref})$, the generalized fast operator $\mathcal{R}_{\ell,\hbar}$ acts only on the p-variable. By construction, $\mathcal{A}_p = \nabla_p^\dagger \nabla_p$ is nonnegative and self-adjoint on $L^2(M(p)dp)$, hence $\mathcal{R}_{\ell,\hbar} = D_{pp}\mathcal{N}_{\ell,\hbar}$ is obtained from $\mathcal{A}_p$ by (self-adjoint) functional calculus on any smooth core. In particular, $\mathcal{R}_{\ell,\hbar}$ is self-adjoint, non-positive, and has a spectral gap on $\mathcal{H}_F$, with kernel $\mathcal{H}_S$ consisting of p-independent functions. Moreover, with the Gibbs weight above (a function of $H_0 = \frac{|p|^2}{2m} + U(q)$), the Liouville operator $\mathcal{V}_{\rm skew}$ is skew-adjoint on a natural smooth core. For $\phi \in \mathcal{H}_S$ we have $\mathcal{V}_{\rm skew}\phi = \frac{p}{m}\cdot\nabla_q\phi$, hence $P_S\mathcal{V}_{\rm skew}P_S = 0$ because $\int p\, M(p)dp = 0$.

For Assumption IV.1.3, let $\psi \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S$ and set $g_\psi = P_F\mathcal{V}_{\rm skew}P_S\psi = \frac{p}{m}\cdot\nabla_q\psi$. Since $\mathcal{A}_p$ is diagonal on the Hermite basis, each linear function p_i is an eigenfunction of $\mathcal{A}_p$ with eigenvalue D_{pp}^{-1} (the first Hermite sector). With $\mathcal{R}_{\ell,\hbar} = D_{pp}\mathcal{N}_{\ell,\hbar}$ and $\mathcal{N}_{\ell,\hbar} = -\mathcal{A}_p - \frac{\hbar^2}{4\ell^2}\mathcal{A}_p^2$, we obtain the explicit eigen-relation

$$\mathcal{R}_{\ell,\hbar}(p_i) = -\kappa_1(\ell,\hbar)\, p_i, \qquad \kappa_1(\ell,\hbar) := 1 + \frac{\hbar^2}{4\ell^2 D_{pp}}. \tag{143}$$

Hence $(-\mathcal{R}_{\ell,\hbar}^+)g_\psi = \kappa_1(\ell,\hbar)^{-1}g_\psi$ and the flux conditioning bound (22) holds with the (sharp) choice $M_R := \kappa_1(\ell,\hbar)$. Moreover,

$$P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}_{\ell,\hbar}^+)g_\psi = P_F\mathcal{V}_{\rm skew}\Big(\frac{1}{\kappa_1(\ell,\hbar)}\,\frac{p}{m}\cdot\nabla_q\psi\Big) = \frac{1}{\kappa_1(\ell,\hbar)}\,\frac{1}{m^2}\sum_{i,j}\big(p_ip_j - D_{pp}\,\delta_{ij}\big)\,\partial_{q_iq_j}\psi, \tag{144}$$

and (as in Example VI A) the Gaussian moment bound in p together with ellipticity on the slow space yield (24) with some finite constant $\kappa_{\rm ell}$ depending only on U, m, D_{pp}, d.

Condition 4(a)–4(b): Perturbation decomposition and energy bounds. Here we keep the same forcing as in Example VI A, $\mathcal{V}_{\rm pert} = F(q)\cdot\nabla_p$. The key point is that the present fast operator dominates the Ornstein–Uhlenbeck Dirichlet form: since $-\mathcal{N}_{\ell,\hbar} = \mathcal{A}_p + \frac{\hbar^2}{4\ell^2}\mathcal{A}_p^2 \geq \mathcal{A}_p$, we have

$$\langle f, (-\mathcal{R}_{\ell,\hbar})f\rangle_{L^2(\mu_{\rm ref})} = D_{pp}\,\langle f, (-\mathcal{N}_{\ell,\hbar})f\rangle \geq D_{pp}\,\langle f, \mathcal{A}_pf\rangle = D_{pp}\,\|\nabla_pf\|^2_{L^2(\mu_{\rm ref})}. \tag{145}$$

Equivalently, on the energy domain $\mathrm{dom}((-\mathcal{R}_{\ell,\hbar})^{1/2})$,

$$\|\nabla_pf\|_{L^2(\mu_{\rm ref})} \leq D_{pp}^{-1/2}\,\|(-\mathcal{R}_{\ell,\hbar})^{1/2}f\|_{L^2(\mu_{\rm ref})}. \tag{146}$$

With this replacement (and the same Gaussian integration-by-parts estimate controlling $\||p|f\|$ by $\|f\|+\|\nabla_pf\|$), one obtains Assumption IV.1.4(a)–4(b) exactly as in Example VI A (with β replaced by m/D_{pp}), i.e. there exist constants $a_1, b_1, a_2, b_2 \geq 0$ depending only on $\|F\|_{L^\infty}$, D_{pp} and d such that

$$\|\mathcal{K}_{\rm pert}f\| \leq a_1\,\|(-\mathcal{R}_{\ell,\hbar})^{1/2}f\| + b_1\,\|f\|, \qquad \|\mathcal{S}_{\rm pert}f\| \leq a_2\,\|(-\mathcal{R}_{\ell,\hbar})^{1/2}f\| + b_2\,\|f\|. \tag{147}$$

□

Numerical verification (Galerkin spectral gap). We perform the same Galerkin spectral check as in Example VI A, now with the WFP coefficients and the D_{pp}-weighted reference measure. We work in $d = 1$ with the rotor potential $U(q) = U_0(1-\cos q)$ (here $U_0 = 1$), mass $m = 1$, quantum diffusion parameter $D_{pp} = 0.8$, correlation length $\ell = 1.5$, and $\hbar = 1$. The basis is $\psi_{nk}(q,p) = G_k(q)H_n(p)$ with $G_k(q) \propto e^{ikq+\frac{m}{2D_{pp}}U(q)}$ and Hermite polynomials orthonormal in the Gaussian momentum marginal with variance D_{pp}. The

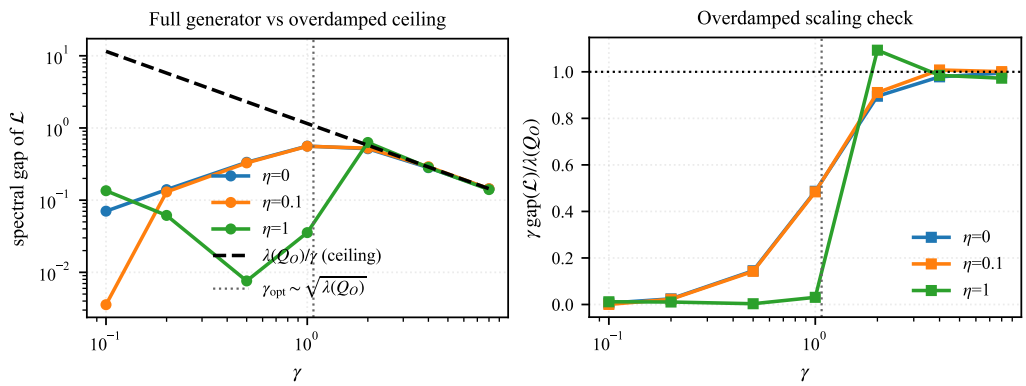

Figure 2. Numerical verification of Example VI B (Galerkin spectral gap). *Left:* $\mathrm{gap}(\mathcal{L})$ vs. γ with the overdamped ceiling $\lambda(Q_O)/\gamma$ and $\gamma_{\rm opt} \sim \sqrt{\lambda(Q_O)}$. *Right:* scaled gap $\gamma\,\mathrm{gap}(\mathcal{L})/\lambda(Q_O)$ approaching a constant for large γ. Parameters: $U_0 = 1$, $m = 1$, $D_{pp} = 0.8$, $\ell = 1.5$, $\hbar = 1$, $K = 8$, $N = 16$, $K_{\rm ovd} = 24$, and $\eta \in \{0, 0.1, 1\}$.

full generator $\mathcal{L}$ is represented by the finite Galerkin matrix of the exact coefficients (obtained by replacing β with m/D_{pp} in the transport part and using the fast spectrum $\kappa_n(\ell,\hbar) = \frac{2\ell^2 D_{pp}}{\hbar^2}(1 - e^{-\frac{n\hbar^2}{2\ell^2 D_{pp}}})$ on the nth Hermite level), while the slow reference operator is $Q_O = -(D_{pp}/(m^2\kappa_1))D^\dagger D$ on $L^2(\nu)$ with $D = \partial_q + (m/2D_{pp})U'(q)$.

We take truncations $K = 8$ (Fourier) and $N = 16$ (Hermite) for $\mathcal{L}$, and $K_{\rm ovd} = 24$ for Q_O. Friction is scanned over $\gamma \in \{0.1, 0.2, 0.5, 1, 2, 4, 8\}$ and forcing over $\eta \in \{0, 0.1, 1\}$ (with $F \equiv 1$). The computed macroscopic gap is $\lambda(Q_O) \approx 1.1522$, hence the predicted optimal friction is $\gamma_{\rm opt} \approx \sqrt{\lambda(Q_O)} \approx 1.073$.

The figures confirm the same scaling statements for the WFP setting. For $\eta = 0$, the underdamped ratio $\mathrm{gap}(\mathcal{L})/\gamma$ is nearly constant at small friction (e.g., 0.70 at $\gamma = 0.1$ and 0.70 at $\gamma = 0.2$), while the overdamped ratio $\gamma\,\mathrm{gap}(\mathcal{L})/\lambda(Q_O)$ approaches 1 at large friction (0.979 at $\gamma = 4$ and 0.995 at $\gamma = 8$), confirming the diffusive ceiling. The spectral gap peaks near $\gamma \approx 1$, consistent with $\gamma_{\rm opt} \approx 1.073$. For $\eta = 0.1$ the same trends persist with small prefactor shifts (the smallest γ point is unstable). For $\eta = 1$ the four smallest frictions show positive real eigenvalues (loss of stability outside the perturbative regime), while the remaining points still exhibit the overdamped γ^{-1} scaling.

C. Example 3: A finite-dimensional boundary-driven toy model

We present a minimal finite-dimensional example (in the same spirit as boundary-driven/Zeno-limit settings[11,14,32]) that satisfies Assumption IV.1 and for which the effective slow generator $\mathcal{L}_O$ is generically *non-self-adjoint* on the slow space.

Setup.
Hilbert space and reference state. Let the fast ancilla be a qubit $\mathfrak{h}_F \cong \mathbb{C}^2$ and the slow system be a qubit $\mathfrak{h}_S \cong \mathbb{C}^2$. We work on the observable space $\mathcal{H} = L^2(\sigma)$ associated with the product reference state

$$\sigma := \pi_F \otimes \pi_S, \qquad \pi_F := \tfrac{1}{2}\mathbf{1}_F, \qquad \pi_S := \tfrac{1}{2}\mathbf{1}_S. \tag{148}$$

Since σ is proportional to the identity, the KMS inner product $\langle X, Y\rangle_\sigma := \mathrm{tr}(\sigma^{1/2}X^\dagger\sigma^{1/2}Y)$ is proportional to the Hilbert–Schmidt inner product; we use this $L^2(\sigma)$ structure throughout.

Slow/fast splitting. We take the slow subspace

$$\mathcal{H}_S := \mathbf{1}_F \otimes \mathcal{B}(\mathfrak{h}_S), \qquad P_S(X) := \mathbf{1}_F \otimes \mathrm{tr}_F(\pi_F X), \qquad P_F := \mathrm{id} - P_S. \tag{149}$$

Then P_S is the $L^2(\sigma)$-orthogonal projection (conditional expectation) onto $\mathcal{H}_S$.

Generator decomposition. We consider

$$\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\,\mathcal{R} + \eta\,\mathcal{V}_{\text{pert}}. \tag{150}$$

1. **Fast dissipator ($\mathcal{R}$).** We fix an explicit primitive Lindbladian acting only on the ancilla,

$$\mathcal{R}_F X := \Gamma \sum_{j\in\{x,y,z\}} \mathcal{D}^\dagger[\sigma_j]X, \qquad \Gamma > 0, \tag{151}$$

and set $\mathcal{R} := \mathcal{R}_F \otimes \mathrm{id}_S$. Here σ_j are the Pauli matrices and, for any operator L, the Heisenberg dissipator is

$$\mathcal{D}^\dagger[L]X := L^\dagger X L - \tfrac{1}{2}\big(L^\dagger L\, X + X\, L^\dagger L\big). \tag{152}$$

Since each σ_j is unitary, $\mathcal{D}^\dagger[\sigma_j]X = \sigma_j X \sigma_j - X$. In particular, $\mathcal{R}_F$ (hence $\mathcal{R}$) is self-adjoint and non-positive on $L^2(\sigma)$, with $\ker(\mathcal{R}) = \mathcal{H}_S$. Moreover, for each Pauli matrix σ_j one has $\mathcal{R}_F(\sigma_j) = -4\Gamma\,\sigma_j$, so the spectral gap on the traceless (fast) subspace is $\lambda_R = 4\Gamma$.

2. **Hamiltonian transport ($\mathcal{V}_{\text{skew}}$).** We choose the skew-adjoint transport generated by a Hamiltonian commutator,

$$\mathcal{V}_{\text{skew}} X := -i[H_0, X], \qquad H_0 := g\,\sigma_z \otimes \sigma_z, \qquad g \in \mathbb{R}. \tag{153}$$

3. **Perturbation ($\mathcal{V}_{\text{pert}}$).** To obtain an effective slow generator that is generically neither self-adjoint nor skew-adjoint, we choose a bounded centered perturbation

$$\mathcal{V}_{\text{pert}} := \mathcal{L}_z \otimes \mathcal{G}_S, \qquad (\mathcal{L}_z Z_F) := \sigma_z Z_F, \tag{154}$$

where $\mathcal{G}_S$ is a fixed bounded generator on the slow qubit that is not self-adjoint in $L^2(\pi_S)$. For concreteness, take the (Heisenberg) amplitude-damping type map

$$\mathcal{G}_S(X) := \mathcal{D}^\dagger[\sigma_-]X, \qquad \sigma_- := \tfrac{1}{2}(\sigma_x - i\sigma_y), \qquad \sigma_+ := \sigma_-^\dagger = \tfrac{1}{2}(\sigma_x + i\sigma_y). \tag{155}$$

Since $\mathrm{tr}(\pi_F \sigma_z) = 0$, both couplings map $\mathcal{H}_S$ into $\mathcal{H}_F$. Indeed, for any $X_S \in \mathcal{B}(\mathfrak{h}_S)$,

$$\mathcal{V}_{\text{skew}}(\mathbf{1}_F \otimes X_S) = -ig\,\sigma_z \otimes [\sigma_z, X_S], \qquad \mathcal{V}_{\text{pert}}(\mathbf{1}_F \otimes X_S) = \sigma_z \otimes \mathcal{G}_S(X_S), \tag{156}$$

and hence $P_S \mathcal{V}_{\text{skew}} P_S = 0$ and $P_S \mathcal{V}_{\text{pert}} P_S = 0$.

Effective generator $\mathcal{L}_O$ (explicit formula). Let $X = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$.
Step 1 (first excursion to fast space).

$$(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = \sigma_z \otimes \Big(-ig\,[\sigma_z, X_S] + \eta\,\mathcal{G}_S(X_S)\Big) \in \mathcal{H}_F. \tag{157}$$

Step 2 (Poisson equation on the fast ancilla). In finite dimensions, the Moore–Penrose pseudoinverse can be written as

$$\mathcal{R}^+ := -\int_0^\infty \big(e^{s\mathcal{R}} - P_S\big)\, ds, \tag{158}$$

so that $\mathcal{R}\mathcal{R}^+ = \mathrm{id} - P_S = P_F$ and $P_S\mathcal{R}^+ = \mathcal{R}^+ P_S = 0$. In particular, since $\mathcal{R}_F(\sigma_z) = -\kappa\,\sigma_z$ with $\kappa := 4\Gamma$, we have $\mathcal{R}^+(\sigma_z \otimes Y) = \mathcal{R}_F^+(\sigma_z) \otimes Y = -(1/\kappa)\sigma_z \otimes Y$. Therefore

$$\mathcal{R}^+(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = -\frac{1}{\kappa}\sigma_z \otimes \Big(-ig\,[\sigma_z, X_S] + \eta\,\mathcal{G}_S(X_S)\Big). \tag{159}$$

Step 3 (return map and projection). Since $\mathcal{V}_{\rm skew}(\sigma_z \otimes Y) = -ig\,\mathbf{1}_F \otimes [\sigma_z, Y]$ and $\mathcal{V}_{\rm pert}(\sigma_z \otimes Y) = \mathbf{1}_F \otimes \mathcal{G}_S(Y)$, we compute

$$(\mathcal{V}_{\rm skew} + \eta\mathcal{V}_{\rm pert})\,\mathcal{R}^+(\mathcal{V}_{\rm skew} + \eta\mathcal{V}_{\rm pert})X \tag{160}$$

$$= \frac{1}{\kappa}\mathbf{1}_F \otimes \Big(g^2\,[\sigma_z,[\sigma_z,X_S]] + i\eta g\,\big[\sigma_z,\mathcal{G}_S(X_S)\big] + i\eta g\,\mathcal{G}_S([\sigma_z,X_S]) - \eta^2\,\mathcal{G}_S^2(X_S)\Big). \tag{161}$$

Projecting with $P_S(\mathbf{1}_F \otimes Y) = \mathbf{1}_F \otimes Y$ yields

$$\mathcal{L}_O(\mathbf{1}_F \otimes X_S) = -\frac{1}{\kappa}\,\mathbf{1}_F \otimes \Big(g^2\,[\sigma_z,[\sigma_z,X_S]] + i\eta g\,\big[\sigma_z,\mathcal{G}_S(X_S)\big] + i\eta g\,\mathcal{G}_S([\sigma_z,X_S]) - \eta^2\,\mathcal{G}_S^2(X_S)\Big). \tag{162}$$

The reference operator is the $\eta = 0$ part,

$$Q_O(\mathbf{1}_F \otimes X_S) = -\frac{g^2}{\kappa}\,\mathbf{1}_F \otimes [\sigma_z,[\sigma_z,X_S]]. \tag{163}$$

Non-self-adjointness on the slow space. Since $\pi_S \propto \mathbf{1}_S$, the $L^2(\pi_S)$ inner product is proportional to the Hilbert–Schmidt inner product, and one checks that $(\mathcal{D}^\dagger[\sigma_-])^\dagger = \mathcal{D}^\dagger[\sigma_+] \neq \mathcal{D}^\dagger[\sigma_-]$. Thus $\mathcal{G}_S$ is not self-adjoint on $L^2(\pi_S)$. Consequently, the $\mathcal{O}(\eta)$ mixed operator

$$\mathcal{B} := ig\,\big(\,{\rm ad}_{\sigma_z} \circ \mathcal{G}_S + \mathcal{G}_S \circ {\rm ad}_{\sigma_z}\,\big), \qquad {\rm ad}_{\sigma_z}(X) := [\sigma_z, X], \tag{164}$$

is generically not self-adjoint either (unless additional commutation/self-adjointness relations hold), so $\mathcal{L}_O$ is generically non-self-adjoint for $\eta \neq 0$.

Verification of Assumption IV.1. All operators are bounded on the finite-dimensional space $\mathcal{H}$.

1. Assumption IV.1.1 holds for $\mathcal{R}$ by construction, with $\lambda_R = 4\Gamma$.
2. Assumption IV.1.2 holds since both $\mathcal{V}_{\rm skew}$ and $\mathcal{V}_{\rm pert}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$.
3. Assumption IV.1.3 holds with $M_R := \|\mathcal{R}\|$. Moreover, in this toy model the flux regularity bound (24) can be checked explicitly. For $\psi = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$ one has $g_\psi = P_F\mathcal{V}_{\rm skew}P_S\psi = -ig\,\sigma_z \otimes [\sigma_z, X_S]$ and $(-\mathcal{R}^+)g_\psi = (1/\kappa)g_\psi$. Therefore

$$P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\psi = 0, \tag{165}$$

$$(-Q_O)\psi = P_S\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\psi = \frac{g^2}{\kappa}\,\mathbf{1}_F \otimes [\sigma_z,[\sigma_z,X_S]]. \tag{166}$$

   Since the left-hand side is identically zero, (24) holds (e.g. with $\kappa_{\rm ell} := 1$).
4. For Assumption IV.1.4, decompose $\mathcal{V}_{\rm pert} = \mathcal{K}_{\rm pert} + \mathcal{S}_{\rm pert}$ by $\mathcal{K}_{\rm pert} := \frac{1}{2}(\mathcal{V}_{\rm pert} - \mathcal{V}_{\rm pert}^\dagger)$ and $\mathcal{S}_{\rm pert} := \frac{1}{2}(\mathcal{V}_{\rm pert} + \mathcal{V}_{\rm pert}^\dagger)$. Since $\mathcal{V}_{\rm pert}$ is bounded, the energy bounds (27)–(28) hold with $a_1 = a_2 = 0$ and $b_1 = \|\mathcal{K}_{\rm pert}\|$, $b_2 = \|\mathcal{S}_{\rm pert}\|$.

Numerical verification (spectral gap). We construct the Liouville-space Lindbladian for the two-qubit toy model and compute its spectral gap on the mean-zero subspace. Friction is scanned over $\gamma \in \{0.1, 0.2, 0.5, 1, 2, 4, 8\}$ and forcing over $\eta \in \{0, 0.1, 1\}$. The computed macroscopic gap is $\lambda(Q_O) \approx 1.25$, hence the predicted optimal friction is $\gamma_{\rm opt} \approx \sqrt{\lambda(Q_O)} \approx 1.118$.

For $\eta = 0$, the underdamped ratio ${\rm gap}(\mathcal{L})/\gamma$ is nearly constant at small friction (e.g., ≈ 1.6 at $\gamma = 0.1$ and $\gamma = 0.2$), while the overdamped ratio $\gamma\,{\rm gap}(\mathcal{L})/\lambda(Q_O)$ approaches 1 at large γ (approximately 1.026 at $\gamma = 4$ and 1.006 at $\gamma = 8$), confirming the diffusive ceiling and the predicted $\gamma_{\rm opt}$. For $\eta = 0.1$ and $\eta = 1$ the scan exhibits numerical instabilities (positive real eigenvalues) across the sampled range, indicating a breakdown of the perturbative regime for those perturbation strengths; where the spectrum is reliable, the overdamped γ^{-1} scaling remains visible at larger friction.

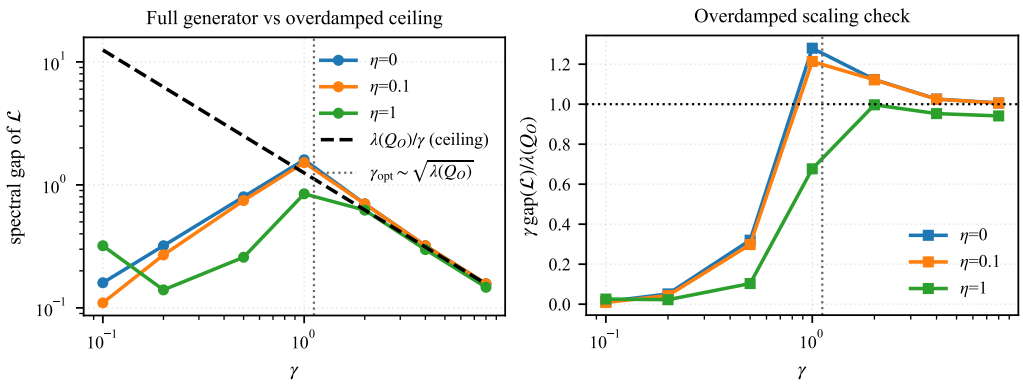

Figure 3. Numerical verification of Example VI C (spectral gap). *Left:* gap($\mathcal{L}$) vs. γ with the overdamped ceiling $\lambda(Q_O)/\gamma$ and $\gamma_{\rm opt} \sim \sqrt{\lambda(Q_O)}$. *Right:* scaled gap $\gamma\,\mathrm{gap}(\mathcal{L})/\lambda(Q_O)$ approaching a constant for large γ. Parameters: $\Gamma = 0.8$, $g = 1$, $\kappa = 3.2$, and $\eta \in \{0, 0.1, 1\}$.

VII. CONCLUSION

In this work, we have established a unified operator-theoretic framework linking the physical theory of adiabatic elimination (AE) with the mathematical theory of lifting for Markov processes. We demonstrated that generators of the form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\rm skew} + \eta\mathcal{V}_{\rm pert}$ can be analyzed as *robust hypocoercive lifts* of their effective slow dynamics: the Zeno-limit reduced generator is $\mathcal{L}_O$, while the quantitative acceleration mechanism is governed by the self-adjoint reference operator Q_O extracted from $\mathcal{V}_{\rm skew}$. This perspective effectively inverts the logic of the Quantum Zeno effect for state preparation tasks: rather than driving the system to the overdamped limit ($\gamma \to \infty$) to obtain a robust but kinetically frozen evolution, we identify an intermediate dissipation regime ($\gamma \sim \sqrt{\lambda(Q_O)}$) that maximizes the convergence rate to the Non-Equilibrium Steady State (NESS).

The theoretical core of our approach lies in the construction of the **operator-modified Lyapunov functional** $\mathcal{E}_{\alpha,\mu}(f)$. By introducing the **macroscopic solver** $\mathcal{A}_\mu$, defined via the resolvent of the effective generator, we explicitly capture the coherent response of the fast degrees of freedom to the slow manifold dynamics. This construction allows us to control entropy production cross-terms without relying on detailed balance. Our main result, Theorem V.1, establishes that this structure is robust against non-conservative perturbations, provided their strength η satisfies a friction-dependent stability condition $\eta < \eta^*(\gamma)$. Crucially, at the optimal lift, this stability threshold scales with the accelerated gap, $\eta^* \sim \sqrt{\lambda(Q_O)}$, proving that the "diffusive-to-ballistic" quadratic speedup is a resilient feature available to open quantum systems.

We validated the universality of this framework across three distinct physical regimes:

1. **Classical Kinetic Theory:** We recovered the optimal scaling of Langevin dynamics, showing that our L^2 spectral framework reproduces the sharp rates derived via entropic methods, provided the perturbation is physically saturated at high energies.[19]

2. **Semiclassical Phase Space:** We derived the regularization required for the Wigner-Fokker-Planck equation and showed that lifting accelerates relaxation even in the presence of vacuum fluctuation corrections to the diffusion tensor.[27]

3. **Finite-dimensional boundary driving:** We constructed a minimal toy model motivated by boundary-driven/Zeno-limit settings, whose effective slow generator is typically non-self-adjoint[11,14] and hence lies outside the reversible lifting results in[24].

Limitations and physical validity. Our spectral analysis relies on structural assumptions—specifically energy-norm relative bounds on the perturbations—that require careful physical

interpretation in infinite-dimensional spaces. In concrete implementations, the formal "optimal" lifting/friction parameters predicted by the effective model are meaningful only within the regime where the underlying reduced description (e.g., perturbative, Markovian, or dispersive approximations) remains valid. In practice, the admissible parameter range is bounded from above by non-perturbative effects such as ionization or Measurement-Induced State Transitions (MIST).[33] Thus, our framework provides the *intrinsic* kinetic limit of the effective model, which must be matched against the *extrinsic* breakdown scales of the specific hardware.

Outlook. Looking forward, this work suggests several promising directions. The concept of *iterative lifting*—recursively identifying hierarchies of fast degrees of freedom—offers a potential pathway for simulating ultra-slow quantum dynamics in hierarchical reservoirs. Additionally, the interplay between our operator-modified entropy method and recent developments in linear response theory and dilation theory could yield tighter bounds for specific classes of quantum master equations with degenerate dissipation.[34,35] Ultimately, by bridging the gap between the geometry of hypocoercivity and the physics of the Zeno effect, this framework provides a rigorous foundation for the optimal control and reservoir engineering of complex quantum systems out of equilibrium.

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Appendix A: Proofs for Preliminaries

In this appendix, we provide the rigorous proofs for the spectral bounds and hypocoercivity characterizations presented in Section II. We work within the weighted Hilbert space $L^2(\sigma)$ restricted to the mean-zero subspace $\mathcal{H}_0 := \{X \in \mathcal{B}(\mathcal{H}) : \mathrm{tr}(\sigma X) = 0\}$.

1. Proof of Lemma II.3

We first clarify the geometric relationship between the fixed point space $\mathcal{F}(\mathcal{L}) = \mathbb{C}\mathbf{1}$ and the kernel of the symmetric part $\mathcal{S}$.

Proposition A.1 (Kernel Inclusion)**.** *For any quantum Markov semigroup, the fixed point space is contained in the kernel of the symmetric part:* $\ker(\mathcal{L}) \subseteq \ker(\mathcal{S})$.

Proof. Let $X \in \ker(\mathcal{L})$. Then $\mathcal{L}(X) = 0$. Consequently,

$$\langle X, \mathcal{S}(X)\rangle_\sigma = \Re\langle X, \mathcal{L}(X)\rangle_\sigma = 0. \tag{A1}$$

Since $\mathcal{S}$ is a dissipative operator (negative semi-definite, $\mathcal{S} \leq 0$), the equality $\langle X, \mathcal{S}X\rangle_\sigma = 0$ implies $\mathcal{S}X = 0$ (essentially due to the Cauchy-Schwarz inequality for the semi-inner product defined by $-\mathcal{S}$). Thus, $X \in \ker(\mathcal{S})$. □

Proof of Lemma II.3. We prove the coercivity characterization and show that, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$, then no exponential decay estimate can hold with prefactor $C = 1$.

Direction 1 (⇐): Assume $\ker(\mathcal{S}) = \ker(\mathcal{L}) = \mathbb{C}\mathbf{1}$. Restricting operators to the mean-zero subspace $\mathcal{H}_0$, the condition implies that $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$. Since we are in finite dimensions, the self-adjoint operator $-\mathcal{S}|_{\mathcal{H}_0}$ is strictly positive definite. Let $\lambda_S > 0$ be the symmetric spectral gap. For any $X_t = \mathcal{P}_t X_0 \in \mathcal{H}_0$, the evolution of the norm satisfies:

$$\frac{1}{2}\frac{d}{dt}\|X_t\|_\sigma^2 = \Re\langle X_t, \mathcal{L}X_t\rangle_\sigma = \langle X_t, \mathcal{S}X_t\rangle_\sigma \leq -\lambda_S\|X_t\|_\sigma^2. \tag{A2}$$

Integrating this differential inequality yields $\|X_t\|_\sigma \leq e^{-\lambda_S t}\|X_0\|_\sigma$. This is an exponential decay with prefactor $C = 1$ and rate λ_S. Thus, the system is coercive, not strictly hypocoercive.

Direction 2 (⇒): Assume the system is coercive ($C = 1$). By definition, there exists $\nu > 0$ such that $\|\mathcal{P}_t X\|_\sigma \leq e^{-\nu t}\|X\|_\sigma$ for all $X \in \mathcal{H}_0$. The Hille-Yosida theorem (or simple differentiation at $t = 0$) implies that the generator satisfies the dissipativity condition:

$$\Re\langle X, \mathcal{L}X\rangle_\sigma \leq -\nu\|X\|_\sigma^2, \quad \forall X \in \mathcal{H}_0. \tag{A3}$$

Substituting $\Re\langle X, \mathcal{L}X\rangle_\sigma = \langle X, \mathcal{S}X\rangle_\sigma$, we obtain:

$$\langle X, \mathcal{S}X\rangle_\sigma \leq -\nu\|X\|_\sigma^2. \tag{A4}$$

Now, suppose there exists a non-trivial vector $Y \in \ker(\mathcal{S}) \cap \mathcal{H}_0$. Then $\langle Y, \mathcal{S}Y\rangle_\sigma = 0$. However, the inequality above requires $\langle Y, \mathcal{S}Y\rangle_\sigma \leq -\nu\|Y\|_\sigma^2 < 0$, a contradiction. Therefore, $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$, which implies $\ker(\mathcal{S}) = \mathbb{C}\mathbf{1} = \ker(\mathcal{L})$.

Finally, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$ then the above argument shows that coercivity ($C = 1$ with some positive rate) is impossible. In that case, any exponential decay estimate (when it holds) must necessarily come with some prefactor $C > 1$. □

Appendix B: Proofs for Adiabatic Elimination

In this appendix, we provide a rigorous derivation of the error bounds for the effective dynamics in the Zeno limit, employing the variation of constants formula and standard Grönwall estimates.

1. Auxiliary Lemmas

Lemma B.1 (Iterative Dyson Formula)**.** *Let $\mathcal{L} = \mathcal{A} + \mathcal{B}$ generate a contraction semigroup $\mathcal{P}_t$. Then:*

$$\mathcal{P}_t = e^{t\mathcal{A}} + \int_0^t e^{(t-s)\mathcal{A}}\mathcal{B}\mathcal{P}_s\, ds. \tag{B1}$$

Lemma B.2 (Volterra-Grönwall Bound)**.** *Let $u(t) \geq 0$ satisfy*

$$u(t) \leq A + B\int_0^t e^{-\mu(t-s)}u(s)\, ds$$

with $\mu > B \geq 0$. Then $u(t)$ is uniformly bounded: $u(t) \leq \frac{A\mu}{\mu - B}$.

2. Proof of Theorem III.2

We consider the full dynamics on the microscopic timescale t, generated by $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}$. We define the fast decay rate $\lambda_R > 0$ such that $\|e^{t\mathcal{R}}P_F\| \leq e^{-\lambda_R t}$. Throughout this

appendix, $\|\cdot\|$ denotes the $L^2(\sigma)$ norm (i.e. $\|\cdot\|_\sigma$), and for superoperators we use the induced operator norm on $L^2(\sigma)$. We will also use the σ–GNS norm $\|X\|_{\sigma,\mathrm{GNS}} := \sqrt{\mathrm{tr}(X^\dagger X\,\sigma)}$.

In this appendix we estimate the slow/fast components in the $L^2(\sigma)$ norm. Since $\mathcal{L}$ is the generator of a (finite-dimensional) QMS in the Heisenberg picture, $\mathcal{P}_t := e^{t\mathcal{L}}$ is unital and completely positive. By the Kadison–Schwarz inequality, for all observables X we have

$$\mathcal{P}_t(X)^\dagger \mathcal{P}_t(X) \le \mathcal{P}_t(X^\dagger X). \tag{B2}$$

Using invariance of σ in the Schrödinger picture, i.e. $\mathcal{P}_t^\dagger(\sigma) = \sigma$, this yields contractivity in the σ–GNS norm:

$$\|\mathcal{P}_t(X)\|_{\sigma,\mathrm{GNS}}^2 = \mathrm{tr}\big(\mathcal{P}_t(X)^\dagger \mathcal{P}_t(X)\,\sigma\big) \le \mathrm{tr}\big(\mathcal{P}_t(X^\dagger X)\,\sigma\big) = \mathrm{tr}\big(X^\dagger X\,\mathcal{P}_t^\dagger(\sigma)\big) = \mathrm{tr}(X^\dagger X\,\sigma) = \|X\|_{\sigma,\mathrm{GNS}}^2. \tag{B3}$$

Finally, the KMS and GNS norms are equivalent in finite dimensions, and one may bound them in terms of the extremal eigenvalues of σ as follows:

$$\sqrt{\lambda_{\min}(\sigma)}\,\|X\|_{\sigma,\mathrm{GNS}} \le \|X\|_\sigma \le \sqrt{\lambda_{\max}(\sigma)}\,\|X\|_{\sigma,\mathrm{GNS}}. \tag{B4}$$

Combining these inequalities, we obtain the uniform bound

$$\|\mathcal{P}_t(X)\|_\sigma \le \sqrt{\kappa(\sigma)}\,\|X\|_\sigma, \qquad t \ge 0, \tag{B5}$$

where $\kappa(\sigma) := \lambda_{\max}(\sigma)/\lambda_{\min}(\sigma)$. We denote $C_{\mathrm{bd}} := \sqrt{\kappa(\sigma)}$.

Proof. We analyze the coupled dynamics of the slow component $u(t) = P_S X(t)$ and the fast component $v(t) = P_F X(t)$ in microscopic time t, governed by the generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}$.

Use of the centered coupling condition. Throughout this proof we use the centered coupling condition (11), i.e. $P_S\mathcal{V}P_S = 0$. This eliminates the potentially leading drift term $P_S\mathcal{V}u(t)$ in the slow equation. (If $P_S\mathcal{V}P_S \neq 0$, then $\dot{u}$ contains an additional $\mathcal{O}(1)$ term in microscopic time, which becomes an $\mathcal{O}(\gamma)$ drift on the macroscopic time scale $\tau = t/\gamma$ and invalidates the diffusive Zeno/AE scaling used in Theorem III.2.)

The evolution equations are derived from the projections

$$P_S\mathcal{L} = P_S\mathcal{V}P_F, \tag{B6}$$
$$P_F\mathcal{L} = \gamma\mathcal{R} + P_F\mathcal{V}(P_S + P_F), \tag{B7}$$

where we used $P_S\mathcal{R} = 0$ and $P_S\mathcal{V}P_S = 0$:

$$\dot{u}(t) = P_S\mathcal{V}v(t), \tag{B8a}$$
$$\dot{v}(t) = \gamma\mathcal{R}v(t) + P_F\mathcal{V}(u(t) + v(t)). \tag{B8b}$$

Uniform bound on fast leakage. We first bound the magnitude of the excursion into the fast subspace. Integrating (B8b) with $v(0) = 0$:

$$v(t) = \int_0^t e^{\gamma\mathcal{R}(t-s)} P_F\mathcal{V}(u(s) + v(s))\,ds. \tag{B9}$$

Taking norms and using the spectral gap bound $\|e^{\gamma\mathcal{R}\tau}P_F\| \le e^{-\gamma\lambda_R\tau}$:

$$\|v(t)\| \le \|\mathcal{V}\| \int_0^t e^{-\gamma\lambda_R(t-s)}(\|u(s)\| + \|v(s)\|)\,ds. \tag{B10}$$

Using $\|u(s)\| \le \|X(s)\| \le C_{\mathrm{bd}}\,\|X_0\|$, we apply the Volterra–Grönwall lemma with source term $A = \frac{\|\mathcal{V}\|}{\gamma\lambda_R}\,C_{\mathrm{bd}}\,\|X_0\|$ and kernel parameters $\mu = \gamma\lambda_R$ and $B = \|\mathcal{V}\|$. The bound yields:

$$\|v(t)\| \le \frac{\left(\frac{\|\mathcal{V}\|}{\gamma\lambda_R}\,C_{\mathrm{bd}}\,\|X_0\|\right)\cdot\gamma\lambda_R}{\gamma\lambda_R - \|\mathcal{V}\|} = \frac{\|\mathcal{V}\|}{\gamma\lambda_R - \|\mathcal{V}\|}\,C_{\mathrm{bd}}\,\|X_0\|. \tag{B11}$$

Imposing the stability condition $\gamma\lambda_R \geq 2\|\mathcal{V}\|$, we have $\gamma\lambda_R - \|\mathcal{V}\| \geq \frac{1}{2}\gamma\lambda_R$. Thus, we obtain the strict uniform bound:

$$\|v(t)\| \leq \frac{2\|\mathcal{V}\|}{\gamma\lambda_R}\, C_{\mathrm{bd}}\, \|X_0\|. \tag{B12}$$

It follows immediately from (B8a) that the slow velocity is bounded by:

$$\|\dot{u}(t)\| \leq \|P_S\mathcal{V}\|\|v(t)\| \leq \frac{2\|\mathcal{V}\|^2}{\gamma\lambda_R}\, C_{\mathrm{bd}}\, \|X_0\|. \tag{B13}$$

Expansion of the effective dynamics. Substituting the integral expression for $v(t)$ back into (B8a):

$$\dot{u}(t) = P_S\mathcal{V}\int_0^t e^{\gamma\mathcal{R}(t-s)}P_F\mathcal{V}u(s)\,ds + \mathcal{E}_{noise}(t), \tag{B14}$$

where the **feedback noise** is $\mathcal{E}_{noise}(t) = P_S\mathcal{V}\int_0^t e^{\gamma\mathcal{R}(t-s)}P_F\mathcal{V}v(s)\,ds$. Using the bound (B12):

$$\|\mathcal{E}_{noise}(t)\| \leq \|\mathcal{V}\|^2\int_0^t e^{-\gamma\lambda_R(t-s)}\|v(s)\|ds \leq \frac{\|\mathcal{V}\|^2}{\gamma\lambda_R}\sup\|v\| \leq \frac{2\|\mathcal{V}\|^3}{(\gamma\lambda_R)^2}\, C_{\mathrm{bd}}\, \|X_0\|. \tag{B15}$$

To extract the generator, we integrate by parts on the fast subspace. In the bounded-operator setting, $\mathcal{R}|_{\mathcal{H}_F}$ is invertible and $\mathcal{R}^+|_{\mathcal{H}_F} = (\mathcal{R}|_{\mathcal{H}_F})^{-1}$. Moreover, for $\tau \geq 0$ we have

$$\frac{d}{d\tau}\, e^{\gamma\mathcal{R}\tau} = \gamma\,\mathcal{R}\, e^{\gamma\mathcal{R}\tau}, \qquad \mathcal{R}^+\,\mathcal{R} = P_F, \tag{B16}$$

and therefore

$$e^{\gamma\mathcal{R}\tau}P_F = \mathcal{R}^+\mathcal{R}\, e^{\gamma\mathcal{R}\tau} = \frac{1}{\gamma}\,\mathcal{R}^+\,\frac{d}{d\tau}\, e^{\gamma\mathcal{R}\tau}. \tag{B17}$$

Using (B17) with $\tau = t - s$ yields the integration-by-parts formula

$$\begin{aligned}\int_0^t e^{\gamma\mathcal{R}(t-s)}P_F\mathcal{V}u(s)\,ds &= \left[-\frac{1}{\gamma}e^{\gamma\mathcal{R}(t-s)}\mathcal{R}^+\mathcal{V}u(s)\right]_0^t + \frac{1}{\gamma}\int_0^t e^{\gamma\mathcal{R}(t-s)}\mathcal{R}^+\mathcal{V}\dot{u}(s)\,ds \\ &= -\frac{1}{\gamma}\mathcal{R}^+\mathcal{V}u(t) + \frac{1}{\gamma}e^{\gamma\mathcal{R}t}\mathcal{R}^+\mathcal{V}u(0) + \mathcal{I}_{rem}(t).\end{aligned} \tag{B18}$$

Multiplying by $P_S\mathcal{V}$ and identifying $\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$, the equation of motion becomes:

$$\dot{u}(t) = \frac{1}{\gamma}\mathcal{L}_O u(t) + \mathcal{R}_{init}(t) + \mathcal{R}_{int}(t) + \mathcal{E}_{noise}(t). \tag{B19}$$

The remainder terms are bounded using $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$:

1. **Initial Layer ($\mathcal{R}_{init}$):**

$$\|\mathcal{R}_{init}(t)\| = \|\frac{1}{\gamma}P_S\mathcal{V}e^{\gamma\mathcal{R}t}\mathcal{R}^+\mathcal{V}u(0)\| \leq \frac{\|\mathcal{V}\|^2}{\gamma\lambda_R}e^{-\gamma\lambda_R t}\, C_{\mathrm{bd}}\, \|X_0\|. \tag{B20}$$

2. **Integral Remainder ($\mathcal{R}_{int}$):** Using the bound on $\|\dot{u}\|$ from (B13):

$$\begin{aligned}\|\mathcal{R}_{int}(t)\| &= \|\frac{1}{\gamma}P_S\mathcal{V}\int_0^t e^{\gamma\mathcal{R}(t-s)}\mathcal{R}^+\mathcal{V}\dot{u}(s)ds\| \\ &\leq \frac{\|\mathcal{V}\|^2}{\gamma\lambda_R}\left(\frac{1}{\gamma\lambda_R}\right)\left(\frac{2\|\mathcal{V}\|^2}{\gamma\lambda_R}\, C_{\mathrm{bd}}\, \|X_0\|\right) = \frac{2\|\mathcal{V}\|^4}{(\gamma\lambda_R)^3}\, C_{\mathrm{bd}}\, \|X_0\|.\end{aligned} \tag{B21}$$

Macroscopic error accumulation. We compare $u(t)$ to the reference trajectory $u_{ref}(t) := e^{t\mathcal{L}_O/\gamma}X_0$. Let $\Delta(t) := u(t) - u_{ref}(t)$. Since $\dot{u} = \gamma^{-1}\mathcal{L}_O u + (\mathcal{R}_{init} + \mathcal{R}_{int} + \mathcal{E}_{noise})$ and $\dot{u}_{ref} = \gamma^{-1}\mathcal{L}_O u_{ref}$ with $\Delta(0) = 0$, the variation-of-constants formula yields

$$\Delta(t) = \int_0^t e^{(t-s)\mathcal{L}_O/\gamma}\big(\mathcal{R}_{init}(s) + \mathcal{R}_{int}(s) + \mathcal{E}_{noise}(s)\big)\, ds. \tag{B22}$$

In particular, for $t = \gamma\tau$ we have

$$\|\Delta(\gamma\tau)\| \le \sup_{0\le r\le\tau} \|e^{r\mathcal{L}_O}\| \int_0^{\gamma\tau} (\|\mathcal{R}_{init}(s)\| + \|\mathcal{R}_{int}(s)\| + \|\mathcal{E}_{noise}(s)\|)\, ds. \tag{B23}$$

By the additional assumption in Theorem III.2 that $\mathcal{L}_O$ generates a QMS leaving σ invariant, the same Kadison–Schwarz argument as above yields

$$\|e^{r\mathcal{L}_O}Y\|_\sigma \le C_{\rm bd}\, \|Y\|_\sigma, \qquad \forall r \ge 0. \tag{B24}$$

Therefore $\sup_{0\le r\le\tau} \|e^{r\mathcal{L}_O}\| \le C_{\rm bd}$.

1. Initial Layer Contribution:

$$\int_0^{\gamma\tau} \|\mathcal{R}_{init}(s)\| ds \le \frac{\|\mathcal{V}\|^2}{\gamma\lambda_R} \int_0^\infty e^{-\gamma\lambda_R s} ds\, C_{\rm bd}\, \|X_0\| = \frac{\|\mathcal{V}\|^2}{(\gamma\lambda_R)^2}\, C_{\rm bd}\, \|X_0\|. \tag{B25}$$

This term yields the C_{init} component, scaling as $O(\gamma^{-2})$.

2. Drift Contribution: The terms $\|\mathcal{E}_{noise}\|$ and $\|\mathcal{R}_{int}\|$ are uniformly bounded. Their integral scales linearly with duration $\gamma\tau$:

$$\begin{aligned}\int_0^{\gamma\tau} (\|\mathcal{E}_{noise}\| + \|\mathcal{R}_{int}\|) ds &\le \gamma\tau \left[\frac{2\|\mathcal{V}\|^3}{(\gamma\lambda_R)^2} + \frac{2\|\mathcal{V}\|^4}{(\gamma\lambda_R)^3}\right] C_{\rm bd}\, \|X_0\| \\ &= \frac{\tau}{\gamma}\frac{2\|\mathcal{V}\|^3}{\lambda_R^2}\left[1 + \frac{\|\mathcal{V}\|}{\gamma\lambda_R}\right] C_{\rm bd}\, \|X_0\|. \end{aligned} \tag{B26}$$

Using the condition $\frac{\|\mathcal{V}\|}{\gamma\lambda_R} \le \frac{1}{2}$, the bracket term is bounded by $\frac{3}{2}$. Thus:

$$\text{Drift Error} \le \frac{\tau}{\gamma}\frac{3\|\mathcal{V}\|^3}{\lambda_R^2}\, C_{\rm bd}\, \|X_0\|. \tag{B27}$$

Combining terms yields the result:

$$\|\Delta(\gamma\tau)\| \le C_{\rm bd}^2 \left(\frac{1}{\gamma^2}C_{init} + \frac{\tau}{\gamma}C_{drift}\right) \|X_0\|, \tag{B28}$$

with $C_{init} = \frac{\|\mathcal{V}\|^2}{\lambda_R^2}$ and $C_{drift} = \frac{3\|\mathcal{V}\|^3}{\lambda_R^2}$. □

Appendix C: Proof of Lemma IV.5

In this appendix, we establish the structural bounds for the macroscopic solver $\mathcal{A}_\mu$.

Proof. Recall the definition of the reference effective dynamics $Q_O = -P_S\mathcal{V}_{\rm skew}\mathcal{R}^+\mathcal{V}_{\rm skew}P_S$. By Assumption IV.1.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$, hence

$$\langle g, -\mathcal{R}^+ g\rangle \le \frac{1}{\lambda_R}\|g\|^2, \quad \forall g \in \mathcal{H}_F. \tag{C1}$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (22), which applies to fluxes of the form $g = g_\psi = P_F\mathcal{V}_{\rm skew}P_S\psi$.

1. Proof of Properties 1 and 2 (Energy and Duality)

Solver boundedness and forward regularity. Let $f \in \mathcal{H}$ and define $u := \mathcal{A}_\mu f \in \mathcal{H}_S$. The defining equation is $(\mu - Q_O)u = P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$:

$$\mu\|u\|^2 + \langle u, -Q_O u\rangle = \langle u, P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f\rangle = -\langle \mathcal{V}_{\text{skew}} u, (-\mathcal{R}^+)P_F f\rangle. \tag{C2}$$

Let $v := \mathcal{V}_{\text{skew}} u$. Since $P_S \mathcal{V}_{\text{skew}} P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using the definition of Q_O and the pseudoinverse bound (C1):

$$\langle u, -Q_O u\rangle = \langle v, -\mathcal{R}^+ v\rangle = \|(-\mathcal{R}^+)^{1/2} v\|^2. \tag{C3}$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$\mu\|u\|^2 + \|(-\mathcal{R}^+)^{1/2} v\|^2 \le \|(-\mathcal{R}^+)^{1/2} v\|\, \|(-\mathcal{R}^+)^{1/2} P_F f\|. \tag{C4}$$

Writing $a := \|(-\mathcal{R}^+)^{1/2} v\|$ and $b := \|(-\mathcal{R}^+)^{1/2} P_F f\|$, we have $\mu\|u\|^2 + a^2 \le ab$. Completing the square,

$$a^2 - ab = \left(a - \tfrac{1}{2}b\right)^2 - \tfrac{1}{4}b^2 \ge -\tfrac{1}{4}b^2, \tag{C5}$$

so $\mu\|u\|^2 \le \frac{1}{4}b^2$, hence

$$\|u\| \le \frac{1}{2}\,\mu^{-1/2}\,\|(-\mathcal{R}^+)^{1/2} P_F f\| \le \frac{1}{2\sqrt{\lambda_R}}\,\mu^{-1/2}\,\|f\|. \tag{C6}$$

Moreover, $a \le b$. In the bounded-operator setting one has $\text{dom}(Q_O) = \mathcal{H}_S$, so $u \in \text{dom}(Q_O)$. Then the vector $v = \mathcal{V}_{\text{skew}} u$ is a slow-induced flux of the form g_ψ with $\psi \in \text{dom}(Q_O)$; we may therefore apply the flux conditioning bound (22) to v to obtain

$$\|v\|^2 \le M_R\,\langle v, -\mathcal{R}^+ v\rangle = M_R a^2 \le M_R b^2 \le \frac{M_R}{\lambda_R}\,\|f\|^2, \tag{C7}$$

so $\|v\| \le \sqrt{M_R/\lambda_R}\,\|f\|$. This proves $\|\mathcal{A}_\mu\| \le k_{\mathcal{A}}\mu^{-1/2}$ and $\|\mathcal{V}_{\text{skew}}\mathcal{A}_\mu\| \le k_{\text{reg}}$, with

$$k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}, \qquad k_{\text{reg}} := \sqrt{M_R/\lambda_R}. \tag{C8}$$

Dual regularity. We consider the operator $\mathcal{T} := \mathcal{A}_\mu \mathcal{V}_{\text{skew}} P_F$. Its norm equals that of its adjoint $\mathcal{T}^\dagger$. We compute the adjoint of the solver:

$$\mathcal{A}_\mu^\dagger = \left((\mu - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F\right)^\dagger = -P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}} P_S(\mu - Q_O)^{-1}. \tag{C9}$$

(Using $Q_O^\dagger = Q_O$ and $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$). Therefore:

$$\mathcal{T}^\dagger = P_F \mathcal{V}_{\text{skew}}^\dagger \mathcal{A}_\mu^\dagger = P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}} P_S(\mu - Q_O)^{-1}. \tag{C10}$$

Let $g \in \mathcal{H}_S$ and $\psi := (\mu - Q_O)^{-1} g$. Define the macroscopic flux

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{C11}$$

Then $\mathcal{T}^\dagger g = P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) g_\psi$. Invoking the **Flux Regularity** condition from Assumption IV.1.3:

$$\|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) g_\psi\| \le \kappa_{\text{ell}}\|(-Q_O)\psi\|. \tag{C12}$$

Substituting ψ back:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\text{ell}}\|(-Q_O)(\mu - Q_O)^{-1} g\|. \tag{C13}$$

Since Q_O is self-adjoint and non-positive, the operator $(-Q_O)(\mu - Q_O)^{-1}$ is a spectral contraction ($\|\cdot\| \le 1$). Thus:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\text{ell}}\|g\|. \tag{C14}$$

This proves $\|\mathcal{A}_\mu \mathcal{V}_{\text{skew}} P_F\| \le \kappa_{\text{ell}}$.

2. Proof of Property 3 (Perturbation Control)

We use the structural decomposition $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ from Assumption IV.1.4. We also recall the centered mapping property (40), which will be used later to sharpen the perturbation bookkeeping in the dissipation matrix. Recall the bound for the skew part (Assumption IV.1.4(a)):

$$\|\mathcal{K}_{\text{pert}} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{C15}$$

For the symmetric part, we use the energy bound (Assumption IV.1.4(b)):

$$\|\mathcal{S}_{\text{pert}} f\| \le a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{C16}$$

Reverse estimate ($\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu$). Let $f \in \mathcal{H}$ and $u := \mathcal{A}_\mu f$. The adjoint is $\mathcal{V}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$. Applying the triangle inequality and the bounds above:

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le \|\mathcal{K}_{\text{pert}} u\| + \|\mathcal{S}_{\text{pert}} u\| \tag{C17}$$

$$\le b_1 \|u\| + b_2 \|u\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} u\|. \tag{C18}$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_\mu$), one has $(-\mathcal{R})^{1/2} u = 0$, hence

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le (b_1 + b_2) \|u\|. \tag{C19}$$

Substituting the Solver Bound ($\|u\| \le k_{\mathcal{A}} \mu^{-1/2} \|f\|$):

$$\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f\| \le (b_1 + b_2) k_{\mathcal{A}} \mu^{-1/2} \|f\|. \tag{C20}$$

This establishes (37a) with $k_P(\mu) := (b_1 + b_2) k_{\mathcal{A}} \mu^{-1/2}$.

Forward estimate ($\mathcal{A}_\mu \mathcal{V}_{\text{pert}}$). We aim to bound the operator norm of $\mathcal{W}_{\text{fwd}} := \mathcal{A}_\mu \mathcal{V}_{\text{pert}}$. By adjoints on Hilbert spaces, $\|\mathcal{W}_{\text{fwd}}\| = \|\mathcal{W}_{\text{fwd}}^\dagger\| = \|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu^\dagger\|$. Let $g \in \mathcal{H}_S$ be arbitrary with $\|g\| = 1$ and set $w := \mathcal{A}_\mu^\dagger g$. Using the bounds on $\mathcal{K}_{\text{pert}}$ and the energy bound on $\mathcal{S}_{\text{pert}}$,

$$\begin{aligned} \|\mathcal{V}_{\text{pert}}^\dagger w\| &\le \|\mathcal{K}_{\text{pert}} w\| + \|\mathcal{S}_{\text{pert}} w\| \\ &\le (b_1 + b_2) \|w\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} w\|. \end{aligned} \tag{C21}$$

We compute the norms on the right-hand side.

1. **Solver term ($\|w\|$):**

$$\|w\| = \|\mathcal{A}_\mu^\dagger g\| \le \|\mathcal{A}_\mu\| \, \|g\| \le k_{\mathcal{A}} \, \mu^{-1/2}. \tag{C22}$$

2. **Energy term ($\|(-\mathcal{R})^{1/2} w\|$):** Using $w = \mathcal{A}_\mu^\dagger g$ and the functional calculus for the non-positive self-adjoint operator Q_O,

$$\|(-\mathcal{R})^{1/2} w\| = \|(-\mathcal{R})^{1/2} \mathcal{A}_\mu^\dagger g\| = \|(-\mathcal{R})^{-1/2} P_F \mathcal{V}_{\text{skew}} P_S (\mu - Q_O)^{-1} g\|. \tag{C23}$$

Setting $\psi := (\mu - Q_O)^{-1} g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$\|(-\mathcal{R})^{-1/2} P_F \mathcal{V}_{\text{skew}} P_S \psi\|^2 = \left\langle P_F \mathcal{V}_{\text{skew}} P_S \psi, \, -\mathcal{R}^+ \, P_F \mathcal{V}_{\text{skew}} P_S \psi \right\rangle = -\langle \psi, Q_O \psi \rangle = \|(-Q_O)^{1/2} \psi\|^2, \tag{C24}$$

hence

$$\|(-\mathcal{R})^{1/2} w\| = \|(-Q_O)^{1/2} (\mu - Q_O)^{-1} g\| \le \frac{1}{2} \, \mu^{-1/2} \, \|g\|. \tag{C25}$$

(The last step follows from the scalar bound $\sup_{x \ge 0} \frac{\sqrt{x}}{\mu + x} = \frac{1}{2\sqrt{\mu}}$.)

Substituting these estimates into (C21) yields

$$\|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_{\mu}^{\dagger}g\| \leq \left[\left((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\right)\mu^{-1/2}\right]\|g\|. \tag{C26}$$

Since g is arbitrary, this establishes (37b) with $\tilde{k}_P(\mu) := \left((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\right)\mu^{-1/2}$.

□

Appendix D: Proof of Theorem V.1

In this appendix, we explicitly derive the dissipation matrix governing the hypocoercive convergence. We analyze the modified Lyapunov functional defined with a resolvent shift $\mu > 0$ and a mixing weight $\alpha > 0$:

$$\mathcal{E}_{\alpha,\mu}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle\mathcal{A}_{\mu}f, f\rangle, \tag{D1}$$

where $\mathcal{A}_{\mu} = (\mu - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^{+})P_F$ is the macroscopic solver associated with the reference effective dynamics Q_O.

1. Preliminaries

We strictly utilize the structural estimates established in Lemma IV.5. We recall the constants derived therein, depending solely on the parameters from Assumption IV.1 $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$. For convenience in the Lyapunov dissipation bounds below, we introduce the derived symmetric-perturbation coefficient

$$b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1+\lambda_R). \tag{D2}$$

- **Solver Bound:** $\|\mathcal{A}_{\mu}\| \leq k_{\mathcal{A}}\mu^{-1/2}$, with $k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}$.
- **Regularity:** $\|\mathcal{V}_{\text{skew}}\mathcal{A}_{\mu}\| \leq k_{\text{reg}}$ with $k_{\text{reg}} := \sqrt{M_R/\lambda_R}$, and $\|\mathcal{A}_{\mu}\mathcal{V}_{\text{skew}}P_F\| \leq \kappa_{\text{ell}}$.
- **Perturbation Control:**

$$\|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_{\mu}\| \leq k_P(\mu) = (b_1+b_2)k_{\mathcal{A}}\,\mu^{-1/2}, \tag{D3a}$$

$$\|\mathcal{A}_{\mu}\mathcal{V}_{\text{pert}}\| \leq \tilde{k}_P(\mu) = \left((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\right)\mu^{-1/2}. \tag{D3b}$$

We denote the spectral gap of the self-adjoint operator $-Q_O$ by $\lambda(Q_O)$.

2. Derivation of Differential Estimates

Standing regularity for the Lyapunov computation. In the bounded-operator setting, for any initial datum $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}}f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L}f_t$. In unbounded settings, the computations below are performed on a common invariant core $\mathcal{C} \subset \text{dom}(\mathcal{L}) \cap \text{dom}(\mathcal{A}_{\mu})$, and then extended to the relevant energy domain by density.

Lemma D.1 (Energy identity for the fast dissipator)**.** *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \text{dom}(\mathcal{R})$,*

$$\Re\langle\mathcal{R}f, f\rangle = \langle\mathcal{R}f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2. \tag{D4}$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$\Re\langle\mathcal{R}f, f\rangle = \Re\langle\mathcal{R}f_F, f_F\rangle. \tag{D5}$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \geq 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle \mathcal{R}f, f\rangle = -\langle(-\mathcal{R})f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. Moreover $\mathcal{R}f_S = 0$ and $\mathcal{R}f = \mathcal{R}f_F$. □

Let f_t satisfy $\partial_t f_t = \mathcal{L}f_t$. By differentiating (42) along the flow and using that $\mathcal{A}_\mu$ is time-independent, we obtain

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\mu}(f_t) = \Re\langle \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\mu \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f_t, \partial_t f_t\rangle \tag{D6}$$

$$= \Re\langle \mathcal{L}f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{L}f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f_t, \mathcal{L}f_t\rangle. \tag{D7}$$

We now split the right-hand side into the "plain" dissipation coming from the generator itself and the hypocoercive correction generated by the cross term:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\mu}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\mathrm{mod}}(f_t), \tag{D8}$$

where we define

$$\mathcal{D}_I(f) := \Re\langle \mathcal{L}f, f\rangle, \tag{D9}$$

$$\mathcal{D}_{\mathrm{mod}}(f) := \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{L}f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{L}f\rangle. \tag{D10}$$

Expanding $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$\mathcal{D}_I = \gamma\Re\langle \mathcal{R}f, f\rangle + \Re\langle \mathcal{V}_{\mathrm{skew}}f, f\rangle + \eta\Re\langle \mathcal{V}_{\mathrm{pert}}f, f\rangle. \tag{D11}$$

Using Assumption IV.1.1 ($\mathcal{R}$ coercivity) and Assumption IV.1.4 (Perturbation decomposition $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$):

- $\gamma\Re\langle \mathcal{R}f, f\rangle \leq -\gamma\lambda_R\|f_F\|^2$.
- $\Re\langle \mathcal{V}_{\mathrm{skew}}f, f\rangle = 0$ and $\Re\langle \mathcal{K}_{\mathrm{pert}}f, f\rangle = 0$ (Skew-adjointness).
- For the symmetric part, using (28) and Cauchy–Schwarz,

$$\eta\Re\langle \mathcal{S}_{\mathrm{pert}}f, f\rangle \leq \eta\|\mathcal{S}_{\mathrm{pert}}f\|\,\|f\| \leq \eta\Big(b_2\|f\|^2 + a_2\,\|(-\mathcal{R})^{1/2}f\|\,\|f\|\Big) \tag{D12}$$

$$\leq \eta\Big(\big(b_2 + \tfrac{a_2}{2}\big)\|f\|^2 + \tfrac{a_2}{2}\,\|(-\mathcal{R})^{1/2}f\|^2\Big). \tag{D13}$$

  Combining with the dissipator contribution, note that by Lemma D.1, $\Re\langle \mathcal{R}f, f\rangle = \Re\langle \mathcal{R}f_F, f_F\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. Hence

$$\gamma\Re\langle \mathcal{R}f, f\rangle + \eta\Re\langle \mathcal{S}_{\mathrm{pert}}f, f\rangle \leq -\big(\gamma - \tfrac{\eta a_2}{2}\big)\,\|(-\mathcal{R})^{1/2}f\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f\|^2 \tag{D14}$$

$$\leq -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta b_{2,\mathrm{eff}}\,\|f_S\|^2, \tag{D15}$$

  where in the last step we used $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and $b_{2,\mathrm{eff}} := b_2 + \frac{a_2}{2}(1+\lambda_R) \geq b_2 + \frac{a_2}{2}$.

Thus:

$$\mathcal{D}_I \leq -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta b_{2,\mathrm{eff}}\,\|f_S\|^2. \tag{D16}$$

Hypocoercive correction ($\mathcal{D}_{\mathrm{mod}}$). Recall

$$\mathcal{D}_{\mathrm{mod}}(f) = \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{L}f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{L}f\rangle. \tag{D17}$$

Using $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\rm skew} + \eta\mathcal{V}_{\rm pert}$ and linearity, we regroup $\mathcal{D}_{\rm mod}$ according to which part of the generator appears:

$$\mathcal{D}_{\rm mod}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f), \tag{D18}$$

where

$$T_{\mathcal{R}}(f) := \alpha\gamma\,\Re\langle \mathcal{A}_\mu \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\mu f, \mathcal{R} f\rangle, \tag{D19}$$

$$T_{\mathcal{V}}(f) := \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\rm skew} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\rm skew} f\rangle, \tag{D20}$$

$$T_{\mathcal{P}}(f) := \alpha\eta\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\rm pert} f, f\rangle + \alpha\eta\,\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\rm pert} f\rangle. \tag{D21}$$

a. Interaction with the fast dissipator. We start from the regrouped definition

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\mu \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\mu f, \mathcal{R} f\rangle. \tag{D22}$$

Since $\mathcal{R} f = \mathcal{R} f_F$ and $\mathcal{A}_\mu f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$\Re\langle \mathcal{A}_\mu f, \mathcal{R} f\rangle = \Re\langle \mathcal{R}\mathcal{A}_\mu f, f\rangle = 0. \tag{D23}$$

Moreover $\mathcal{A}_\mu \mathcal{R} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\mu \mathcal{R} f_F, f_S\rangle. \tag{D24}$$

Using $\mathcal{A}_\mu = (\mu - Q_O)^{-1} P_S \mathcal{V}_{\rm skew}(-\mathcal{R}^+) P_F$ and the pseudoinverse identity $\mathcal{R}^+\mathcal{R} = P_F$ on $\mathrm{dom}(\mathcal{R})$, we obtain

$$\mathcal{A}_\mu \mathcal{R} f_F = (\mu - Q_O)^{-1} P_S \mathcal{V}_{\rm skew}(-\mathcal{R}^+)\mathcal{R} f_F = -(\mu - Q_O)^{-1} P_S \mathcal{V}_{\rm skew} f_F. \tag{D25}$$

Therefore, by skew-adjointness of $\mathcal{V}_{\rm skew}$,

$$|T_{\mathcal{R}}| = \alpha\gamma\left|\Re\langle (\mu - Q_O)^{-1} P_S \mathcal{V}_{\rm skew} f_F, f_S\rangle\right| = \alpha\gamma\left|\Re\langle f_F, \mathcal{V}_{\rm skew}(\mu - Q_O)^{-1} f_S\rangle\right| \tag{D26}$$

$$\leq \alpha\gamma\,\|f_F\|\,\|\mathcal{V}_{\rm skew}(\mu - Q_O)^{-1} f_S\|. \tag{D27}$$

To bound $\|\mathcal{V}_{\rm skew}(\mu - Q_O)^{-1} f_S\|$, set $\psi := (\mu - Q_O)^{-1} f_S \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$g_\psi := P_F \mathcal{V}_{\rm skew} P_S \psi \in \mathcal{H}_F. \tag{D28}$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S \mathcal{V}_{\rm skew}\psi = 0$, hence $g_\psi = \mathcal{V}_{\rm skew}\psi = \mathcal{V}_{\rm skew}(\mu - Q_O)^{-1} f_S$. Applying the flux conditioning bound (22) yields

$$\|\mathcal{V}_{\rm skew}(\mu - Q_O)^{-1} f_S\|^2 = \|g_\psi\|^2 \leq M_R\,\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle. \tag{D29}$$

The "extra Q_O" comes from rewriting the right-hand side using the definition $Q_O := -P_S \mathcal{V}_{\rm skew}\mathcal{R}^+\mathcal{V}_{\rm skew} P_S$: indeed, for $\psi \in \mathcal{H}_S$,

$$\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = \langle \mathcal{V}_{\rm skew}\psi, (-\mathcal{R}^+)\mathcal{V}_{\rm skew}\psi\rangle = \langle \psi, (-Q_O)\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2. \tag{D30}$$

Therefore

$$\|\mathcal{V}_{\rm skew}(\mu - Q_O)^{-1} f_S\| \leq \sqrt{M_R}\,\|(-Q_O)^{1/2}(\mu - Q_O)^{-1} f_S\|. \tag{D31}$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \geq 0$,

$$\|(-Q_O)^{1/2}(\mu - Q_O)^{-1}\| = \sup_{x\geq 0} \frac{\sqrt{x}}{\mu + x} = \frac{1}{2\sqrt{\mu}}, \tag{D32}$$

which implies

$$\|\mathcal{V}_{\rm skew}(\mu - Q_O)^{-1} f_S\| \leq \frac{\sqrt{M_R}}{2}\,\mu^{-1/2}\,\|f_S\|. \tag{D33}$$

Hence

$$|T_{\mathcal{R}}| \leq \alpha\gamma\,\frac{\sqrt{M_R}}{2}\,\mu^{-1/2}\,\|f_F\|\,\|f_S\|. \tag{D34}$$

b. Interaction with transport. Starting from the regrouped definition

$$T_{\mathcal{V}}(f) = \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\text{skew}} f\rangle, \tag{D35}$$

we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\mu$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\mu = \mathcal{A}_\mu P_F$ (since P_F is on the right in (32)). Hence $\mathcal{A}_\mu f = \mathcal{A}_\mu f_F$. Expanding in the *first* bracket gives

$$\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f, f\rangle = \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_F\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_F\rangle. \tag{D36}$$

The last term vanishes because $\mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S \in \mathcal{H}_S$, we have $\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_F\rangle = 0$. Therefore

$$\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f, f\rangle = \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle. \tag{D37}$$

In the *second* bracket we use $\mathcal{A}_\mu f = \mathcal{A}_\mu f_F$ and expand only $\mathcal{V}_{\text{skew}} f$:

$$\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\text{skew}} f\rangle = \Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_S\rangle + \Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_F\rangle. \tag{D38}$$

Putting the pieces together, we obtain the explicit decomposition

$$T_{\mathcal{V}} = \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle \tag{D39}$$

$$+ \alpha\,\Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_F\rangle. \tag{D40}$$

- **Main Coercivity:** Since $f_S \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\text{skew}} P_S = 0$, we have $P_F \mathcal{V}_{\text{skew}} f_S = \mathcal{V}_{\text{skew}} f_S$. By the definition of $\mathcal{A}_\mu$ and Q_O this yields

$$\alpha\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle = \alpha\langle (\mu - Q_O)^{-1} Q_O f_S, f_S\rangle = -\alpha\,\langle (\mu - Q_O)^{-1}(-Q_O) f_S, f_S\rangle \tag{D41}$$

$$\leq -\alpha\,\frac{\lambda(Q_O)}{\mu + \lambda(Q_O)}\,\|f_S\|^2. \tag{D42}$$

- **Cross Terms:** Using Regularity estimates:

$$\alpha\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle + \alpha\Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_S\rangle \tag{D43}$$

$$\leq \alpha(\|\mathcal{A}_\mu \mathcal{V}_{\text{skew}} P_F\| + \|\mathcal{V}_{\text{skew}} \mathcal{A}_\mu\|)\|f_F\|\|f_S\| \tag{D44}$$

$$\leq \alpha(\kappa_{\text{ell}} + k_{\text{reg}})\|f_F\|\|f_S\|. \tag{D45}$$

- **Fast–fast remainder.** Using $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$ and Cauchy–Schwarz,

$$\big|\alpha\Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_F\rangle\big| = \alpha\big|\Re\langle -\mathcal{V}_{\text{skew}} \mathcal{A}_\mu f_F, f_F\rangle\big| \leq \alpha\,\|\mathcal{V}_{\text{skew}} \mathcal{A}_\mu\|\,\|f_F\|^2 \leq \alpha\,k_{\text{reg}}\,\|f_F\|^2. \tag{D46}$$

c. Interaction with perturbation.

$$T_{\mathcal{P}} = \alpha\eta\Re\langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f\rangle + \alpha\eta\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{pert}} f, f\rangle. \tag{D47}$$

We estimate each term using Lemma IV.5 and the centered mapping property (40).

- **Reverse term.** Since $\mathcal{A}_\mu f \in \mathcal{H}_S$ and $\mathcal{V}_{\text{pert}}^\dagger \mathcal{H}_S \subset \mathcal{H}_F$, we have $\langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f\rangle = \langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f_F\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f\rangle\big| \leq \alpha\eta\,\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f\|\,\|f_F\| \leq \alpha\eta\,k_P(\mu)\,\|f\|\,\|f_F\| \tag{D48}$$

$$\leq \alpha\eta\,k_P(\mu)\,\big(\|f_F\|^2 + \|f_F\|\,\|f_S\|\big). \tag{D49}$$

- **Forward term.** Since $\mathcal{A}_\mu \mathcal{H} \subset \mathcal{H}_S$, we have $\langle \mathcal{A}_\mu \mathcal{V}_{\text{pert}} f, f\rangle = \langle \mathcal{A}_\mu \mathcal{V}_{\text{pert}} f, f_S\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{pert}} f, f\rangle\big| \leq \alpha\eta\,\|\mathcal{A}_\mu \mathcal{V}_{\text{pert}} f\|\,\|f_S\| \leq \alpha\eta\,\tilde{k}_P(\mu)\,\|f\|\,\|f_S\| \tag{D50}$$

$$\leq \alpha\eta\,\tilde{k}_P(\mu)\,\big(\|f_S\|^2 + \|f_F\|\,\|f_S\|\big). \tag{D51}$$

3. Dissipation Matrix Construction

Summing all estimates and using Cauchy–Schwarz bounds of the form $|\langle Xf_F, f_S\rangle| \leq \|X\|\,\|f_F\|\,\|f_S\|$, we obtain a closed differential inequality of the form

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\mu}(f_t) \leq -\mathbf{v}(t)^T\mathbf{M}\,\mathbf{v}(t), \qquad \mathbf{v}(t) := \begin{pmatrix} \|P_F f_t\| \\ \|P_S f_t\| \end{pmatrix}. \tag{D52}$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α,μ). The entries of $\mathbf{M}$ are explicitly decomposed by powers of μ below.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$\begin{aligned} M_{11} &= (\gamma\lambda_R - \eta b_{2,\text{eff}}) - \alpha k_{\text{reg}} - \alpha\eta k_P(\mu) \\ &= \gamma\lambda_R - \eta b_{2,\text{eff}} - \alpha k_{\text{reg}} - \alpha\eta(b_1+b_2)k_{\mathcal{A}}\,\mu^{-1/2}. \end{aligned} \tag{D53}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$\begin{aligned} M_{22} &= \alpha\frac{\lambda(Q_O)}{\mu+\lambda(Q_O)} - \eta b_{2,\text{eff}} - \alpha\eta\,\tilde{k}_P(\mu) \\ &= \alpha\frac{\lambda(Q_O)}{\mu+\lambda(Q_O)} - \eta b_{2,\text{eff}} - \alpha\eta\left[\left((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\right)\mu^{-1/2}\right]. \end{aligned} \tag{D54}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$\begin{aligned} M_{12} &= -\frac{1}{2}\alpha\left[\gamma\frac{\sqrt{M_R}}{2}\,\mu^{-1/2} + (k_{\text{reg}}+\kappa_{\text{ell}}) + \eta\big(k_P(\mu)+\tilde{k}_P(\mu)\big)\right] \\ &= -\frac{1}{2}\alpha\left[k_{\text{reg}}+\kappa_{\text{ell}} + \left(\gamma\frac{\sqrt{M_R}}{2} + \eta\Big(2(b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\right)\mu^{-1/2}\right]. \end{aligned} \tag{D55}$$

This explicit form allows for direct asymptotic analysis by choosing $\mu \sim \lambda(Q_O)$, confirming the stability for small η.

4. Optimization of the Decay Rate

We now derive the explicit convergence rate by simultaneously optimizing the resolvent shift μ and the mixing weight α. We consider the unperturbed regime ($\eta = 0$) to establish the dominant hypocoercive rate.

Let $\mathbf{M}_0(\alpha,\mu)$ denote the dissipation matrix for $\eta = 0$. Since $\mathbf{M}_0$ is a 2×2 symmetric positive definite matrix (verified below), its smallest eigenvalue $\lambda_{\min}$ satisfies the rigorous algebraic bound:

$$\lambda_{\min} \geq \frac{\det(\mathbf{M}_0)}{\operatorname{Tr}(\mathbf{M}_0)} = \frac{M_{11}^0 M_{22}^0 - (M_{12}^0)^2}{M_{11}^0 + M_{22}^0}. \tag{D56}$$

The entries of $\mathbf{M}_0$ are:

$$M_{11}^0 = \gamma\lambda_R - \alpha k_{\text{reg}}, \tag{D57}$$

$$M_{22}^0 = \alpha\mathcal{Y}(\mu), \quad \text{where } \mathcal{Y}(\mu) := \frac{\lambda(Q_O)}{\mu+\lambda(Q_O)}, \tag{D58}$$

$$M_{12}^0 = -\frac{1}{2}\alpha\mathcal{Z}(\mu,\gamma), \quad \text{where } \mathcal{Z}(\mu,\gamma) := \gamma\frac{\sqrt{M_R}}{2}\,\mu^{-1/2} + (k_{\text{reg}}+\kappa_{\text{ell}}). \tag{D59}$$

Choice of the resolvent shift μ. To maximize the structural efficiency of the rate, we select μ to align with the macroscopic spectral gap. A simple way to see the relevant μ-dependence is to look at the large-friction leading balance in the unperturbed matrix: the cross term contributes $\mathcal{Z}(\mu,\gamma) \sim \gamma\Theta\,\mu^{-1/2}$, while the slow coercivity contributes $\mathcal{Y}(\mu) = \frac{\lambda(Q_O)}{\mu+\lambda(Q_O)}$. In the overdamped-dominated part of the optimized bound one therefore obtains (up to γ-independent structural factors)

$$\Lambda_{\text{hypo}}(\gamma) \;\sim\; \frac{\gamma\,\mathcal{Y}(\mu)^2}{\gamma^2\,\mu^{-1}} \;=\; \frac{1}{\gamma}\,\mu\left(\frac{\lambda(Q_O)}{\mu+\lambda(Q_O)}\right)^2 \;=\; \frac{\lambda(Q_O)^2}{\gamma}\,\frac{\mu}{(\mu+\lambda(Q_O))^2}. \tag{D60}$$

Thus the "effective diffusion strength" is governed by the scalar function $\mu(\mu+\lambda(Q_O))^{-2}$, whose maximizer is $\mu = \lambda(Q_O)$. Therefore we fix:

$$\mu_* := \lambda(Q_O). \tag{D61}$$

At this value, the structural coefficients simplify to:

$$\mathcal{Y}_* := \mathcal{Y}(\mu_*) = \frac{1}{2}, \tag{D62}$$

$$\mathcal{Z}_*(\gamma) := \mathcal{Z}(\mu_*,\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}, \tag{D63}$$

where $\Theta := \frac{\sqrt{M_R}}{2}$ and $K_{\text{geom}} := k_{\text{reg}} + \kappa_{\text{ell}}$.

Optimization of the mixing weight α. We aim to maximize the determinant in the numerator of (D56), which is the dominant factor for the rate. The determinant as a function of α is:

$$D(\alpha) := M_{11}^0 M_{22}^0 - (M_{12}^0)^2 = \gamma\lambda_R\mathcal{Y}_*\alpha - \Big(\mathcal{Y}_* k_{\text{reg}} + \frac{1}{4}\mathcal{Z}_*(\gamma)^2\Big)\alpha^2. \tag{D64}$$

This is a concave parabola maximizing at:

$$\alpha_* := \frac{\gamma\lambda_R\mathcal{Y}_*}{2\left(\mathcal{Y}_* k_{\text{reg}} + \frac{1}{4}\mathcal{Z}_*(\gamma)^2\right)} = \frac{2\gamma\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_* k_{\text{reg}}}. \tag{D65}$$

At this optimal weight, the maximum determinant is:

$$D(\alpha_*) = \frac{(\gamma\lambda_R\mathcal{Y}_*)^2}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_* k_{\text{reg}}}. \tag{D66}$$

General lower bound. We substitute α_* into the full lower bound expression (D56). First, we compute the trace at the optimal weight:

$$\operatorname{Tr}(\mathbf{M}_0)|_{\alpha_*} = M_{11}^0 + M_{22}^0 = \big(\gamma\lambda_R - \alpha_* k_{\text{reg}}\big) + \alpha_*\mathcal{Y}_* \le \gamma\lambda_R + \alpha_*\mathcal{Y}_* = \gamma\lambda_R\left(1 + \frac{2\mathcal{Y}_*^2}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_* k_{\text{reg}}}\right). \tag{D67}$$

The convergence rate Λ is bounded by the ratio $D(\alpha_*)/\operatorname{Tr}(\mathbf{M}_0)|_{\alpha_*}$:

$$\Lambda \ge \frac{\frac{(\gamma\lambda_R\mathcal{Y}_*)^2}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_* k_{\text{reg}}}}{\gamma\lambda_R\left(1+\frac{2\mathcal{Y}_*^2}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_* k_{\text{reg}}}\right)} = \frac{\gamma\lambda_R\mathcal{Y}_*^2}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_* k_{\text{reg}} + 2\mathcal{Y}_*^2}. \tag{D68}$$

Substituting the explicit forms of $\mathcal{Y}_*$ and $\mathcal{Z}_*(\gamma)$:

$$\Lambda_{\text{hypo}}(\gamma) := \frac{\frac{1}{4}\,\gamma\,\lambda_R}{\big(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}\big)^2 + 2k_{\text{reg}} + \frac{1}{2}}. \tag{D69}$$

Using $k_{\rm reg} = \sqrt{M_R/\lambda_R}$, we may rewrite

$$\Theta = \frac{\sqrt{M_R}}{2}, \qquad K_{\rm geom} = k_{\rm reg} + \kappa_{\rm ell} = \sqrt{M_R/\lambda_R} + \kappa_{\rm ell}, \tag{D70}$$

so that

$$\Lambda_{\rm hypo}(\gamma) = \frac{\frac{1}{4}\,\gamma\,\lambda_R}{\left[\frac{\sqrt{M_R}}{2}\,\lambda(Q_O)^{-1/2}\,\gamma + \left(\sqrt{M_R/\lambda_R} + \kappa_{\rm ell}\right)\right]^2 + 2\sqrt{M_R/\lambda_R} + \frac{1}{2}}. \tag{D71}$$

This result provides a rigorous, explicit lower bound valid for all $\gamma > 0$ (provided η is small).

Asymptotic scaling analysis. Equation (D69) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.

- **Overdamped Limit ($\gamma \to \infty$):** The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term.

$$\lim_{\gamma\to\infty} \gamma\Lambda_{\rm hypo}(\gamma) = \frac{\lambda_R\,\mathcal{Y}_*^2\,\lambda(Q_O)}{\Theta^2} = \lambda(Q_O)\,\frac{\lambda_R}{M_R}. \tag{D72}$$

  Thus, $\Lambda_{\rm hypo}(\gamma) = \lambda(Q_O)\,\frac{\lambda_R}{M_R}\,\gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\rm hypo}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$.

- **Underdamped/Intermediate Regime:** For small or order-one γ, the denominator is controlled by the γ-independent structural constants $K_{\rm geom}$, $2k_{\rm reg}$, and $\frac{1}{2}$. These terms originate from the slow–fast coupling through $\mathcal{V}_{\rm skew}$ (captured by the solver bounds and regularity constants) and therefore prevent the estimate from degenerating even though $\mathcal{R}$ does not directly dissipate on $\mathcal{H}_S$. In other words, the transport mixes slow directions into dissipative fast directions, yielding a strictly positive hypocoercive lower bound in the small/intermediate-friction regime.

We now rigorously establish the stability condition by explicitly calculating the norm of the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ, explicitly tracking the dependency on the macroscopic spectral gap $\mu_* = \lambda(Q_O)$.

Explicit norm of the perturbation matrix. With the sharpened perturbation bookkeeping above, the perturbation matrix $\mathbf{E}$ has entries

$$E_{11} = b_{2,\rm eff} + \alpha_*\,k_P(\mu_*), \tag{D73}$$

$$E_{22} = b_{2,\rm eff} + \alpha_*\,\tilde{k}_P(\mu_*), \tag{D74}$$

$$E_{12} = \frac{1}{2}\,\alpha_*\left(k_P(\mu_*) + \tilde{k}_P(\mu_*)\right). \tag{D75}$$

Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$\|\mathbf{E}(\gamma)\| \le \max\{E_{11} + |E_{12}|,\, E_{22} + |E_{12}|\} = E_{22} + |E_{12}|. \tag{D76}$$

Therefore,

$$\|\mathbf{E}(\gamma)\| \le b_{2,\rm eff} + \alpha_*(\gamma)\left[\tfrac{1}{2}\,k_P(\mu_*) + \tfrac{3}{2}\,\tilde{k}_P(\mu_*)\right]. \tag{D77}$$

Recall from Lemma IV.5 that the bounds k_P and $\tilde{k}_P$ depend on $\mu_*^{-1/2}$. We define the *strictly constant* structural coefficients (independent of μ and γ):

$$C_{\rm sing} := 2(b_1 + b_2)k_{\mathcal{A}} + \frac{3}{4}(a_1 + a_2). \tag{D78}$$

Substituting the explicit optimal mixing weight $\alpha_*(\gamma) = \frac{2\gamma\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_*k_{\text{reg}}}$ derived above, the bound becomes:

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \frac{2\gamma\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}}}\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}. \tag{D79}$$

This expression is now fully explicit, revealing the interplay between the mixing weight (vanishing at large γ) and the macroscopic singularity of the solver.

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ requires that the perturbative spectral shift does not exceed the unperturbed gap. A sufficient condition is $\eta\|\mathbf{E}(\gamma)\| \leq \frac{1}{2}\Lambda_{\text{hypo}}(\gamma)$. This defines the explicit stability threshold function $\eta^*(\gamma)$:

$$\eta^*(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|}. \tag{D80}$$

Substituting the explicit form of $\Lambda_{\text{hypo}}(\gamma)$ from Eq. (D69) and the perturbation bound derived above:

$$\eta^*(\gamma) = \frac{\frac{\gamma\lambda_R\mathcal{Y}_*^2}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_*k_{\text{reg}}+2\mathcal{Y}_*^2}}{2b_{2,\text{eff}} + \frac{4\gamma\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_*k_{\text{reg}}}\, C_{\text{sing}}\lambda(Q_O)^{-1/2}}. \tag{D81}$$

Multiplying numerator and denominator by $\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}}$ simplifies the expression to

$$\eta^*(\gamma) = \frac{\gamma\lambda_R\mathcal{Y}_*^2}{2b_{2,\text{eff}}\big(\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}}\big) + 4\gamma\lambda_R\mathcal{Y}_*\, C_{\text{sing}}\lambda(Q_O)^{-1/2}} \tag{D82}$$

$$\times\, \frac{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}}}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}} + 2\mathcal{Y}_*^2}. \tag{D83}$$

Recalling $\mathcal{Y}_* = \frac{1}{2}$ and rewriting yields the more explicit form

$$\eta^*(\gamma) = \frac{\frac{1}{4}\,\gamma\lambda_R}{2b_{2,\text{eff}}\big(\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}}\big) + 2\gamma\lambda_R\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}} \times \frac{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}}}{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}}. \tag{D84}$$

Asymptotic analysis of stability. We analyze the scaling of $\eta^*(\gamma)$ in the limiting regimes. Note that $\mathcal{Z}_*(\gamma) \asymp \gamma$.

- **Overdamped Limit ($\gamma \to \infty$):** The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\text{eff}}\mathcal{Z}_*(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$.
  - If $b_{2,\text{eff}} > 0$ (symmetric perturbation present): The $\mathcal{O}(\gamma^2)$ term dominates the denominator. Thus, $\eta^*(\gamma) \asymp \gamma^{-1}$.
  - If $b_{2,\text{eff}} = 0$ (purely skew perturbation): The denominator is dominated by the skew term $\mathcal{O}(\gamma)$. Thus, $\eta^*(\gamma) \asymp \mathcal{O}(1)$.
- **Optimal Lift ($\gamma \sim \sqrt{\lambda(Q_O)}$):** At the optimal friction γ_{opt} (will be derived the optimization section later), we have $\mathcal{Z}_*(\gamma) \asymp \mathcal{O}(1)$. Consider the singular term in the denominator:

  $$\gamma_{\text{opt}} \cdot C_{\text{sing}}\lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1). \tag{D85}$$

  Crucially, the singularity cancels out exactly. This implies that at the optimal lift, the stability threshold is well-behaved:

  $$\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{D86}$$

This explicit derivation confirms that the stability estimates are robust and structurally consistent with the macroscopic limit.

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step involves explicit algebraic substitution to derive the final decay rate, ensuring that the singular scaling of the perturbation is correctly propagated.

Uniform norm equivalence. Recall the distortion parameter $\delta(\gamma)$ defined by

$$\delta(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\mu_*}\|. \tag{D87}$$

Using $\|\mathcal{A}_{\mu_*}\| \le k_{\mathcal{A}}\,\mu_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2+2k_{\mathrm{reg}}}$, we obtain the crude but uniform estimate

$$\delta(\gamma) \le 2\,\frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2}\,k_{\mathcal{A}}\,\mu_*^{-1/2} = \frac{\gamma\sqrt{\lambda_R}\,\mu_*^{-1/2}}{\left(\gamma\Theta\mu_*^{-1/2}+K_{\mathrm{geom}}\right)^2}. \tag{D88}$$

Setting $x := \gamma\Theta\mu_*^{-1/2}$ gives

$$\delta(\gamma) \le \frac{\sqrt{\lambda_R}}{\Theta}\,\frac{x}{(x+K_{\mathrm{geom}})^2} \le \frac{\sqrt{\lambda_R}}{4\,\Theta\,K_{\mathrm{geom}}} =: \delta_{\max}. \tag{D89}$$

Since $K_{\mathrm{geom}} \ge \sqrt{M_R/\lambda_R}$ and $\Theta = \sqrt{M_R}/2$, we obtain $\delta_{\max} \le \frac{\lambda_R}{2M_R}$. In particular, after enlarging M_R if necessary (which only weakens constants and does not affect the scaling conclusions), we may assume $M_R \ge \lambda_R$ and hence $\delta_{\max} \le \frac{1}{2}$. This implies that the condition number of the norm equivalence is uniformly bounded for all γ:

$$C_\gamma := \sqrt{\frac{1+\delta(\gamma)}{1-\delta(\gamma)}} \le \sqrt{\frac{1+1/2}{1-1/2}} \tag{D90}$$

$$= \sqrt{3}. \tag{D91}$$

Thus, we have the uniform equivalence bounds:

$$\frac{1}{3}\|f\|^2 \le \frac{1}{2}(1-\delta(\gamma))\|f\|^2 \le \mathcal{E}_{\alpha_*,\mu_*}(f) \le \frac{1}{2}(1+\delta(\gamma))\|f\|^2 \le \frac{3}{4}\|f\|^2. \tag{D92}$$

Explicit calculation of the decay rate. Set

$$c(\gamma,\eta) := \Lambda_{\mathrm{hypo}}(\gamma) - \eta\,\|\mathbf{E}(\gamma)\|. \tag{D93}$$

The algebraic dissipation estimate reads

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le -c(\gamma,\eta)\,\|f_t\|^2. \tag{D94}$$

Next we use the norm equivalence from Proposition IV.7 in the convenient form

$$\frac{1}{2}(1-\delta(\gamma))\,\|f\|^2 \le \mathcal{E}_{\alpha_*,\mu_*}(f) \le \frac{1}{2}(1+\delta(\gamma))\,\|f\|^2, \tag{D95}$$

where $\delta(\gamma) := 2\alpha_*(\gamma)\,\|\mathcal{A}_{\mu_*}\| \in (0,1)$. The upper bound implies

$$\|f\|^2 \ge \frac{2}{1+\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\mu_*}(f). \tag{D96}$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le -\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t). \tag{D97}$$

By Grönwall's inequality,

$$\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le \exp\Big(-\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}\,t\Big)\,\mathcal{E}_{\alpha_*,\mu_*}(f_0). \tag{D98}$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$\|f_t\|^2 \le \frac{2}{1-\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le \frac{1+\delta(\gamma)}{1-\delta(\gamma)}\,e^{-\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}t}\,\|f_0\|^2. \tag{D99}$$

Therefore

$$\|f_t\| \le \sqrt{\frac{1+\delta(\gamma)}{1-\delta(\gamma)}}\,e^{-\lambda_\eta t}\,\|f_0\|, \qquad \lambda_\eta := \frac{c(\gamma,\eta)}{1+\delta(\gamma)}. \tag{D100}$$

We now substitute the explicit expressions derived in the previous steps. For the unperturbed rate, using $\mathcal{Y}_* = \frac{1}{2}$:

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\gamma}{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}}, \quad \text{with } \mathcal{Z}_*(\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}. \tag{D101}$$

For the perturbation norm, we use the sharp bound (D77) at $\mu_* = \lambda(Q_O)$ to handle the macroscopic singularity:

$$\|\mathbf{E}(\gamma)\| \le b_{2,\text{eff}} + \underbrace{\frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}}}}_{\alpha_*(\gamma)}\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}. \tag{D102}$$

Substituting these into the expression for λ_η and using the uniform bound $1+\delta(\gamma) \le 3/2$:

$$\lambda_\eta \ge \frac{2}{3}\left[\frac{\frac{1}{4}\lambda_R\,\gamma}{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}} - \eta\left(b_{2,\text{eff}} + \frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}}}\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}\right)\right]. \tag{D103}$$

This expression constitutes the rigorous lower bound. It explicitly demonstrates that the term driving the instability ($C_{\text{sing}}\lambda(Q_O)^{-1/2}$) is multiplied by the mixing weight factor $\frac{\gamma}{\mathcal{Z}_*^2}$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal lift, ensuring the boundedness of the perturbation.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate:

$$\|e^{t\mathcal{L}}f\|_{\mathcal{H}} \le \sqrt{3}\,e^{-\lambda_\eta t}\|f\|_{\mathcal{H}}. \tag{D104}$$

This completes the proof.

5. Optimization over Friction γ: The Kinetic Acceleration

We finally determine the optimal friction coefficient γ_{opt} that maximizes the hypocoercive convergence rate. This optimization reveals the "quadratic speedup" phenomenon, where the rate is improved from the diffusive scaling $\mathcal{O}(\lambda(Q_O))$ to the kinetic scaling $\mathcal{O}(\sqrt{\lambda(Q_O)})$.

Recall the explicit lower bound derived in Eq. (D69):

$$\Lambda_{\text{hypo}}(\gamma) = \frac{C_{\text{num}}\gamma}{(C_1\gamma + C_0)^2 + C_{\text{tr}}}, \tag{D105}$$

where the structural constants (independent of γ) are defined as:

$$C_{\text{num}} := \frac{1}{4}\,\lambda_R, \tag{D106}$$

$$C_1 := \Theta\lambda(Q_O)^{-1/2} = \frac{\sqrt{M_R}}{2}\,\lambda(Q_O)^{-1/2}, \tag{D107}$$

$$C_0 := K_{\text{geom}} = \sqrt{M_R/\lambda_R} + \kappa_{\text{ell}}, \tag{D108}$$

$$C_{\text{tr}} := 2k_{\text{reg}} + \frac{1}{2} = 2\sqrt{M_R/\lambda_R} + \frac{1}{2}. \tag{D109}$$

Optimal friction coefficient. The function $f(\gamma) = \frac{\gamma}{(C_1\gamma+C_0)^2+C_{\text{tr}}}$ attains its global maximum on $(0,\infty)$ when the derivative vanishes. Computing the critical point:

$$\frac{d}{d\gamma}\log f(\gamma) = \frac{1}{\gamma} - \frac{2C_1(C_1\gamma + C_0)}{(C_1\gamma + C_0)^2 + C_{\text{tr}}} = 0. \tag{D110}$$

This implies the balance condition $(C_1\gamma + C_0)^2 + C_{\text{tr}} = 2C_1\gamma(C_1\gamma + C_0)$. Simplifying this quadratic equation yields $C_1^2\gamma^2 = C_0^2 + C_{\text{tr}}$. Thus, the optimal friction is explicitly:

$$\gamma_{\text{opt}} = \frac{\sqrt{C_0^2 + C_{\text{tr}}}}{C_1} = \frac{\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}}}{\Theta}\sqrt{\lambda(Q_O)}. \tag{D111}$$

We define the structural prefactor $C_{\text{lift}} := \Theta^{-1}\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}}$. The optimal lift scales as the square root of the macroscopic spectral gap:

$$\gamma_{\text{opt}} = C_{\text{lift}}\sqrt{\lambda(Q_O)}. \tag{D112}$$

We evaluate the maximal rate $\Lambda_{\max} := \Lambda_{\text{hypo}}(\gamma_{\text{opt}})$. Substituting the optimality condition $C_1\gamma_{\text{opt}} = \sqrt{C_0^2 + C_{\text{tr}}}$ into the denominator of (D105):

$$\text{Denom} = (\sqrt{C_0^2 + C_{\text{tr}}} + C_0)^2 + C_{\text{tr}} = 2\left(C_0^2 + C_{\text{tr}} + C_0\sqrt{C_0^2 + C_{\text{tr}}}\right). \tag{D113}$$

The numerator is $C_{\text{num}}\gamma_{\text{opt}} = C_{\text{num}}C_1^{-1}\sqrt{C_0^2 + C_{\text{tr}}}$. Thus:

$$\Lambda_{\max} = \frac{C_{\text{num}}}{2C_1(\sqrt{C_0^2 + C_{\text{tr}}} + C_0)}. \tag{D114}$$

Substituting the scaling $C_1 = \Theta\lambda(Q_O)^{-1/2}$, we isolate the dependency on $\lambda(Q_O)$:

$$\Lambda_{\max} = \left[\frac{C_{\text{num}}}{2\Theta(\sqrt{C_0^2 + C_{\text{tr}}} + C_0)}\right]\sqrt{\lambda(Q_O)}. \tag{D115}$$

This explicitly confirms the quadratic speedup:

$$\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)} \quad \text{vs.} \quad \Lambda_{\text{overdamped}} \asymp \lambda(Q_O). \tag{D116}$$

Stability threshold at optimal lift. We determine the admissible perturbation range $\eta^*(\gamma_{\text{opt}})$ at this accelerated regime. From Eq. (D81), the threshold depends on the operator norm of the perturbation matrix $\|\mathbf{E}(\gamma)\|$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at γ_{opt}:

$$\alpha_*(\gamma_{\text{opt}}) = \frac{2\gamma_{\text{opt}}\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma_{\text{opt}})^2 + 4\mathcal{Y}_*k_{\text{reg}}}. \tag{D117}$$

Using the scaling $\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}$:

- Numerator: $\gamma_{\rm opt}\lambda_R \asymp \sqrt{\lambda(Q_O)}$.
- Denominator: $\mathcal{Z}_*(\gamma_{\rm opt}) = C_1\gamma_{\rm opt} + C_0 = \sqrt{C_0^2 + C_{\rm tr}} + C_0 = \text{const}$ (independent of $\lambda(Q_O)$).

Thus, $\alpha_*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$. We now apply the perturbation bound (D77), which accounts for the macroscopic singularity:

$$\|\mathbf{E}(\gamma_{\rm opt})\| \le b_{2,\rm eff} + \alpha_*(\gamma_{\rm opt})\, C_{\rm sing}\, \lambda(Q_O)^{-1/2}. \tag{D118}$$

The key observation is the cancellation of the singularity by the mixing weight:

$$\alpha_*(\gamma_{\rm opt}) \cdot \lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \frac{1}{\sqrt{\lambda(Q_O)}} \asymp \mathcal{O}(1). \tag{D119}$$

Consequently, the perturbation matrix norm is uniformly bounded at the optimal lift:

$$\|\mathbf{E}(\gamma_{\rm opt})\| \le b_{2,\rm eff} + K_{\rm robust}, \quad \text{where } K_{\rm robust} \asymp \mathcal{O}(1). \tag{D120}$$

The stability threshold $\eta^* = \Lambda_{\max}/(2\|\mathbf{E}\|)$ therefore scales as:

$$\eta^*(\gamma_{\rm opt}) \asymp \frac{\sqrt{\lambda(Q_O)}}{b_{2,\rm eff} + K_{\rm robust}} \asymp \sqrt{\lambda(Q_O)}. \tag{D121}$$

This proves that the accelerated convergence retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ threshold even in the presence of singular macroscopic interactions.

Appendix E: Upper bound via singular value gap

This appendix proves Theorem V.4.

1. Exponential decay implies a singular value-gap bound

For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$s(T) := \inf_{f \in \mathrm{dom}(T)\cap\mathcal{H}_0\setminus\{0\}} \frac{\|Tf\|}{\|f\|}. \tag{E1}$$

Lemma E.1 (From exponential decay to a singular value-gap bound)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *If there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0, \tag{E2}$$

then 0 *belongs to the resolvent set of* $\mathcal{L}|_{\mathcal{H}_0}$ *and*

$$\nu \le (1 + \log C)\, s(\mathcal{L}). \tag{E3}$$

Proof. Set $T := \frac{\log C}{\nu}$. For $f \in \mathcal{H}_0$, define

$$u := -\int_0^\infty e^{t\mathcal{L}} f\, dt. \tag{E4}$$

Since $\{e^{t\mathcal{L}}\}_{t\ge 0}$ is a (subspace) contraction semigroup on $\mathcal{H}_0$ in our L^2 setting, we have $\|e^{t\mathcal{L}} f\| \le \|f\|$ for all $t \ge 0$. Together with the assumed exponential bound, this implies that

the Bochner integral converges in $\mathcal{H}_0$ and

$$\|u\| \le \int_0^T \|e^{t\mathcal{L}} f\|\, dt + \int_T^\infty \|e^{t\mathcal{L}} f\|\, dt \tag{E5}$$

$$\le T\,\|f\| + \int_T^\infty C e^{-\nu t}\, dt\, \|f\| \tag{E6}$$

$$= \left(\frac{\log C}{\nu} + \frac{1}{\nu}\right) \|f\|. \tag{E7}$$

Moreover, $u \in \mathrm{dom}(\mathcal{L})$ and differentiating under the integral gives

$$\mathcal{L}u = -\int_0^\infty \mathcal{L} e^{t\mathcal{L}} f\, dt = -\int_0^\infty \partial_t\big(e^{t\mathcal{L}} f\big)\, dt = f. \tag{E8}$$

Thus $(\mathcal{L}|_{\mathcal{H}_0})^{-1}$ exists and satisfies $\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le (1+\log C)/\nu$, hence $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge \nu/(1+\log C)$, i.e. $\nu \le (1+\log C)\, s(\mathcal{L})$. □

Lemma E.2 (Relaxation time is bounded below by the singular value gap)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *Define the* L^2 *relaxation time*

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{t \ge 0\,;\ \|e^{t\mathcal{L}} f\| \le e^{-1}\,\|f\| \text{ for all } f \in \mathcal{H}_0\Big\}. \tag{E9}$$

Then

$$t_{\mathrm{rel}}(\mathcal{L}) \ge \frac{1}{2\, s(\mathcal{L})}. \tag{E10}$$

This L^2 *bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains; compare*[26].

Proof. Since $\{e^{t\mathcal{L}}\}_{t\ge 0}$ is a (subspace) contraction semigroup on $\mathcal{H}_0$ in our L^2 setting, we have $\|e^{t\mathcal{L}}\|_{\mathcal{H}_0\to\mathcal{H}_0} \le 1$ for all $t \ge 0$. By the definition of $t_{\mathrm{rel}}(\mathcal{L})$, for all $f \in \mathcal{H}_0$ and $t \ge t_{\mathrm{rel}}(\mathcal{L})$ we can write $t = n\, t_{\mathrm{rel}}(\mathcal{L}) + r$ with $n \in \mathbb{N}$ and $r \in [0, t_{\mathrm{rel}}(\mathcal{L}))$, and obtain from the semigroup property

$$\|e^{t\mathcal{L}}\| \le \|e^{r\mathcal{L}}\|\, \|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\|^n \le e^{-n}.$$

As in the proof of Lemma E.1, the inverse on $\mathcal{H}_0$ is given by $(\mathcal{L}|_{\mathcal{H}_0})^{-1} = -\int_0^\infty e^{t\mathcal{L}}\, dt$, hence

$$\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le \int_0^\infty \|e^{t\mathcal{L}}\|\, dt \le \sum_{n=0}^\infty \int_0^{t_{\mathrm{rel}}(\mathcal{L})} e^{-n}\, dr = \frac{t_{\mathrm{rel}}(\mathcal{L})}{1-e^{-1}} \le 2\, t_{\mathrm{rel}}(\mathcal{L}).$$

Therefore $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge (2\, t_{\mathrm{rel}}(\mathcal{L}))^{-1}$. □

2. Test function and completion of the proof

Proof of Theorem V.4. Let $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0 \cap \mathcal{K}^\perp$ be a normalized eigenfunction of Q_O with

$$Q_O \phi = -\lambda(Q_O)\, \phi, \qquad \|\phi\| = 1. \tag{E11}$$

Define the first-order corrector

$$\psi := \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} \phi \in \mathcal{H}_F, \tag{E12}$$

and the test function $f_\gamma := \phi - \gamma^{-1}\psi$. Since $\phi \in \mathcal{H}_S$ and $\psi \in \mathcal{H}_F$ we have $\langle \phi, \psi \rangle = 0$ and therefore

$$\|f_\gamma\|^2 = \|\phi\|^2 + \gamma^{-2}\|\psi\|^2 \geq 1. \tag{E13}$$

Using $\mathcal{R}\phi = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\text{skew}}\phi \in \mathcal{H}_F$, we obtain

$$\mathcal{R}\psi = \mathcal{R}\mathcal{R}^+\mathcal{V}_{\text{skew}}\phi = P_F\mathcal{V}_{\text{skew}}\phi = \mathcal{V}_{\text{skew}}\phi. \tag{E14}$$

Consequently,

$$\mathcal{L}f_\gamma = \big(\gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}\big)\big(\phi - \gamma^{-1}\psi\big) \tag{E15}$$

$$= \underbrace{\mathcal{V}_{\text{skew}}\phi - \mathcal{R}\psi}_{=0} - \gamma^{-1}\mathcal{V}_{\text{skew}}\psi \;+\; \eta\mathcal{V}_{\text{pert}}f_\gamma. \tag{E16}$$

We decompose the remaining transport term into slow/fast components:

$$\mathcal{V}_{\text{skew}}\psi = P_S\mathcal{V}_{\text{skew}}\psi + P_F\mathcal{V}_{\text{skew}}\psi. \tag{E17}$$

For the slow part, using $Q_O := -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$ and $P_S\phi = \phi$, we have

$$P_S\mathcal{V}_{\text{skew}}\psi = P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}\phi = -Q_O\phi = \lambda(Q_O)\,\phi. \tag{E18}$$

For the fast part, apply the flux regularity estimate (24) with the slow input ϕ. Writing $g_\phi := P_F\mathcal{V}_{\text{skew}}P_S\phi = \mathcal{V}_{\text{skew}}\phi$ and noting that $(-\mathcal{R}^+)g_\phi = -\psi$ on $\mathcal{H}_F$, we obtain

$$\|P_F\mathcal{V}_{\text{skew}}\psi\| = \|P_F\mathcal{V}_{\text{skew}}P_F\psi\| = \|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\phi\| \leq \kappa_{\text{ell}}\,\|(-Q_O)\phi\| = \kappa_{\text{ell}}\,\lambda(Q_O). \tag{E19}$$

Combining the slow and fast contributions yields

$$\|\mathcal{V}_{\text{skew}}\psi\| \leq (1 + \kappa_{\text{ell}})\,\lambda(Q_O). \tag{E20}$$

Hence

$$\|\mathcal{L}f_\gamma\| \leq \frac{(1 + \kappa_{\text{ell}})\,\lambda(Q_O)}{\gamma} + \eta\,\|\mathcal{V}_{\text{pert}}f_\gamma\|. \tag{E21}$$

To bound the perturbation, use Assumption IV.1.4 to estimate

$$\|\mathcal{V}_{\text{pert}}f\| \leq \|\mathcal{K}_{\text{pert}}f\| + \|\mathcal{S}_{\text{pert}}f\| \tag{E22}$$

$$\leq (a_1 + a_2)\,\|(-\mathcal{R})^{1/2}f\| + (b_1 + b_2)\,\|f\|, \qquad f \in \text{dom}((-\mathcal{R})^{1/2}). \tag{E23}$$

Since $\phi \in \ker(\mathcal{R})$ and $f_\gamma = \phi - \gamma^{-1}\psi$, we have

$$\|(-\mathcal{R})^{1/2}f_\gamma\| = \gamma^{-1}\,\|(-\mathcal{R})^{1/2}\psi\|. \tag{E24}$$

Finally, using $\psi = \mathcal{R}^+\mathcal{V}_{\text{skew}}\phi$ and the definition of Q_O,

$$\|(-\mathcal{R})^{1/2}\psi\|^2 = \langle(-\mathcal{R})\psi, \psi\rangle = \langle\mathcal{V}_{\text{skew}}\phi, (-\mathcal{R}^+)\mathcal{V}_{\text{skew}}\phi\rangle = -\langle Q_O\phi, \phi\rangle = \lambda(Q_O). \tag{E25}$$

Therefore $\|(-\mathcal{R})^{1/2}f_\gamma\| = \sqrt{\lambda(Q_O)}/\gamma$, and

$$\|\mathcal{V}_{\text{pert}}f_\gamma\| \leq (b_1 + b_2)\,\|f_\gamma\| + (a_1 + a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{E26}$$

Dividing by $\|f_\gamma\| \geq 1$ gives

$$\frac{\|\mathcal{L}f_\gamma\|}{\|f_\gamma\|} \leq \frac{(1 + \kappa_{\text{ell}})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1 + b_2) + \eta\,(a_1 + a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{E27}$$

By definition of $s(\mathcal{L})$, this implies $s(\mathcal{L}) \leq \|\mathcal{L}f_\gamma\|/\|f_\gamma\|$. Combining with Lemma E.1 yields the claimed bound on ν. □